

々 Imperative commands (single line commands to create the objects)

▲ **kubectl run my-pod --image=nginx** [here *run* is used, not *create*]

✧ **restartPolicy**: always

always: restart pod even if the container exits with exit code 0

onFailure: restart pod only if container's exit code is non-zero

never: never restart the pod even if container stops.

✧ **namespace**: default

it is *logical* grouping of a cluster to separate logically like dev, prod, ..etc. You can set the resource allocation according to namespace as well.

✧ **labels**: *run: my-pod*

✧ **port**: not exposed

[port is never exposed from container though, means from Service you can route to any port of the pod you want; *port* only there for ease of querying pods according to port or to display the usable port of the container to user when they get *kubectl describe pod <pod name>*]

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: my-pod
    name: my-pod
spec:
  containers:
  - image: nginx
    name: my-pod
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

(default configs of the command)

▲ **kubectl create deployment my-deploy --image=nginx** [here *create* is used]

✧ Deployment name: my-deploy

✧ ReplicaSet: my-deploy-*<hash>*

✧ Pod: my-deploy-*<hash>*-*<random>*

- labels: **app:my-deploy**
- These are the things kubectl create commands can create. pod cannot be created using kubectl create command

clusterrole	Create a cluster role
clusterrolebinding	Create a cluster role binding for a particular cluster role
configmap	Create a config map from a local file, directory or literal value
cronjob	Create a cron job with the specified name
deployment	Create a deployment with the specified name
ingress	Create an ingress with the specified name
job	Create a job with the specified name
namespace	Create a namespace with the specified name
poddisruptionbudget	Create a pod disruption budget with the specified name
priorityclass	Create a priority class with the specified name
quota	Create a quota with the specified name
role	Create a role with single rule
rolebinding	Create a role binding for a particular role or cluster role
secret	Create a secret using a specified subcommand
service	Create a service using a specified subcommand
serviceaccount	Create a service account with the specified name
token	Request a service account token

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: my-deployment
  name: my-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: my-deployment
  strategy: {}
  template:
    metadata:
      labels:
        app: my-deployment
    spec:
      containers:
        - image: nginx
          name: nginx
          resources: {}
status: {}
```

(default configs of the command)

labels are automatically added.

- **ReplicaSet** cannot be created directly using imperative way, because Deployment is there to create and manage replicaset.
- **kubectl expose** takes replication controller, service, deployment, or pod and creates a Service to expose that.

```
vagrant@controlplane:~$ kubectl get svc,pod
NAME                                TYPE                CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/kubernetes                  ClusterIP           10.96.0.1     <none>         443/TCP    36m

NAME    READY   STATUS    RESTARTS   AGE
pod/my-pod  1/1     Running   0          34s
vagrant@controlplane:~$ kubectl expose pod my-pod --port 80
service/my-pod exposed
vagrant@controlplane:~$ kubectl get svc,pod
NAME                                TYPE                CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/kubernetes                  ClusterIP           10.96.0.1     <none>         443/TCP    37m
service/my-pod                      ClusterIP           10.96.68.232  <none>         80/TCP     6s

NAME    READY   STATUS    RESTARTS   AGE
pod/my-pod  1/1     Running   0          76s
```

you can see here, It created a service named *my-pod*. Its of type **ClusterIP** which is default.

- If you give **--type=NodePort** flag, then it'll create a service of type NodePort

```
vagrant@controlplane:~$ kubectl expose pod my-pod --port 80 --type NodePort
service/my-pod exposed
vagrant@controlplane:~$ kubectl get svc,pod
NAME                                TYPE                CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/kubernetes                  ClusterIP           10.96.0.1     <none>         443/TCP    39m
service/my-pod                      NodePort            10.96.119.232 <none>         80:32530/TCP 3s
```

The collage shows the following commands in terminal windows:

- `$ kubectl run --image=nginx nginx`
- `$ kubectl expose deployment nginx --port 80`
- `$ kubectl scale deployment nginx --replicas=5`
- `$ kubectl create -f nginx.yaml`
- `$ kubectl delete -f nginx.yaml`
- `$ kubectl create deployment --image=nginx nginx`
- `$ kubectl edit deployment nginx`
- `$ kubectl set image deployment nginx nginx=nginx:1.18`
- `$ kubectl replace -f nginx.yaml`

[these are the imperative commands]

create -f is imperative because if already the same config present it'll fail, where as **apply -f** will not fail.

🔗 Important trick

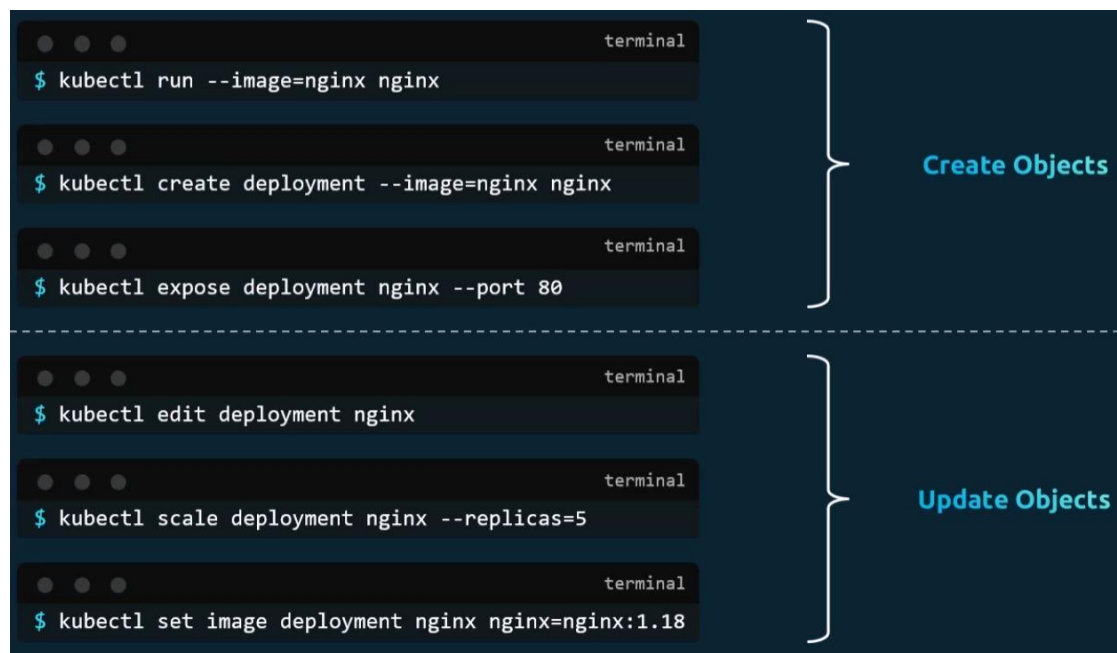
- **kubectl create deployment my-deploy --image=nginx --dry-run=client -o yaml**

kubectl run my-pod --image=nginx --dry-run=client -o yaml

- it'll give the yaml file of default deployment, remove the unnecessary fields, customize that, and create deployment
- Because of the flag **--dry-run=client** it'll not create the deployment, just simulate and give the output as yaml.

➤ **kubectl get pod my-pod -o yaml**

- It'll give the running config of the pod as an yaml.
- Can be done with any types of objects i.e. pod, replica set, container runtime, deployment ..etc
- But it'll contain a lot of things, that needs to be removed.



々

々 **edit vs replace** command

- **kubectl edit** command open the running status of the object (pod, replicaset, replication controller, deployment, service ..etc) in a yaml file.

This configurations are there in the memory, not stored in the yaml file that was being used to create the object.

So, if you do **kubectl edit** then it'll not have any track of the changes. If someone later want to update the object, and he sees the yaml file present (not the running configuration yaml) and try to update, it might do some wrong changes in the running object.

- **kubectl replace** command uses the physical yaml file to update the running object.

It does **dumb** update in the running object. Whatever there in the physical yaml, it'll update all the things and delete the configs which are not present in that physical yaml file.

By default it doesn't delete the current object (if anything doesn't match with physical yaml), to replace force fully you can give the flag **--force**.

```
vagrant@controlplane:~$ kubectl replace --force -f pd.yaml
pod "my-pod" deleted from default namespace
pod/my-pod replaced
```

It looks similar to **kubectl apply**, but **kubectl apply** does **intelligent** update. It finds the **diff** between running object and the physical yaml (used to update the object) and update those only.

It **doesn't delete any configs** which are not present in the current yaml but there in the running object.

kubectl apply is **declarative** command.

- ✧ If there are multiple files out of which objects will have to be created, then just give the path of the directory where those are present to **kubectl apply**, it'll create all those.

kubectl apply -f <path where all files are present>

--- kubectl apply internals ---

々 [here I'll take an example of a deployment named *my-dep*]

々 There are 3 things.

yaml (file that you write)

last applied config (when you execute *kubectl get deploy my-dep -o yaml*)

[JSON format]

live config (when you execute *kubectl describe deploy my-dep*)

[*deploy* is shorthand of *deployment*]

▲ Whenever you do *declarative* change using a *yaml* i.e.

kubectl apply -f my-dep.yaml

it changes both *live config*, *last applied config*. and obviously *yaml* is changed by you only.

▲ Whenever you do *imperative* change using the command i.e.

kubectl scale deployment my-dep --replicas=5

it changes only *live config*. *Last applied config* will not be changed. And of course, *yaml* has not been changed.

command type	yaml	last applied config	live config
declarative	Y	Y	Y
imperative	N	N	Y

々 There are a few commands [*deployment is used for example; can be used for any objects*]:

▲ *kubectl diff -f <yaml file name>*

compare the *yaml* with the *live config* along with *last applied* and display how it'll look like after you execute *kubectl apply*.

▲ *kubectl apply view-last-updated deployment <deployment name>*

show the last updated configs [which is present actually in JSON] in *yaml* format cleanly.

▲ *kubectl get deployment <deployment name> -o yaml*

kubectl get -f <deployment yaml file name> -o yaml [it fetch from the file]

Both will display the *live configs* of the deployment object.

Also, there will be a JSON structure under *annotations* field where you can see *last updated config*.

々 Stage-1: [create a deployment from yaml]

▲ Initial yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16.1
        ports:
        - containerPort: 80
```

▲ **kubectl diff -f <yaml file name>** will display like below

```
vagrant@controlplane:~$ kubectl diff -f simple deployment.yaml
diff -u -N /tmp/LIVE-3981462885/apps.v1.Deployment.default.ng-d
--- /tmp/LIVE-3981462885/apps.v1.Deployment.default.ng-dep
+++ /tmp/MERGED-1106384729/apps.v1.Deployment.default.ng-dep
@@ -0,0 +1,41 @@
+apiVersion: apps/v1
+kind: Deployment
+metadata:
+  creationTimestamp: "2026-04-30T21:32:08Z"
+  generation: 1
+  name: ng-dep
+  namespace: default
```

[cropped]

▲ Run the deployment using: *kubectl apply -f <yaml file name>*

- ⚡ **kubectl apply view-last-applied deployment <deployment name>**

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  name: ng-dep
  namespace: default
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

々 Stage 2: [change image using imperative command]

- ⚡ **kubectl set image deploy <deployment name> <container name>=<new image>**

kubectl set image deploy ng-dep nginx=nginx:1.14.2

```
vagrant@controlplane:~$ kubectl set image deploy ng-dep nginx=nginx:1.14.2
deployment.apps/ng-dep image updated
```

- ⚡ **kubectl diff -f <deployment file name>**

```
vagrant@controlplane:~$ kubectl diff -f simple_deployment.yaml
diff -u -N /tmp/LIVE-2986659376/apps.v1.Deployment.default.ng-dep
--- /tmp/LIVE-2986659376/apps.v1.Deployment.default.ng-dep
+++ /tmp/MERGED-1145231250/apps.v1.Deployment.default.ng-dep
@@ -6,7 +6,7 @@
   kubernetes.io/last-applied-configuration: |
     {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"name":"ng-dep","namespace":"default","resourceVersion":"33813"},"spec":{"replicas":3,"selector":{"matchLabels":{"app":"nginx"},"template":{"metadata":{"labels":{"app":"nginx"},"spec":{"containers":[{"image":"nginx:1.14.2","name":"nginx","ports":[{"containerPort":80}]}]}}}}}
-   - image: nginx:1.14.2
+   - image: nginx:1.16.1
     imagePullPolicy: IfNotPresent
```

[you can see, yaml file has *nginx:1.16.1* but live config has *nginx:1.14.2*]

[so, its showing - on *nginx:1.14.2* and + on *nginx:1.16.1*]

⚡ **kubectl apply view-last-applied deploy <deployment name>**

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  name: ng-dep
  namespace: default
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

[its showing image as *nginx:1.16.1* because imperative change doesn't effect last applied config]

々 Stage-3: [update yaml, remove replica field, then declarative update]

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16.1
        ports:
        - containerPort: 80
```

```
vagrant@controlplane:~$ kubectl diff -f simple_deployment.yaml
diff -u -N /tmp/LIVE-3508972230/apps.v1.Deployment.default.ng-dep
--- /tmp/LIVE-3508972230/apps.v1.Deployment.default.ng-dep
+++ /tmp/MERGED-2827279498/apps.v1.Deployment.default.ng-dep
@@ -6,14 +6,14 @@
   kubectl.kubernetes.io/last-applied-configuration: |
     {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"name":"ng-dep","namespace":"default","resourceVersion":"33813","uid":"92189358-0dd5-4f8f-8144-1936af5322ab"},"spec":{"selector":{"matchLabels":{"app":"nginx"},"template":{"metadata":{"labels":{"app":"nginx"},"spec":{"containers":[{"image":"nginx:1.16.1","name":"nginx","ports":[{"containerPort":80}]}]}}}}}
-   generation: 2
+   generation: 3
   name: ng-dep
   namespace: default
   resourceVersion: "33813"
   uid: 92189358-0dd5-4f8f-8144-1936af5322ab
 spec:
   progressDeadlineSeconds: 600
-   replicas: 3
+   replicas: 1
   revisionHistoryLimit: 10
   selector:
     matchLabels:
@@ -29,7 +29,7 @@
     app: nginx
   spec:
     containers:
-     - image: nginx:1.14.2
+     - image: nginx:1.16.1
```

[its showing if I update, *replica* and *image* will be updated]

- Now in last applied config, *replicas* field would have been removed because we deleted that field in yaml.

```
vagrant@controlplane:~$ kubectl apply view-last-applied -f simple_deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  name: ng-dep
  namespace: default
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

[replicas is not there, image is *nginx:1.16.1*]

♣ *live config*

```
spec:
  progressDeadlineSeconds: 600
  replicas: 1
```

```
spec:
  containers:
  - image: nginx:1.16.1
    imagePullPolicy: IfNotPresent
```

[the image and replicas got changed]

々 Its looking like, why there is need of *last applied config* because when we apply, the live config is becoming same as the yaml.

々 Stage-1: [create declarative from yaml]

♣ See the yaml now

```
vagrant@controlplane:~$ cat simple_deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
  labels:
    app: ng-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16.1
        ports:
        - containerPort: 80
```

[added one label only]

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  labels:
    app: ng-deployment
  name: ng-dep
  namespace: default
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

[last applied config will be same as yaml because of declarative change]

々 Stage-2: [add one more label in imperative way; *kubectl edit* command]

```
labels:
  app: ng-deployment
  keyx: valx
```

[live config changed]

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  labels:
    app: ng-deployment
  name: ng-dep
```

[last applied config will be same because it was imperative update]

々 Stage-3: [update the replicas using declarative]

```
vagrant@controlplane:~$ cat simple_deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
  labels:
    app: ng-deployment
spec:
  replicas: 3
  selector:
```

[updated yaml

file]


```

creationTimestamp: "2026-04-16T16:07:16Z"
- generation: 1
+ generation: 2
labels:
  app: ng-deployment
  keyx: valx
@@ -16,7 +16,7 @@
uid: 5071163c-87e2-49db-a1a0-000000000000
spec:
  progressDeadlineSeconds: 600
- replicas: 1
+ replicas: 3
revisionHistoryLimit: 10

```

⚡ [ignore generation]

[`kubectl diff -f <deployment file name>`]

you can see, its not showing *keyx: valx* label to be deleted when it is not there in our current yaml file.

⚡ `kubectl apply -f <deployment file name>`

```

vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  labels:
    app: ng-deployment
  name: ng-dep
  namespace: default
spec:
  replicas: 3

```

⚡

[last applied config]

```

creationTimestamp: "2026-04-16T16:07:16Z"
generation: 2
labels:
  app: ng-deployment
  keyx: valx
name: ng-dep
namespace: default
resourceVersion: "37619"
uid: 5071163c-87e2-49db-a1a0-000000000000
spec:
  progressDeadlineSeconds: 600
  replicas: 3

```

⚡

[live config, replicas got increased but that label (*keyx: valx*) is still there]

♪ So, how does ***kubectl apply*** work then?

⚡ It does the below

Sets fields, that appear in the configuration (yaml) file, in the live

configuration.

Clears fields, that removed from the configuration (yal) file, in the live configuration.

- It does the above after comparing with *last applied field*.
- If any field is there in [either of *yaml* or *last applied config*] (or) [the field is present in both *yaml* and *last applied config*] then that field will be updated in the *live config*.
- In the previous example. The label (key: valx) was not there in both *yaml* and *last applied config*, so it didn't track that.

🔗 In Simple Words it does the below

It tracks the configs which are present in either *yaml* or *last applied config* and updates the *live config* accordingly. If something is there in *live config* but not there in both *yaml* and *last applied config* then it'll skip that.

Merging changes to primitive fields

Primitive fields are replaced or cleared.

Note: - is used for "not applicable" because the value is not used.

Field in object configuration file	Field in live object configuration	Field in last-applied-configuration	Action
Yes	Yes	-	Set live to configuration file value.
Yes	No	-	Set live to local configuration.
No	-	Yes	Clear from live configuration.
No	-	No	Do nothing. Keep live value.

Merging changes to map fields

Fields that represent maps are merged by comparing each of the subfields or elements of the map:

Note: - is used for "not applicable" because the value is not used.

Key in object configuration file	Key in live object configuration	Field in last-applied-configuration	Action
Yes	Yes	-	Compare sub fields values.
Yes	No	-	Set live to local configuration.
No	-	Yes	Delete from live configuration.
No	-	No	Do nothing. Keep live value.

yaml	last applied config	live config	RES
Y	Y	Y/N	yaml : Y last : update to yaml live : update to yaml
Y	N	Y/N	yaml : Y last : update to yaml live : update to yaml
N	Y	Y/N	yaml : N last : remove live : remove
N	N	Y	yaml : N last : N live : no update

Namespaces

♪ Consider the below analogy



- There are 2 houses,
In one house (house-A), one person called *Mark Smith* is there.
In another house (house-B), one person called *Mark Williams* is there.
- In house-A, people can call *Mark Smith* as only *Mark* because there is no other guy having this name in that house.
Similarly in house-B, people can call *Mark Williams* as *Mark* only.
- But, if people of house-A wants to call *Mark Williams* then they need to call by full-name i.e. *Mark Williams*. Not only *Mark*.
Similarly for people of house-B as well. They need to call *Mark Smith*.
- A person who neither belong to house-A nor house-B, needs to call those by their full name only i.e. *Mark Smith* or *Mark Williams*.
- Consider these **houses as namespaces**.
Inside a namespace, resources having **same names cannot be present** (you can't create 2 pods having exactly same name). but, in multiple namespaces, resources of **same name can be present**.
- For example, if you want multiple environments like *dev*, *prod* and you don't want to update any *production* pods while testing during *development* or something like that, **namespaces** help in that.
You can have exact same resources in both the namespaces.

♪ Example:

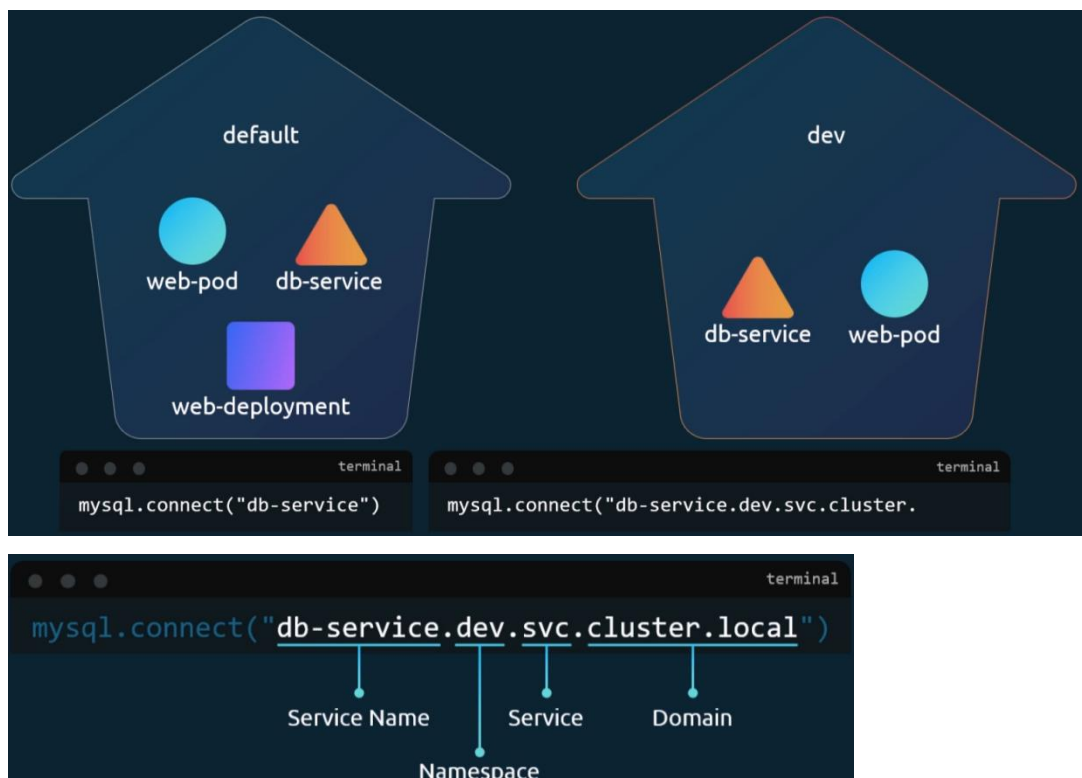
- Consider a single namespace scenario:
 - Some pods are running DB. And service name (ClusterIP) is **db**.

- ✧ The application pods who want to use database can use **db** directly to access the database.
 - ✧ Consider a multiple namespace scenario:
 - ✧ In the same example as above, lets imagine 2 namespaces are there.
 - ✧ Now, the application pods of one namespace can directly use **db** to access the database running in same namespace.
- But, if they want to use **db** of the other namespace then they need to give the full DNS path i.e. **db.<namespace name>** or **db.<namespace name>.svc.cluster.local** [you don't need to give this, **db.<namespace name>** is converted to this internally]

✧ **db.<namespace name>.svc.cluster.local**

in here, **svc** means don't access the pod directly, rather do using the **db** service.

cluster.local is the internal DNS of the cluster. [.local is a **top-level-domain** just like .com, .in ..etc but it is internal]



- ✧ If you want to get or create the resources present in a specific namespace then

kubectl get <resource type like pod, rs, deploy> -n <namespace name>

kubectl create -f <yaml file name> -n <namespace name>

[**NOTE** You need to write '**create**', not '**apply**' when you are creating something from the yaml **first time (still have to verify this)**]

[if the **namespace** key is present in the *yaml* file already, then no need to give **--namespace** flag]

- Or, in *yaml* you can write **namespace** inside metadata section.

```
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
  namespace: dev
```

- To create a *namespace*, the yaml file format will be like this

```
vagrant@controlplane:~$ kubectl create namespace test --dry-run=client -o yaml
apiVersion: v1
kind: Namespace
metadata:
  name: test
```

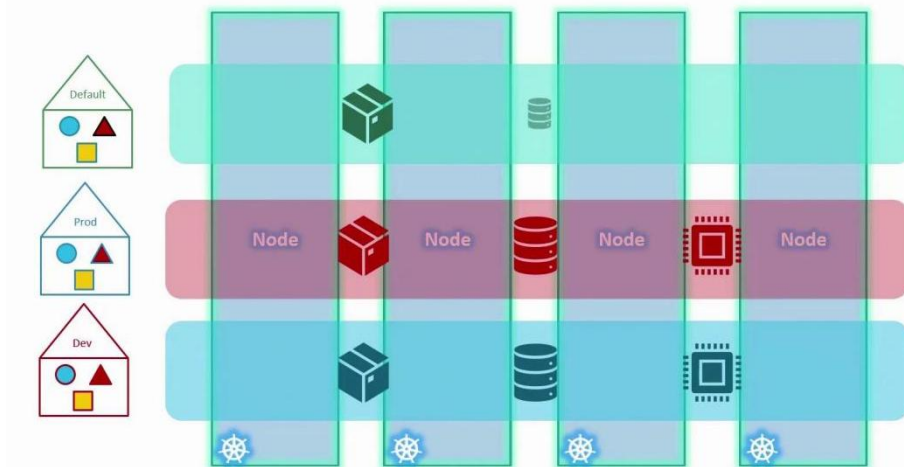
```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
```

々 All the **namespace** configures while setup:

- When a kubernetes *cluster* is first setup, **default** namespace is created. Whatever resources you create, it is created inside this.
- Kubernetes creates a set of Pods and Services for its internal purpose. These all stays in the namespace called **kube-system**.
- There is another namespace called **kube-public** where resources available to all users are created.

- 々 You can create **quota** (long format: **ResourceQuota**) to the namespaces (setting the limits)

Namespace – Resource Limits



- 々 If you want to change the default (not talking about the namespace having name *default*) namespace

```
kubectl config set-context $(kubectl config current-context) --  
namespace=<namespace name which has to be made default>
```

kubeconfig

々 The config file path is `~/.kube/config`.

```
apiVersion: v1
clusters:
- cluster:
    certificate-authority-data: LS0tLS1CRUdJTiB1c2Vybm9keSB0b290aW50
    server: https://192.168.56.11:6443
    name: kubernetes
contexts:
- context:
    cluster: kubernetes
    namespace: test
    user: kubernetes-admin
    name: kubernetes-admin@kubernetes
current-context: kubernetes-admin@kubernetes
kind: Config
users:
- name: kubernetes-admin
  user:
    client-certificate-data: LS0tLS1CRUdJTiB1c2Vybm9keSB0b290aW50
    client-key-data: LS0tLS1CRUdJTiB1c2Vybm9keSB0b290aW50
```

々

```
contexts:
- context:
    cluster: kubernetes
    user: kubernetes-admin
    name: kubernetes-admin@kubernetes
current-context: kubernetes-admin@k
```

[if namespace is not there]

it looks something like this (its content format is of yaml, but the filename is `config` only, not `config.yaml`)

々 It mainly contains three things: **cluster**, **contexts**, **users**

々 **clusters**

- ♣ its a list of clusters.
- ♣ Each cluster contains *name*, *cluster* (cluster contains *certificate-authority-data*, *server*) fields.
- ♣ **name**
 - ♣ its just an identifier that will be used later inside *context* (you can see inside *context* field).
 - ♣ every operation i.e. create pod, get svc ..etc **hits this URL**.

kubectrl → API Server → Cluster

- cluster.server

it's the end-point of **API Server** where all **kubectl** command goes.

- cluster.certificate-authority-data

- Ensures you're talking to the real cluster
- Prevents man-in-the-middle attacks

々 users

- ✦ It contains the list of users.
- ✦ It defines “Who are you when talking to the Cluster”
- ✦ Each entries of *user* contain the below fields **name**, **user** (*user* contains *client-certificate-data*, *client-key-data*) fields.

name

just a identifier to the user. Which is referenced inside *context* field (you can see in the yaml).

- 1 user.client-certificate-data & user.client-key-data (common in local
setup)

certificate is like ID card; **key** is like signature (Analogy)

- In real-world, instead of certificate and key, **token** is used.

token: eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXLTJ5IiwiaWF0IjoiMTY1MjY0MjY0In0=

- it is used in ServiceAccounts, CI/CD, API access.

- In cloud provider cases, external Auth is present.

- 4 All-In-One, **user** field is for authentication with the **cluster**.

々 contexts

- It is something that reference all the things i.e.

```
context = cluster + user + default namespace
```

--- IMPORTANT ---

- When you switch to a context, all the things will be switched i.e. user, cluster, default namespace

- When you execute any command like

kubectl get pod

it is executed like the below

kubectl get pod --namespace=<default namespace name>

- ✦ If the namespace field is not present in the *config* file (refer the image), then it'll take the namespace called “**default**” by default.

- ♣ But, whatever namespace you give (override) like the below
`kubectl get pod --namespace=<namespace name other than context default one>`
 the **user and cluster will be same only**.

々 **NOTE** namespace field will not be there inside the *config* file. -----

々 You can create as many context you want.

- ♣ **Same cluster + different namespaces** : new context
 (optional, if you want to maintain different contexts; otherwise you can just override the default namespace and move on in the same context using
`--namespace=<other namespace name>`)
- ♣ **Same cluster + different user** : new context
 (its mandatory)
 you cannot switch user independently, only you can switch *context*.
- ♣ **Different cluster** : new context
 (its mandatory)

々 **kubectl config --help**

```
Available Commands:
current-context  Display the current-context
delete-cluster   Delete the specified cluster from the kubeconfig
delete-context   Delete the specified context from the kubeconfig
delete-user      Delete the specified user from the kubeconfig
get-clusters     Display clusters defined in the kubeconfig
get-contexts     Describe one or many contexts
get-users        Display users defined in the kubeconfig
rename-context   Rename a context from the kubeconfig file
set              Set an individual value in a kubeconfig file
set-cluster      Set a cluster entry in kubeconfig
set-context      Set a context entry in kubeconfig
set-credentials  Set a user entry in kubeconfig
unset            Unset an individual value in a kubeconfig file
use-context      Set the current-context in a kubeconfig file
view             Display merged kubeconfig settings or a specified kubeconfig file
```

- ♣ **use-context** is used to set a new **current-context** (see in yaml).

々 If you want to create a new context

kubectl config set-context test-context --cluster=kubernetes --user=kubernetes-admin --namespace=test

```
vagrant@controlplane:~$ kubectl config set-context test-context --cluster=kubernetes --user=kubernetes-admin --namespace=test
Context "test-context" created.
vagrant@controlplane:~$ kubectl config get-contexts
CURRENT  NAME             CLUSTER  AUTHINFO  NAMESPACE
*         kubernetes-admin@kubernetes  kubernetes  kubernetes-admin  test
          test-context      kubernetes  kubernetes-admin  test
```

々 You can use this command to view all the configs.

kubectl config view

```
vagrant@controlplane:~$ kubectl config view
apiVersion: v1
clusters:
- cluster:
  certificate-authority-data: DATA+OMITTED
  server: https://192.168.56.11:6443
  name: kubernetes
contexts:
- context:
  cluster: kubernetes
  user: kubernetes-admin
  name: kubernetes-admin@kubernetes
current-context: kubernetes-admin@kubernetes
```

[cropped, it'll display the whole config

file]

々 If you want to change the **current-context** then

kubectl config use-context <context name>

々 ----- THE END OF CONFIG -----

- 々 There is another type of resource called `ResourceQuota` which is used to set the limit of a specific `namespace`.

```
apiVersion: v1
kind: ResourceQuota
metadata:
  name: compute-quota
  namespace: dev
spec:
  hard:
    pods: "10"
    requests.cpu: "4"
    requests.memory: 5Gi
    limits.cpu: "10"
    limits.memory: 10Gi
```

々

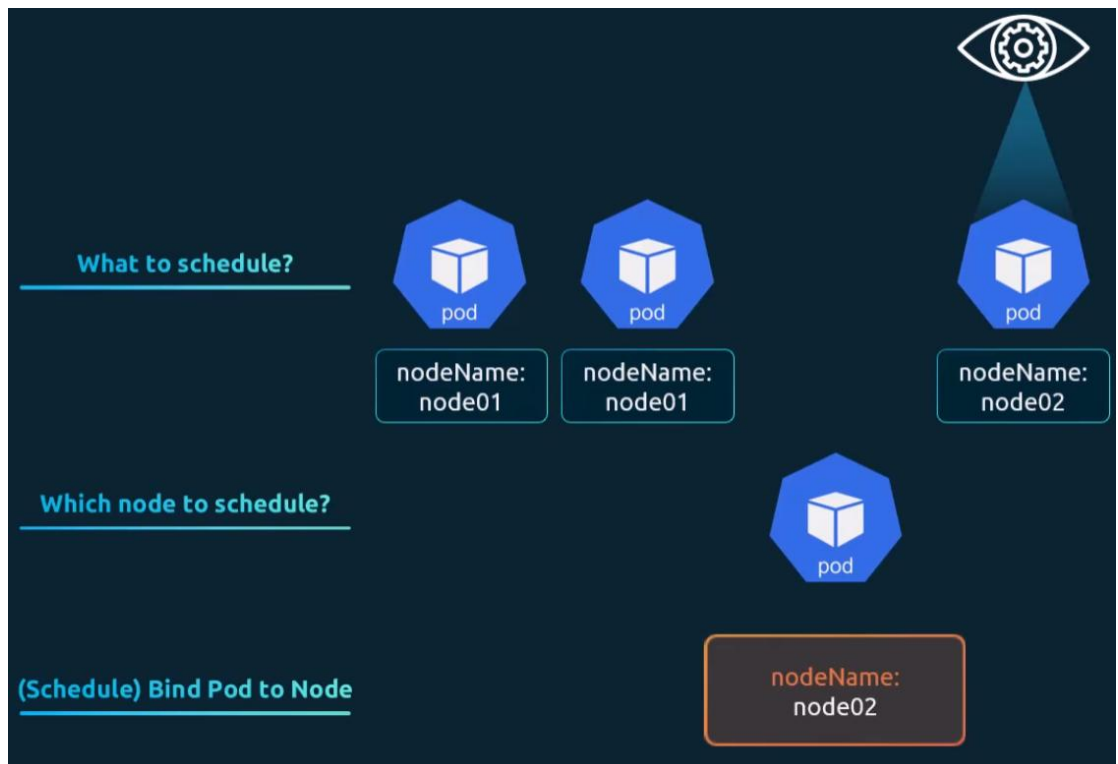
Scheduling

- ♪ In a running pod, when you see it, there will be a field called **nodeName** will be present, which contains the worker node's name where that specific pod is running.

Just execute `kubectl edit pod <pod name>` to see this.

```
enableServiceLinks: true
nodeName: node02
preemptionPolicy: PreemptLowerPriority
```

- ♪ How **scheduling** works?



- ♣ When you run a pod (without specifying *nodeName* field), the pod object is created.
- ♣ Then, *scheduler* sees this, and assign a node to it (***nodeName***).
NOTE: this ***nodeName*** field is under ***spec*** field of *Pod*'s config.
- ♣ Then, one **binding** will be created in the target node which will bind the *pod* to the *target node*.
- ♪ If the Scheduler doesn't exist,
 - ♣ then *nodeName* will not be appended to the pod's config.
Even if you give *nodeName*, **binding** object will not be created in the

target node unless manually send the POST request to the target worker node to create Binding object.

- ✦ In this case, the Pod will be in pending state.

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: test-pod
    name: test-pod
spec:
  containers:
  - image: nginx
    name: test-pod
  nodeName: node03
```

```
Every 2.0s: kubectl get pod

NAME                                READY   STATUS    RESTARTS   AGE
ng-dep-66b9cb6d4d-4r822             1/1     Running   0           38h
ng-dep-66b9cb6d4d-f4gl4             1/1     Running   0           38h
ng-dep-66b9cb6d4d-f52lt             1/1     Running   0           39h
test-pod                             0/1     Pending   0           13s
```

- ✦ I simulated is giving a *nodeName* which doesn't exist inside */etc/hosts* file.

So, the *Binding* object will not be created.

- ✦ After sometime, the pod will be **deleted** because its status couldn't be 'Running'.

✧ Simulating Manual Scheduling of a Pod to a Node

- ✦ First create a yaml config of the node giving *nodeName* field.

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: test-pod
    name: test-pod
spec:
  containers:
  - image: nginx
    name: test-pod
  nodeName: node01
```

- ✦ Now, create the pod

```
vagrant@controlplane:~$ kubectl create -f test-pod.yaml
pod/test-pod created
```

- Now, you need to create a *Binding* yaml file.

```
apiVersion: v1
kind: Binding
metadata:
  name: test-binding
target:
  apiVersion: v1
  kind: Node
  name: node01
```

NOTE: It doesn't contain any data about the Pod, it only contains the details of the target node.

- The yaml is just to understand the key value pairs, in real there is **no such long-lived object called *Binding* exists.**

Its just the JSON payload to be sent in the POST request to bind the pod to the target node.

- Now, you need to trigger the POST request to the **API Server** to bind the pod to the *target worker node*.

NOTE: the worker node doesn't expose any end-points, only API Server in the master node does this.

And, you need to trigger the request to this API Server's end point.

- The POST request looks like this (just an example)

```
curl --header "Content-Type: application/json" \
  --request POST \
  --data '{"apiVersion":"v1", "kind": "Binding", "metadata": {"n
http://$SERVER/api/v1/namespaces/default/pods/$PODNAME/binding
```

- You'll get the \$SERVER from **kubectl config view** command. It'll be in the form

<API server node's IP>:<API Server's port>

```
vagrant@controlplane:~$ kubectl config view
apiVersion: v1
clusters:
- cluster:
  certificate-authority-data: DATA+OMITTED
  server: https://192.168.56.11:6443
```


- \$PODNAME is the Pod's name that you get from **kubectl get pod** command.

```
vagrant@controlplane:~$ kubectl get pod
NAME                                READY   STATUS
ng-dep-66b9cb6d4d-4r822            1/1     Runni
ng-dep-66b9cb6d4d-f4g14            1/1     Runni
ng-dep-66b9cb6d4d-f521t            1/1     Runni
test-pod                            1/1     Runni
```

- Now, trigger the request

```
curl --header "Content-Type: application/json" \
  --request POST \
  --data '{"apiVersion":"v1","kind":"Binding","metadata":{"name":"test-binding"},"target":{"apiVersion":"v1","kind":"Node","name":"node01"}}' \
  https://192.168.56.11:6443/api/v1/namespaces/default/pods/test-pod/binding/
```

- It might not work, because you need to provide token or certificates keys in the URL. WILL SEE LATER.

Labels and Selectors

- ✧ **labels** and **selectors** are used for **grouping** and **filtering**.
- ✧ You can give multiple properties inside **labels** which will be helpful in filtering the resources properly.
- ✦ You can give as many key value pairs inside **labels** field.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
  labels:
    name: ng-deployment
    key1: val1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
        key2: val2
    spec:
      containers:
```

- ✦ (here I have given 2 labels to each deployment and pod template)

- ✦ Then you can select the resources with
--selector key1=val1,key2=val2 (you can give any number of label here)

```
vagrant@controlplane:~$ kubectl get all --selector app=nginx,key2=val2
NAME                                READY   STATUS    RESTARTS   AGE
pod/ng-dep-797565664d-dpkjk        1/1     Running   0           80s
pod/ng-dep-797565664d-qljts        1/1     Running   0           80s
pod/ng-dep-797565664d-zgq4k        1/1     Running   0           80s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/ng-dep-797565664d   3          3         3       80s
```

NOTE: ReplicaSet created by Deployment will have same label as Pods.

- ✧ In case of Deployment or ReplicaSet, whatever you write inside **matchLabels** field will be used to get the Pods.
- ✦ All the fields inside **matchLabels** should be there inside the Pod's labels.

- Pod's labels can contain either *same or extra* fields than *matchLabels*, but *not less*.

If pod doesn't contain all the fields present in *matchLabels* then it'll not be selected.

Annotations

- You can also give a field **annotations** inside *metadata* field.
- Annotations may include *build version, contact information, or other integration data*.

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: simple-webapp
  labels:
    app: App1
    function: Front-end
  annotations:
    buildversion: "1.34"
```

- Annotations are ideal for *adding tool details, version information, or contact data* without impacting how the system groups or selects objects

々

Taints and Tolerations

- 々 Real-life analogy (Assumptions HIT kills cockroaches, but cannot kill Lizards):
 - ♣ You want to prevent cockroaches on your bed. So you sprayed HIT around it.
 - ♣ Now, Cockroaches cannot come because they are *intolerant* to HIT.
 - ♣ But, Lizards might come because HIT cannot kill them.
Or, in other words, lizards are *tolerant* to HIT.
- 々 So, How these concepts being used?
 - ♣ You put taint (HIT in the analogy) on the *nodes*, to prevent the *Pods* being scheduled to it.
 - ♣ Only the tolerant pods (lizard in the analogy) can be scheduled to the *tainted node*.
 - ♣ *Nodes* are *tainted* to prevent the *Pods* to be scheduled to them.
Pods are being *tolerated* (when required) to tolerate the *tainted nodes*.
 - ♣ In simple words,
Taints on nodes repel pods that do not have matching tolerations.
Tolerations in pods allow them to overcome the taint and be scheduled on that node.

々 Taint the Node

kubectl taint nodes <node name> **key=value:**<taint-effect>

[always try to use plural forms i.e. **nodes** not **node**]

Ex: kubectl taint nodes node01 env=prod:NoSchedule

- ♣ to remove the taint of the node, just put a - at the end.

kubectl taint nodes <node name> **key=value:**<taint-effect>-

Ex: kubectl taint nodes node01 env=prod:NoSchedule-

- ♣ Example:

- Initially no *taint* was there.

```
vagrant@controlplane:~$ kubectl describe node node01 | grep -i taint
Taints:                <none>
```

- taint node01

```
vagrant@controlplane:~$ kubectl taint nodes node01 env=prod:NoSchedule
node/node01 tainted
```

- ```
vagrant@controlplane:~$ kubectl describe node node01 | grep -i taint
Taints: env=prod:NoSchedule
```

- ```
vagrant@controlplane:~$ kubectl taint nodes node01 env=prod:NoSchedule-  
node/node01 untainted
```

- ```
vagrant@controlplane:~$ kubectl describe node node01 | grep -i taint
Taints: <none>
```

- it'll remove the taint of the node where the key is matched.

- ```
Taints:      app=product:NoSchedule
             env=dev:NoSchedule
```

々

- ```
spec:
 tolerations:
 - key: "env"
 operator: "Equal"
 value: "prod"
 effect: "NoSchedule"
```

because while giving taint, we wrote *env=prod*

- ```
Taints:      node-role.kubernetes.io/control-plane:NoSchedule
```

and pods don't have *tolation* to this *taint*.

々 Taint Effects -----

々 NoSchedule (we used this previously)

- ⌘ New Pods will **not be scheduled** on this node *if they are **not having the toleration***.
- ⌘ Existing Pods will **remain as it is**. (***even if they doesn't have the toleration***)
- ⌘ In simple terms:
New Pods: ❌
Existing Pods: ✅
- ⌘ In our analogy, you put HIT around of the bed. Cockroaches who are already on the bed will survive but no outside cockroaches can come.

々 PreferNoSchedule

- ⌘ Kubernetes will try to avoid this node to schedule the pods.
- ⌘ But, it might schedule if there is no other option.
- ⌘ In simple terms:
New Pods: ❌✅
Existing Pods: ✅

々 NoExecute

- ⌘ New Pods will not be scheduled to this node (same as *NoSchedule* effect)
- ⌘ Existing Pods will be removed (different from *NoSchedule* effect)
- ⌘ In simple terms:
New Pods: ❌
Existing Pods: ❌
- ⌘ In our analogy, you put HIT around the bed and also on the bed. Now cockroaches will be dead and no outside cockroaches can come. Lizards are safe.

々 NOTE: All the not-scheduling and delete from the node things which are written above, those are for the pods who don't have *tolerance* to that *taint*. If the pods are having *tolerance*, then they (new and existing) will not be affected.

⌘ -----

--

nodeSelector and nodeAffinity

- While *taint* and *tolerations* were to avoid a node to be scheduled for pods, *nodeSelector* and *nodeAffinity* are there to schedule a node for the pods.

nodeSelector

- It works in the same principle *selector*.

- First provide the *labels to the node*. [before we labelling the Pods]

```
kubectl label nodes node1 env=prod
```

- Use *nodeSelector* field to match the Node labels. [ReplicaSet's *selectors* field]

```
spec:
  nodeSelector:
    env: prod
```

- Example

- Created a 2 pod yaml files (one for env=prod, another for env=dev)

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: prod-pod
    name: prod-pod
spec:
  nodeSelector:
    env: prod
  containers:
  - image: nginx
    name: prod-cont
    ports:
    - containerPort: 80
      name: nginx-port
```

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: dev-pod
    name: dev-pod
spec:
  nodeSelector:
    env: dev
  containers:
  - image: nginx
    name: dev-cont
    ports:
    - containerPort: 80
      name: nginx-port
```

- Label *node01* as *env=dev*, and *node02* as *env=prod*

```
vagrant@controlplane:~$ kubectl label nodes node01 env=dev
node/node01 labeled
vagrant@controlplane:~$ kubectl label nodes node02 env=prod
node/node02 labeled
```



```
Labels:      beta.kubernetes.io/arch=amd64
            beta.kubernetes.io/os=linux
            env=dev
```

[node01]

```
Labels:      beta.kubernetes.io/arch=amd64
            beta.kubernetes.io/os=linux
            env=prod
```

[node02]

- Created prod pod and it was scheduled in node02

```
vagrant@controlplane:~$ kubectl create -f prod-pod.yaml
pod/prod-pod created
vagrant@controlplane:~$ kubectl get pods -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE    NOMINATED NODE   READINESS GATES
prod-pod  1/1     Running   0           8s    10.244.140.103 node02    <none>            <none>
```

- Created dev pod and it was scheduled in node01

```
vagrant@controlplane:~$ kubectl get pods -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP            NODE    NOMINATED NODE   READINESS GATES
dev-pod   1/1     Running   0           36s    10.244.196.162 node01    <none>            <none>
```

nodeAffinity

- The *nodeSelector* was very straight forward. You need to give *label* properly
[hard selection just like *matchLabels* in ReplicaSet]

nodeAffinity is having multiple types of label matching. [like
matchExpressions in ReplicaSet]

- What if you want to say like
place the pod *where env=prod or env=test*
place the pod *where env != dev*
in this case, ***nodeAffinity*** can be used.

々 The yaml looks like the below

```
spec:
  containers:
    - name: data-processor
      image: data-processor
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: size
                operator: In
                values:
                  - Large
      preferredDuringSchedulingIgnoredDuringExecution:
        - weight: 5
          preference:
            matchExpressions:
              - key: zone
                operator: In
                values: [us-east-1]
```

々 2 types of nodeAffinity

requiredDuringSchedulingIgnoredDuringExecution

preferredDuringSchedulingIgnoredDuringExecution

々 Pods are having 2 stage: *scheduling, execution*

♣ When the pod is not created in any node, it is first scheduled.

[Scheduling]

♣ After the pod being scheduled, it'll be created in the scheduled node.

[Execution]

♣ Currently, nodeAffinity is only there for *scheduling* phase, you can see its written *Ignored*during*Execution*.

々 *required* one contains **nodeSelectorTerms** whereas *preferred* one contains **weight** and **preference** .

々 In case of **required**DuringSchedulingIgnoredDuringExecution, if the expression doesn't match with any node, then the **Pod will not be scheduled at all.** [**HARD** matching]

In case of preferredDuringSchedulingIgnoredDuringExecution, all the nodes will be checked and weight will be calculated. Highest preferred node will be scheduled for the pod. [**SOFT** matching]

♪ How the *weight* is calculated in case of preferredDuringSchedulingIgnoredDuringExecution ?

- ♣ If you have provided some preferences, all will be checked in all the nodes available.
Each preference will be having some weight (you'll define this)
- ♣ The nodes will be checked for all the preferences. If preference matches in that node then weight for that node will be increased with the weight provided for that preference.

```
preferredDuringSchedulingIgnoredDuringExecution:
- weight: 5
  preference:
    matchExpressions:
      - key: env
        operator: In
        values: [prod]

- weight: 3
  preference:
    matchExpressions:
      - key: disk
        operator: In
        values: [ssd]
```

Node	env=prod	disk=ssd	Score
node1	✓	✓	5 + 3 = 8
node2	✓	✗	5 + 0 = 5
node3	✗	✓	0 + 3 = 3

here, *node1* has both the preference matched so its weight will be 8, so pods will be scheduled to *node1*.

♪ Use **kubectl explain pod.spec.affinity.nodeAffinity --recursive** to get the fields and directly paste inside pod yaml file.

々 Here we used *matchExpressions* to match the label of the nodes, there is another called *matchFields*.

- ♣ It matches any type of fields you want, not just *labels*.

```
matchFields:
- key: metadata.name
  operator: In
  values: [node1]
```

[metadata.name must be node1]

々 There are other 2 types of affinity,

podAffinity, podAntiAffinity

- ♣ *nodeAffinity* checks the labels in the nodes and match that.
- ♣ *podAffinity* and *podAntiAffinity* checks the labels of the pods which are running inside the node.

々 *podAffinity*

- ♣ The pod running on target nodes will be checked with the preferences/nodeSelectorTerms.
if matches then that node will be scheduled.

々 *podAntiAffinity*

- ♣ Opposite of *podAffinity*,
the lowest matched (in case of *preferredDuring*....) will be selected.
or not matched (in case of *requiredDuring*...) will be selected.

々 Some operators of *matchExpressions*

```
- matchExpressions:
  - key: size
    operator: In
    values:
      - Large
```

[In]

```
- matchExpressions:
  - key: size
    operator: Exists
```

[Exists]

- ♣ All the operators are

[kubectl explain](#)

[pod.spec.affinity.nodeAffinity.preferredDuringSchedulingIgnoredDurin](#)

gExecution.preference.matchExpressions

Possible enum values:

- `~"DoesNotExist"~`
- `~"Exists"~`
- `~"Gt"~`
- `~"In"~`
- `~"Lt"~`
- `~"NotIn"~`

々 NodeAffinity configurations -----

- [spec.affinity.nodeAffinity](#) has to be updated.
nodeAffinity support complex queries like DoesNotExist, Exists, Gt, In, Lt, NotIn
- *nodeAffinity* is of 2 types:
 - [requiredDuringSchedulingIgnoredDuringExecution](#) [HARD rule]
 - [preferredDuringSchedulingIgnoredDuringExecution](#) [Scored basis]

```
nodeAffinity
├─ preferredDuringSchedulingIgnoredDuringExecution []
│   └─ preference
│       └─ matchExpressions []
│           └─ matchFields []
│               └─ weight
└─ requiredDuringSchedulingIgnoredDuringExecution
    └─ nodeSelectorTerms []
        └─ matchExpressions []
            └─ matchFields []
```

- [its not ~~match~~**Labels**, its **matchFields**]
- [matchExpressions](#) is to query labels of nodes.
- [matchFields](#) is to query any fields. [Ex: *metadata.name*]
- [matchFields](#) and [matchExpressions](#) [both are list]

```
matchExpressions:
- key: env
  operator: In
  values:
  - prod
- key: disk
  operator: In
  values:
  - ssd
```

```
matchFields:
- key: metadata.name
  operator: In
  values:
  - node01
  - node02
- key: metadata.resourceVersion
  operator: In
  values:
  - "353007"
  - "353008"
```

- [matchExpressions](#): query1 **AND** query2 **AND** ...
- [matchFields](#): query1 **AND** query2 **AND** ...

⚡ *requiredDuringSchedulingIgnoredDuringExecution*

```

requiredDuringSchedulingIgnoredDuringExecution
└─ nodeSelectorTerms [] (OR)
    │   └─ Term 1
    │       │   └─ matchExpressions [] (AND)
    │       │       └─ matchFields [] (AND)
    │       │           ⇒ Term 1 = (matchExpressions AND matchFields)
    │       │
    │       └─ Term 2
    │           │   └─ matchExpressions [] (AND)
    │           │       └─ matchFields [] (AND)
    │           │           ⇒ Term 2 = (matchExpressions AND matchFields)
    │           │
    │           └─ ...
    └─ ...

FINAL:
(Term 1) OR (Term 2) OR ...
= [(matchExpressions AND matchFields)] OR [...]

```

- ⚡ *nodeSelectorTerms*: *term1 OR term2 OR ...*
- ⚡ Each term: *matchFields AND matchExpressions*
matchFields: *query1 AND query2 AND ...*
matchExpressions: *query1 AND query2 AND ...*
- ⚡ Any one term should match.
Means, all the *matchExpressions* and *matchFields* query of that terms should match.
- ⚡ *nodeSelectorTerms* = [{*matchExpressions AND matchFields*} OR {..*}*
OR ..]
- ⚡ If you mark here, *nodeSelectorTerms* is not even required.
requiredDuri...Execution could have made a list instead of a dictionary.

⌘ *preferredDuringSchedulingIgnoredDuringExecution*

```
preferredDuringSchedulingIgnoredDuringExecution [] (scored list)
├─ Item 1
│   └─ preference
│       ├── matchExpressions [] (AND)
│       └─ matchFields [] (AND)
│           ⇒ match = (matchExpressions AND matchFields)
│       └─ weight
│
├─ Item 2
│   └─ preference
│       ├── matchExpressions [] (AND)
│       └─ matchFields [] (AND)
│           ⇒ match = (matchExpressions AND matchFields)
│       └─ weight
│
└─ ...

FINAL (scoring, not filtering):
score(node) = sum(weight of each item that matches)
```

- ⌘ Here, preferredDur....Execution is list. No unnecessary extra child field like the previous.
- ⌘ [
 - {preference: {matchExpressions AND matchFields}. weight: calculate},
 - {...},
 - ...
- ⌘ For a item:
- ⌘ weight=provided value (if matchExpressions AND matchFields) or 0 (zero)
- ⌘ Total weight: item1 weight + item2 weight ...
- ⌘ the node who gets more weight will be selected for the pod.

⌘ **requiredDur....Execution: at least one should match**

preferredDur...Execution: more matches, more score

⌘ -----

Resource Requirements

- 々 There are 2 things, **Request** and **Limit**.
 - ♣ Request: minimum required resource (used for scheduling)
 - ♣ Limit: maximum allowed resource (enforced at runtime)
- 々 The YAML look like this.

```
apiVersion: v1
kind: Pod
metadata:
  name: demo
spec:
  containers:
  - name: app
    image: nginx
    resources:
      requests:
        cpu: "500m"
        memory: "256Mi"
      limits:
        cpu: "1"
        memory: "512Mi"
```

[**resources** is written **inside containers**, not **spec**]

< -----

- 々 **requests** (Scheduling Decision)
 - ♣ If any node has at least this much free resources, then only that node can be scheduled for the pod.
Otherwise it'll not be scheduled.
- 々 **limits** (Runtime enforcement)
 - ♣ CPU: throttled (if exceed than the limit set) [throttled means slow down]
 - ♣ Memory: container is **killed** (**OOMKilled**) (if exceed than the limit set)
- 々 CPU vs Memory behaviour
 - ♣ CPU can exceed request, but Memory cannot.

- ♣ In case of hitting the limit,
CPU: throttled [slow down]
Memory: *container killed*

々 Resource is not there, then by default it takes as 0.

But, if resource is not there and limit is there: resource will be taken same as limit.

If limit is not there, then it'll assume the pod can use *unlimited* resources.

々 1 (one) count of CPU is same as

- ♣ 1 AWS vCPU
- ♣ 1 GCP Core
- ♣ 1 Azure Core
- ♣ 1 Hyperthread

々 Memory

- ♣ 1 G : 1 Gigabyte
- ♣ 1 M : 1 Megabyte
- ♣ 1 K : 1 Kilobyte
- ♣ 1 Gi : 1 Gibibyte
- ♣ 1 Mi : 1 Mebibyte
- ♣ 1 Ki : 1 Kibibyte

[K : 1000 bytes, Ki: 1024 bytes ; K is binary (2's power), Ki is decimal (10's power)]

々 **cpu** lowest unit: **1m** (1 mili CPU or 0.001 CPU) [1 means one full CPU]

memory lowest unit: **1** (1 byte) [only use Gi, Mi, Ki; ~~don't use G, M, K~~]

々 We can have 4 scenarios:

No requests, No limits

No requests, Limits

Requests, Limits

Requests, No limits

- ♣ In the first case, if one pod (container inside pod because requests and limits are set per container, not per pod) is consuming all the CPU, and another pod is also assigned to same node (because if request is not given, it'll be **taken as 0**) which is not able to get the resources.
- ♣ In the second case, request will be **taken as same as limits**.

- In the 3rd case, the pods are having both request and limits. Lets say in this case pod-1 doesn't need extra resources but pod-2 needs.
And we have set the limit for both the pods which will stop *pod-2* to use more resources. Which is not ideal
- In the 4th case, the problem of 3rd case can be solved. Because if a pod is being assigned to a node means the node has required amount of resources available that the pod needs.
And, there is no limit so pod can use as much as it want.

✧ There is something called **LimitRange**, it by default injects the **requests** and **limits** to the pods if not specified in the yaml.

- It works in **namespace level**.

```
apiVersion: v1
kind: LimitRange
metadata:
  name: cpu-resource-constraint
spec:
  limits:
    - default:
        cpu: 500m
      defaultRequest:
        cpu: 500m
      max:
        cpu: "1"
      min:
        cpu: 100m
    type: Container
```

Field	Resource Type
default	Limit
defaultRequest	Request
max	Limit
min	Request

- Multiple LimitRange can be configured in a namespace, but they should not conflict.

Ex: one says max cpu: 500m, another says min cpu: 600m

in this case, pod will not be created because of conflicting values.

In real life: **one LimitRange per namespace is maintained**.

- **default :**
 - ✦ it is the **limits** field.
 - ✦ If the pod yaml doesn't contain **limits** fields, then this LimitRange's *spec.limits.default* will be added there.
- **defaultRequest :**
 - ✦ It is the **requests** field.

- ⌘ If the pod yaml doesn't contain **requests** fields, then this LimitRange's *spec.requests* default will be added there.

⌘ **min**

- ⌘ The **requests** or **limits** should not be less than this..
- ⌘ If the container (inside pod) asks lesser resource than this *min* then it'll be rejected.

⌘ **max :**

- ⌘ The **requests** or **limits** should not be more than this.
- ⌘ The container cannot request more requests than that defined in *max*.

⌘ **type : Container**

々

々 DaemonSet

- ♣ It is similar to *ReplicaSet*
- ♣ It makes sure that each node should contain one copy of the pod specified in *DaemonSet* configuration.
- ♣ Earlier it used to append *nodeName* field and POST request to create the container in the target node.

But now, it uses *nodeAffinity* to assign a specific node to the Pod.

- ♣ The yaml is exact same as that of *ReplicaSet*, just the *kind* will be different.

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: monitoring-daemon
spec:
  selector:
    matchLabels:
      app: monitoring-agent
  template:
    metadata:
      labels:
        app: monitoring-agent
    spec:
      containers:
        - name: monitoring-agent
          image: monitoring-agent
```

♣

々 F

Static Pods

- Pod are created by the command of *kube-apiserver*, which is listened by *kubelet* (in worker nodes) and create and start the container.
what if the worker node is not connected to any Kubernetes cluster, it just has *kubelet* installed. How to run a pod here?
- You need to find the **staticPodPath** in the node where *kubelet* is present.
kubelet constantly see this path and create and update the pod if any *yaml* config is present inside this path.
- NOTE :-** You can only create pods in this way, other resources like *ReplicaSet*, *Deployment* cannot be created because they are completely of *Kubernetes* control plane things.
- To find the **staticPodPath** the below methods are there [all in worker node, not master node]

Method-1

✧ **systemctl cat kubelet**

```
# Note: This dropin only works with kubeadm and kubelet v1.11+
[Service]
Environment="KUBELET_KUBECONFIG_ARGS=--bootstrap-kubeconfig=/etc/kubernetes/bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf"
Environment="KUBELET_CONFIG_ARGS=--config=/var/lib/kubelet/config.yaml"
# This is a file that "kubeadm init" and "kubeadm join" generates at runtime, populating the KUBELET_KUBEADM_ARGS with kubeadm flags
EnvironmentFile=/var/lib/kubelet/kubeadm-flags.env
# This is a file that the user can use for overrides of the kubelet args as a last resort. Preferably, the user should use the .NodeRegistration.KubeletExtraArgs object in the configuration files instead. KUBELET_EXTRA_ARGS should be sourced from this file.
EnvironmentFile=/etc/default/kubelet
ExecStart=
ExecStart=/usr/bin/kubelet $KUBELET_KUBECONFIG_ARGS $KUBELET_CONFIG_ARGS $KUBELET_KUBEADM_ARGS $KUBELET_EXTRA_ARGS
```

(you'll get something like this; this image is cropped)

if you find something like below, then see the **--pod-manifest-path**

```
ExecStart=/usr/local/bin/kubelet \
--container-runtime=remote \
--container-runtime-endpoint=unix:///var/run/containerd.sock \
--pod-manifest-path=/etc/kubernetes/manifests \
--kubeconfig=/var/lib/kubelet/kubeconfig \
--network-plugin=cni \
--register-node=true \
--v=2
```

- ✧ Open that **config.yaml** file present there, you'll find a key called **staticPodPath**

```
staticPodPath: /etc/kubernetes/manifests
```


- ✧ In this path i.e. `/etc/kubernetes/manifests`, if you create a Pod's yaml file, then **kubelet** will create that pod inside that node.
- ✧ The above displayed is **kubelet.service** file (it is same as the above `systemctl cat kubelet`)
- ✧ Its mostly in `/usr/lib/systemd/system/kubelet.service` if not, then find other service file directories like `/etc/systemd/system`, `/var/lib/systemd/system` ..etc
- ✧ There you'll find the **ExecStart** field

```
Unit
Description=kubelet: The Kubernetes Node Agent
Documentation=https://kubernetes.io/docs/
Wants=network-online.target
After=network-online.target

[Service]
ExecStart=/usr/bin/kubelet
Restart=always
StartLimitInterval=0
RestartSec=10

[Install]
WantedBy=multi-user.target
```

✧ Method-2

- ✧ `ps -ef | grep kubelet`

```
vagrant@node01:/usr/local/bin$ ps -ef | grep kubelet
root      3762      1   2 May02 ?        00:32:17 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernetes/
bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf --config=/var/lib/kubelet/config.yaml
--node-ip 192.168.56.21
root      4046      1   2 May02 ?        00:00:05 /usr/bin/node-driver-registry --v=5 --csi-address=/c
```

- ✧ You'll find the **config.yaml** path here.
- ✧ Go to that **yaml** file and get the **staticPodPath**.

✧ To check in the worker node

- ✧ You can use **crictl** command to see if the container is running.
You can't use **kubectl** command there because its not kubernetes.
- ✧ **crictl** is used to talk to **containerd**.
If you get "Permission Denied" then run with **sudo**.
- ✧ **sudo crictl ps**

```
vagrant@node01:~$ sudo crictl ps
```

CONTAINER	IMAGE	CREATED	STATE	NAME	ATTEMPT	POD ID	POD
306e8944e30c3	6c3a6ea6608c8	26 minutes ago	Running	static-cont	0	37b86e808bd54	test-static-pod-node01
09342132410e0	6c3a6ea6608c8	16 hours ago	Running	dev-cont	0	66d3e7016d3e0	dev-pod

- ✧ **kubeadm** creates **static pods** in master node to run the **control plane resources**.

```
vagrant@controlplane:/etc/kubernetes/manifests$ ls
etcd.yaml kube-apiserver.yaml kube-controller-manager.yaml kube-scheduler.yaml
```

- ♣ When you execute ***kubectrl get pods*** in the control plane, you'll see the *static pod* there.

HOW? Because this should be specific to the worker node and kube API Server shouldn't know about this. Read about the ***end-to-end flow*** (next page) for this.

| End-To-End flow from Pod creation command to actual Running of Pod |

----- IMPORTANT -----

々 Below is the flow end-to-end

- ✦ User creates a Pod
 - ✦ `kubectl apply -f my-pod.yaml` (or) `kubectl run ...`
 - ✦ This request goes to API Server.
- ✦ API Server
 - ✦ It validates the request.
 - ✦ Store the *Pod* object (*PodSpec*) in *etcd*.
- ✦ Scheduler
 - ✦ It sees the unscheduled pod in *etcd*.
 - ✦ It schedule any node to this pod [updates ***spec.nodeName***]
- ✦ Kubelet (in the scheduled node)
 - ✦ Watches API Server, and see a pod is assigned to it.
[kubelet opens a *watch* connection. API Server streams updates through it]
 - ✦ Pulls the ***PodSpec***.
- ✦ Kubelet → Container runtime (*containerd*)
 - ✦ Creates Pod sandbox (pause container: otherwise container will get its own IP; instead of using Pod's IP)
 - ✦ Creates all the required containers defined in *PodSpec*.
- ✦ Container Runtime (*containerd*)
 - ✦ Pulls images, & creates the containers.
 - ✦ Assign *container IDs*.
 - ✦ Start the processes.
- ✦ Kubelet (after creation)
 - ✦ Maps *Pod ID* → *Container IDs*
 - ✦ Monitors the containers
running / crashed / restarted
- ✦ Kubelet → API Server (status update)
 - ✦ API server doesn't poll for Pod's status.
Kubelet update API server about the Pod's status.
 - ✦ API Server stores updated status in *etcd*.

```
status:
  phase: Running
  containerStatuses:
    - name: nginx
      containerID: containerd://abc123
      state: running
```

- Control Plane (now updated)
 - ✦ **etcd** contains
 - PodSpec** (desired state)
 - PodStatus** (actual state from Kubelet)
- Controllers (ReplicaSets, etc..)
 - ✦ Watch API Server
 - ✦ Compares
 - Desired state vs Current state
 - ✦ Take action if mismatch.

🔗 Now, the question arises: Does the ReplicaSet or other controllers repeatedly poll API Server about the pod's status?

- No, it starts a **watch event**.
- Its not a *web-hook*, it's a long-lived connection (**streaming connection**)
- Whenever something changes, API Server streams it, so *controllers* get to know about the pod's status.

🔗 *Control Plane* knows nothing about the containers.

It gets the details about the containers (id and status) via Kubectl only.

Component	Knows PodSpec	Knows Container ID	Runs Containers
API Server	✓	⚠ (reported only)	✗
etcd	✓	⚠ (stored only)	✗
kubelet	✓	✓ (authoritative)	✗
containerd	✗	✓ (real)	✓

🔗 To see the details about the containers in *Control Plane* and *Worker Node*

- When you execute
 - kubectl get pod <pod name> -o yaml**
 - (or)
 - kubectl edit pod <pod name>**
 - ✦ It shows the content of **etcd** in yaml format.

♣ **status.containerStatuses**

this field contains the list of containers associated with that pod along with their IDs, image, state (running, etc.), volumes ..ec.

```
containerStatuses:
- containerID: containerd://98243133419e0d62
  image: docker.io/library/nginx:1.29
  imageID: docker.io/library/nginx@sha256:6e
  lastState: {}
  name: dev-cont
  ready: true
  resources: {}
  restartCount: 0
  started: true
  state:
    running:
      startedAt: "2026-05-02T18:10:45Z"
  user:
    linux:
      gid: 0
      supplementalGroups:
        - 0
      uid: 0
  volumeMounts:
    - mountPath: /var/run/secrets/kubernetes.:
```

🔗 **NOTE** -----

- ♣ **kubelet** has a watch connection with API Server.
And, whenever API Server gets some request (either create a new pod or update an existing pod), it updates the **etcd** .
- ♣ When you trigger **kubectrl edit pod <pod name>**
 - ↻ API Server gets the update request, it updates the **etcd** if there is any changes.
 - ↻ **kubelet** already has a *watch connection*, through that API Server sends the modified **PodSpec**.
 - ↻ **kubelet** gets the new **PodSpec** and compare with the existing one.
 - ↻ If any changes, then it recreates the container (if allowed; if not allowed then reject it)
it sends the report of new container to the API Server i.e. new **Container ID**, ..etc.

```

containerStatuses:
- containerID: containerd://3d835f86cfac50fe61ea18455f9b7a9a22f98e73aa22d2c34dd54dc7bf9b76b
  image: docker.io/library/nginx:1.29-perl
  imageID: docker.io/library/nginx@sha256:3d5526f7dc2dfd0b6cbebc664ca2d7ab1e1b5f572a289c6a7607728c5f5b85c
  lastState:
    terminated:
      containerID: containerd://98243133419e0d62baa1134c86c58fa102f93170f3ff7f82e08ebce518078c96
      exitCode: 0
      finishedAt: "2026-05-03T13:08:32Z"
      reason: Completed
      startedAt: "2026-05-02T18:10:45Z"

```

[you can see, *lastState.terminated*; it contains previous containerID here]

[contains only last terminated container details, not all]

- ✦ In case of no change, doesn't do anything.
- ✦ In case of **static pod**, kubelet sends the same report to API Server; but with some extra fields

- ✦ The below is the *metadata* of normal pod (pod created by api server)

```

metadata:
  annotations:
    cni.projectcalico.org/containerID: c6d2a3016d3a055cededa48f1ca2c97a5c69303ce534161abd71d26a4e4c2641
    cni.projectcalico.org/podIP: 10.244.196.162/32
    cni.projectcalico.org/podIPs: 10.244.196.162/32

```

- ✦ The below is the *metadata* of **static pod** (pod created in worker node itself)

```

metadata:
  annotations:
    cni.projectcalico.org/containerID: 37b86e808bd5439f78bf763fa7914144be4b67de4342957d3253bc2412eaa40a
    cni.projectcalico.org/podIP: 10.244.196.163/32
    cni.projectcalico.org/podIPs: 10.244.196.163/32
    kubernetes.io/config.hash: af6e265c669e39311b5f3774274e0832
    kubernetes.io/config.mirror: af6e265c669e39311b5f3774274e0832
    kubernetes.io/config.seen: "2026-05-03T09:51:34.943426976Z"
    kubernetes.io/config.source: file

```

- ✦ You can see 2 key value pairs:

kubernetes.io/config.mirror: <hash>

kubernetes.io/config.source: file

these tells API Server that this is not a Pod created by API Server;

this is a *mirror* of a static pod.

- ✦ *config.source: file* means the pod is created from a *file* present in the *node* itself;
- if this field is not there means the *pod* is created by *API Server* itself.

- ✦ The API server stores mirror pod objects also in etcd, but they are not the source of truth. They are copies of static pods created by kubelet and marked with annotations to indicate their origin.

✦



々

Resources inside Pods vs Resources as normal Processes

♪ As we saw the control plane processes run in **static pods** in the control plane node when you configure the cluster via **kubeadm**, **how the things communicate with each other then?**

And, **how the resources communicate with each other when they are not running inside any pod rather running outside normally as a process?**

♪ First lets see, how they work when run as normal processes.

♣ There are multiple **configuration** files inside **/etc/kubernetes** directory.

```
vagrant@controlplane:/etc/kubernetes$ ls
admin.conf  controller-manager.conf  kubelet.conf  manifests  pki  scheduler.conf  super-admin.conf
```

♣ All these files contains the specific requirements of each processes like **scheduler**, **controller manager** ...etc.

♣ All the **.conf** files are having same structure as the **~/.kube/config** file, but **user**, **certificates**, **permissions** are different.

♣ In the **scheduler.conf** the user is **system:kube-scheduler**

```
users:
- name: system:kube-scheduler
  user:
```

♣ **kubelet**, **kubectl**, **controller**, **scheduler** all talks to API Server.
API Server only directly talks to **etcd**.

```
scheduler —> API Server —> etcd
controller —> API Server
kubelet —> API Server
kubectl —> API Server
```

[No component talks to **etcd** directly except **API Server**]

♣ For example, if you see the **config** files, the **server** will be having **<control plane node's IP>:6443**

because, API Server listens on port 6443.

```
vagrant@controlplane:/etc/kubernetes$ sudo cat admin.conf | grep server
server: https://192.168.56.11:6443
vagrant@controlplane:/etc/kubernetes$ sudo cat scheduler.conf | grep server
server: https://192.168.56.11:6443
vagrant@controlplane:/etc/kubernetes$ sudo cat kubelet.conf | grep server
server: https://192.168.56.11:6443
```

- ⚡ All the server contains this end-point to talk to [because all talk to *API Server* only]

And, all the communication happens **via HTTP Calls**.

- ⚡ This is how the communication and work happens in **control plane** nodes.

🌀 Now, how the resources behaves when all are present inside Pods

- ⚡ As we saw, all the resources communication happens via *HTTP Calls* only.

So, even if they are present inside the pods, there will some port mapping like

6443:6443 (just example)

- ⚡ So, nothing changes. All the end-points remains same.
- ⚡ **kubeadm** keeps the same port mapping from host to pods, so the end-points doesn't change.

In case if you want to bind different host port to the pod, then

configuration files needs to be updated with this.

And, all of the things will work now.

- ⚡ **kubeadm** runs all the resources in **static pods** (not normal pods).

So, *API Server* or *etcd* has nothing to do with this.

[[[as we know, even if the pods are static, when you execute **kubectl get pods** command you'll be able to see the static pods because **kubelet** informs *API Server* about the static servers as well and *API Server* stores the details about the *static pods* inside **etcd**; but the *static pods* are having 2 extra fields inside **annotations** which are config.mirror: <hash> and config.source: file]]]

when you see the details about the *resources pods*, you'll find the fields inside *annotations*.

```
metadata:
  annotations:
    kubernetes.io/config.hash: caf1ce5bf82be95b0c98349010a12fde
    kubernetes.io/config.mirror: caf1ce5bf82be95b0c98349010a12fde
    kubernetes.io/config.seen: "2026-04-30T15:07:40.894345870Z"
    kubernetes.io/config.source: file
```

(kube-scheduler pod)

- ⚡ So, API Server cannot update these pods because *static pod* can only be updated if you change the **yaml** files directly inside *manifest* (or any directory configured) directory.
 - ⚡ The main advantage of running the resources inside *static pods* is if the pods crashes or the resources fails, kubelet will **automatically restarts** the pods by itself.
- 々 **NOTE ::::** *kubelet* doesn't run inside any pod, it is a **systemd** service; So, when you execute **systemctl cat kubelet** you get the *kubelet.service* file content.
- But, it'll not work for kube-apiserver, etcd, kube-scheduler ..etc because they run inside Pods.

Multiple Schedulers

✧ There are 3 different files, that might be confusing but they are different.

♣ **kube-scheduler.yaml**

present inside `/etc/kubernetes/manifest` (Static Pod's yaml) **kind: Pod**

♣ **scheduler.conf**

kube-scheduler config file (same structure as of `~/.kube/config` file)

♣ **scheduler-config.yaml**

[not same as `scheduler.conf`] **kind: KubeSchedulerConfiguration**

it is the yaml file to create **KubeSchedulerConfiguration** resource.

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: default-scheduler
```

♣ In simple words:

`kube-scheduler.yaml` creates the pod to run scheduler within it.

`scheduler.conf` is the configuration file (details and certificates) for that scheduler.

`scheduler-config.yaml` is the extra configuration plugins etc.

✧ When you open the **`kube-scheduler.yaml`** (kind: Pod, where kube-scheduler runs), you'll see a **`command`** field (which is run inside the container).

```
spec:
  containers:
  - command:
    - kube-scheduler
    - --authentication-kubeconfig=/etc/kubernetes/scheduler.conf
    - --authorization-kubeconfig=/etc/kubernetes/scheduler.conf
    - --bind-address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    image: registry.k8s.io/kube-scheduler:v1.36.0
```

[**kube-scheduler** commands comes with that *image* mentioned here]

♣ Here, the **`scheduler-config.yaml`** is not mentioned, rather directly all the things are written here.

- ♣ In modern way, ***scheduler-config.yaml*** is present which is more proper.

```
spec:
  containers:
    - name: kube-scheduler
      image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
      command:
        - kube-scheduler
        - --address=127.0.0.1
        - --kubeconfig=/etc/kubernetes/scheduler.conf
        - --config=/etc/kubernetes/my-scheduler-config.yaml
```

[the name would be *scheduler-config.yaml* in case of default one]

- 々 After creating your custom scheduler, you need to mention this explicitly inside Pod's yaml file to choose the *custom scheduler* instead of the default one.

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
    - name: nginx
      image: nginx
  schedulerName: my-custom-scheduler
```

- 々 F

Scheduler Profiles

々 Till now we saw the below things:

- ♣ One **yaml** file is there to create the Pod containing *scheduler* process (running scheduler binaries) [*/etc/kubernetes/manifest/kube-scheduler.yaml*]
- ♣ One **config** file is there which contains the details about *cluster*, *user*, *context* to authenticate and authorize kube-scheduler.
[*/etc/kubernetes/scheduler.conf*]
- ♣ One **yaml** file is there which contains the *configurations* of the *scheduler* [*scheduler-config.yaml*]

```
# my-scheduler-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
```

[for example]

it's reference will be there in the *command* section of *kube-scheduler.yaml* (in modern systems) or it'll not be present (in older systems the configs are directly given in that command section)

々 The Pod scheduling process comprises of mainly below phases:

- ♣ **Scheduling queue**
all the unscheduled pods will be kept in a queue according to their priority.
- ♣ **Filter phase**
scheduler eliminates the nodes which do not satisfy the requirement of the Pod's resource requirement.
- ♣ **Scoring phase**
nodes that pass *filter phase* are now scored. The node that is scored more will be scheduler.
- ♣ **Binding phase**
the node who got scored more score will be binded to the pod.

々 There are different **plugins** being used during this scheduling process.



♣ **PrioritySort plugin**

during the scheduling queue phase, this plugin orders the Pods based on their priorities.

♣ **NodeResourceFit plugin**

it is used during *filter phase* to exclude nodes.

Also, during *scoring phase* it score the nodes that pass in *filter phase*.

♣ **NodeUnschedulable plugin**

it makes sure that the nodes marked as Unschedulable: true will not get the pod assigned.

```
vagrant@controlplane:/etc/kubernetes$ kubectl describe node node01
Name: node01
Roles: <none>
Labels: beta.kubernetes.io/arch=amd64
        beta.kubernetes.io/os=linux
        env=dev
        kubernetes.io/arch=amd64
        kubernetes.io/hostname=node01
        kubernetes.io/os=linux
Annotations: csi.volume.kubernetes.io/nodeid: {"csi.tigera.io": "node01"}
              node.alpha.kubernetes.io/ttl: 0
              projectcalico.org/IPv4Address: 192.168.56.21/24
              projectcalico.org/IPv4VXLANTunnelAddr: 10.244.1.1
              volumes.kubernetes.io/controller-managed-attach-detach: true
CreationTimestamp: Thu, 30 Apr 2026 15:26:20 +0000
Taints: <none>
Unschedulable: false
```

♣ **ImageLocality plugin**

this just gives a *soft preference* if the node already contains the *required container image*.

♣ *DefaultBinder plugin*

in the binding phase, this plugin finalizes the Pod-to-Node assignment.

♪ In the above image, each stage is further divided into sub-stages like
queueSort

preFilter, filter, postFilter

preScore, score, reserve

permit, preBind, bind, postBind

♣ All these are called **extension points**.

♣ With the *extension points*, the **plugins** are *binded*.

♪ We can create multiple **scheduler pods** and run as many schedulers as we want.

And for each scheduler, we can create a **scheduler configuration** file (yaml) (**kind: KubeSchedulerConfiguration**).

♣ In this *KubeSchedulerConfiguration*, we can choose **which plugins we want to enable/disable for the extension points**.

♣ Let say you have 2 pods to be scheduled, and you want different types of *plugins* will be used in different *extension points* in those.

Then, you can create 2 *scheduler pods* and write 2 *kube scheduler configuration* each having specific plugins enabled/disabled.

```
# my-scheduler-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
```

```
# my-scheduler-2-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler-2
```

♣ But instead of doing this, we can just create one scheduler and attach multiple profiles to it.

[**scheduler** is the **process** that is running inside the Pod.

Profiles are the **configurations** having specific modules

enabled/disabled at the extension points]

- So, instead of creating multiple scheduling processes (pods), just create one scheduling process and attach multiple profiles to it.

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler-2
  plugins:
    score:
      disabled:
        - name: TaintToleration
      enabled:
        - name: MyCustomPluginA
        - name: MyCustomPluginB
- schedulerName: my-scheduler-3
  plugins:
    preScore:
      disabled:
        - name: "*"
    score:
      disabled:
        - name: "*"

```

- [2 scheduler profiles are there]

- The question arises, if we have multiple pods then we'll have concurrency if the pods chooses different profiles, but in this case there will not be having concurrency.

ALL THE PROFILES ATTACHED TO A SCHEDULER RUNS IN **MULTIPLE THREADS.**

- The nesting are like the below:

plugins >> **extension point** >> **disabled/enabled** >> **list of name (of plugins)**

々 Leader Selection

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
  leaderElection:
    leaderElect: true
    resourceNamespace: kube-system
    resourceName: lock-o
```

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
  leaderElection:
    leaderElect: false
```

[have to check if

leaderElection is scheduler level or profile level]

- ♣ If you enable the leader selection in all schedulers, **only one scheduler** will be active at a time.
- ♣ If you disable the leader selection then **all the schedulers** can be active at same time.
- ♣ Lets say you have 2 schedulers and 4 profiles
s1: p1 p2
s2: p3 p4
and all the profiles are different (i.e. the *schedulerName* are different)
 - **leaderElect: true**
in this case lets say your pod wants p4 (scheduler profile) and the leader got selected is **s1**.
as **s1** doesn't contain the required profile i.e. *p4* the pod will remain pending because the selected leader can't schedule the pod.
 - **leaderElect: false**
in this case, both *s1* and *s2* will be active and in this case *s2* (*p4*) will schedule the pod to specific node.
- ♣ If you keep **leaderElect: false**, its obvious race condition will occur (if duplicate *schedulerName* are present across the schedulers),

even in that case, **multiple scheduling doesn't happen in kubernetes**. If one scheduler schedule the node, then another scheduler will not be able to do.

🔗 How to read from the *docs* ? -----

- For all the configuration related things like *KubeSchedulerConfiguration*, ..etc etc, you can directly search in the docs.

kube-scheduler Configuration (v1) - Kubernetes

[Kubernetes](#) › [docs](#) › [reference](#) › [config-api](#) › [kube-scheduler-config](#)

[you'll get links like this]

- For the *Configuration* (you'll NOT get the docs usnig ***kubectl explain*** command because these are not resources, these are just the configurations), you should go to that ***reference > config-api*** path only (most of the cases).
- Inside that you'll see a lot of things like this.

ClientConnectionConfiguration [↗](#)

Appears in:

- [KubeSchedulerConfiguration](#)

ClientConnectionConfiguration contains details for constructing a client.

[you can see here written: *Appears in: KubeSchedulerConfiguration*]

- Then when you click that link (*KubeSchedulerConfiguration*)

leaderElection **[Required]**
[LeaderElectionConfiguration](#)

clientConnection **[Required]**
[ClientConnectionConfiguration](#)

DebuggingConfiguration **[Required]**
[DebuggingConfiguration](#)

you'll see some fields like this.

- It means, the previous field [ClientConnectionConfiguration] values can be written under *clientConnection* field inside *KubeSchedulerConfiguration* .

➤

Security Primitives

✧ RBAC: Roll Based Access Control

ABAC: Attribute Based Access Control

Node Authorization

Webhook Mode

✧ How to secure the cluster?

- ✦ The core component is API Server. It is the single point of contact for all operations, whether it is via *kubectl* or through *API*
 - ✦ Who can access the cluster?
 - ✦ Which actions can they perform once access is granted?

Authentication

✧ In a normal web application backend (what we usually build), the username and password are stored in database and using *JWT* token we verify the user.

But, Kubernetes doesn't have any database to store the user details. So it depends on an external provider like *static files*, *certificate authorities*, *third party identity provider like LDAP* . ---- for authentication **human users**.

[Application level user security is managed within the application itself]

✧ **Static file**

- ✦ You can create a **csv** file and write the user details there.

```
password123,user1,u0001,group1
password123,user2,u0002,group2
password123,user3,u0003,group2
```

If you want **tokens**, you can write the same (write token in place of password column)

```
KpjCvBI7rCFAHYPKByTLzRb7guLcUc4B,user10,u0010,group1
rJjncHmvtXHc6M1WQddhtvNyhgTdxSC,user11,u0011,group1
mjp0FEiFokLgtoikaRNTt59ePtczZSq,user12,u0012,group2
```

[4th column (group) is completely **optional**]

- ✦ Then you can mention this file inside the config.

If kube-apiserver is running as a **systemd** service (not running inside Pod)

```
ExecStart=/usr/local/bin/kube-apiserver \
  --authorization-mode=Node,RBAC \
  --advertise-address=... \
  --basic-auth-file=user-details.csv
```

if *kube-apiserver* is running inside **Pod** then write inside the configuration **yaml** file.

```
spec:
  containers:
  - name: kube-apiserver
    image: k8s.gcr.io/kube-apiserver-amd64:v1.11.3
    command:
      - kube-apiserver
      - --authorization-mode=Node,RBAC
      - --advertise-address=172.17.0.107
      - --basic-auth-file=/path/to/user-details.csv
```

- In case of **username password** the request will be like the below

```
curl -v -k https://master-node-ip:6443/api/v1/pods -u
"user1:password123"
```

in case of **username token** the request will be like below

```
curl -v -k https://master-node-ip:6443/api/v1/pods --header
"Authorization: Bearer df897ejkhefrt"
```

々

Generate Certificates for Cluster

々 One small Analogy about CA

----- IMPORTANT -----

- Lets assume CA as Government Authority.
 - ✦ It first finds its own identity key [government stamp | *ca.key*]
 - ✦ then it creates its own registration form [*ca.csr*]
 - ✦ then, it sign that by itself using that stamp [*ca.crt*]
 - ✦ [because, the authority has to self sign its own document]
- Now, this government certificate has to be made the source of TRUST so that whatever signed by this will be also a source of TRUST.
- Then, one user came with its details to make it verified.
 - ✦ Admin user has its own signature [signature | *admin.key*]
 - ✦ it creates a application form like aadhar card details [*admin.csr*]

- ⌘ Government authority has to approve it with their own stamp
[*admin.crt*]
[here, **ca.key** was used to sign **admin.csr** ; **ca.crt** didn't sign but it is the **TRUST** that this certificate belongs to a trusted authority]



- ⌘ There are multiple tools that provides certificates like OPENSSL, EASYRSA, CFSSL. We'll use **OpenSSL** here.
- ⌘ Will start with **CA** (Certificate Authority) certificate [SELF SIGNED]

🔗 All the components that needs Client or Server or Both certificates.

➤ Components that acts as *ONLY* Client

[these all calls the API Server → needs *Client certificate*]

- ✎ kubectI (admin users)
- ✎ kube-scheduler
- ✎ kube-controller-manager
- ✎ kube-proxy

➤ Components that acts as *BOTH* Client and Server

✎ **kube-apiserver**

- * client cert → talks to etcd, kubelet
- * server cert → serves kubectI / components

✎ **kubelet**

- * client cert → talks to API server
- * server cert → API server calls it (exec/logs)

✎ **etcd**

- * server cert → serves API server
- * client cert → when it talks peer-to-peer (clustered etcd)

🔗 For the clients to validate certificates sent by server and vice-versa, they all need a copy of **ca.crt** (certificate authority).

🔗 Read this Docs to setup SSL for each server and client components.

<https://notes.kodekloud.com/docs/Kubernetes-and-Cloud-Native-Associate-KCNA/Container-Orchestration-Security/TLS-in-Kubernetes-Certificate-Creation/page#generating-server-side-certificates>

🔗 Peer-to-Peer **etcd** communication:

➤ ~~Its not like one etcd server will be there, and another etcd clients.~~

All are etcd servers only.

➤ During communication, one **etcd server** will be chosen as the leader, and other will follow them.

It helps in preventing data inconsistency and duplication.

➤ During **kube-api-server** configuration, all the **etcd** end-points are provided to it [in the **command** section if pod, or **ExecStart** in case of resource]

- ♣ All the **etcd servers** stores the same data [data replicated in all etcd servers]
if one node dies, then also full data will be available.
- ♣ **etcd servers** are **static (never dynamic)**, so their end-points are directly provided in *kube-api-server* command.

```
Command:
  kube-apiserver
  --etcd-servers=https://127.0.0.1:2379
  advertise-address 192.168.56.11
```

- ♣ [if multiple end-points are there, then *comma separated*]

```
--etcd-servers=https://etcd1:2379,https://etcd2:2379,https://etcd3:2379
```

- ♣ kube-api-server just connect to one etcd server's end-point,
etcd cluster handles **leader selection, data consistency, request routing**.
kube-api-server doesn't find the leader to talk to, all that happens
internally within **etcd cluster**.

- ♣ F
- ♣ F
- ♣

々

TLS in Kubernetes

々 Some high-level concepts:

- Kubernetes has one **root authority** (CA)

it signs *client certs*, *server certs*.

- Kubernetes uses **mutual TLS (mTLS)**

Client verifies Server.

Server verifies Client.

[that's why both needs certificates]

Generating CA

```
openssl genrsa -out ca.key 2048
openssl req -new -key ca.key -subj "/CN=KUBERNETES-CA" -out ca.csr
openssl x509 -req -in ca.csr -signkey ca.key -out ca.crt
```

1. create **ca.key** [private key (or) *government's stamp*]

```
-----BEGIN PRIVATE KEY-----
MIIEEwIBADANBgkqhkiG9w0BAQEFAASCBKwwglAgEAAQIBAQCSIG1tgzfbtL91
YV+JFwZMpPd+Jf64ulK55fBgrrL0ce9qsDf7ISyRjqhJaUkIALJlBUGQDeD5ceno
glWSpPhEi817iQD2i/tI TA7ZVnzol602EvITC7YdaRz6H1XrsStii205wEh1+d/
```

2. create **ca.csr** [certificate request (or) *government's self application*]

```
-----BEGIN CERTIFICATE REQUEST-----
MIICXTCCAUAUCAAQAwGDEWMBQGA1UEAwNS1VCRVJ0RVRFUy1DQTCCASIwDQYJKoZI
hvcNAQEBBQADggEPADCCAQoCggEBABKwgbw2DN9u0v3VhX6MXDMyk934l/r16Wtn1
E8ausv0172wN(chl 1G0oE1nSSTAsm1E0ZANAP1y6egaVZ1Erg4Wl zXssAPal +tH
```

3. create **ca.crt** (signing with own *key*) [certificate (or) *government's self authorized certificate* (letter of trust)]

```
-----BEGIN CERTIFICATE-----  
MIICtzCCAZ8CFEdD/85/Rjnbfaf0LyVomsIAIzeOMA0GCSqGS Ib3DQEBCwUAMBgx  
FJAUBgNVBAMMDUtVQkVSTKvURVMtQ0EwHhcNMjYwNTA0MTkxMjJlOWhcNMjYwNzAz  
MTE1MTM6Lm9iaXNyMDEyNDY0ODAwMTU7ZmluZWVudC51bmRpbHRlcjEAMDphbiCo-
```

- **CN** : common name

name of the CA (KUBERNETES-CA in our case)

- CA self signed its own certificate request because CA is the root. No one is above it.

- Final output:

ca.crt, ca.key

```
[ this CA will sign all other certs ]
```

々 Creating **Client** Certificates

- Now we'll create certificates for
admin (kubectl), *scheduler*, *controller manager*, *kube-proxy*

Client Certificate: **Admin**

```
openssl genrsa -out admin.key 2048
```

```
openssl req -new -key admin.key \
-subj "/CN=kube-admin/O=system:masters" \
-out admin.csr
```

[here, CN = username (or) identity (*kube-admin* in here)

O = RBAC group (*system:masters* in here)]

system:masters

built-in super user group having full access to the cluster, and bypass RBAC restrictions.

kubectl get clusterrolebindings cluster-admin -o yaml

```
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: cluster-admin
subjects:
- apiGroup: rbac.authorization.k8s.io
  kind: Group
  name: system:masters
```

[**system:masters** group is cluster

admin]

- Sign the certificate with CA key and certificate.

```
openssl x509 -req -in admin.csr \
-CA ca.crt -CAkey ca.key \
-out admin.crt
```

- Instead of *username + password*

certificate + private key [it'll be used]

Client Certificate: OTHERS

- Same process as **admin**, just the **CN** will be changed for others.
- They'll create *key*, *certificate request* and **CA** will sign those to generate *certificate*.

Using Client Certificate

- Steps:

- Client sends request
- Server asks for certificate
- Client sends admin.crt

- ⌘ Server:
 - * verifies CA
 - * extracts CN (common name) and O (organization or group)
- ⌘ Applies RBAC
- ⌘ To use the certificates to talk to *kube-apiserver*, you have 2 options.
kubectl, *curl* (with key and certificates)

```
apiVersion: v1
clusters:
- cluster:
    certificate-authority: ca.crt
    server: https://kube-apiserver:6443
    name: kubernetes
kind: Config
users:
- name: kubernetes-admin
  user:
    client-certificate: admin.crt
    client-key: admin.key
```

[add in the *config* file and

use *kubectl* to talk to switching to the required **context**]

```
curl https://kube-apiserver:6443/api/v1/pods \
--key admin.key --cert admin.crt \
--cacert ca.crt
```

[or you can

directly pass the *admin key and certificates*, *CA certificate* with *curl*]

々 Creating Server Certificates

々 Server Certificate: *etcd*

⌘ `openssl genrsa -out etcd-server.key 2048`

```
openssl req -new -key etcd-server.key \
-subj "/CN=etcd" \
-out etcd-server.csr
```

```
openssl x509 -req -in etcd-server.csr \
-CA ca.crt -CAkey ca.key \
-out etcd-server.crt
```

- ⌘ As *etcd* is a distributed database across the cluster, so it needs to talk to the *peer etcd* as well.
 One *etcdserver.crt*, *etcdserver.key* will be there. And remaining will be for *etcdpeer* i.e.

etcdpeer1.crt, etcdpeer1.key

etcdpeer2.crt, etcdpeer2.key

.... etc

```
- etcd
- --advertise-client-urls=https://127.0.0.1:2379
- --key-file=/path-to-certs/etcdserver.key
- --cert-file=/path-to-certs/etcdserver.crt
- --client-cert-auth=true
- --data-dir=/var/lib/etcd
- --initial-advertise-peer-urls=https://127.0.0.1:2380
- --initial-cluster=master=https://127.0.0.1:2380
- --listen-client-urls=https://127.0.0.1:2379
- --listen-peer-urls=https://127.0.0.1:2380
- --name=master
- --peer-cert-file=/path-to-certs/etcdpeer1.crt
- --peer-client-cert-auth=true
- --peer-key-file=/etc/kubernetes/pki/etcd/peer.key
- --peer-trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
- --snapshot-count=10000
- --trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
```

[it

takes the keys and certs like this (*server key & crt, peer key & crt, etcd CA crt*) in the *etcd* yaml file]

NOTE: Kubernetes CA is a root, and etcd CA is a root. They don't trust each other.

✧ Server Certificate: **kube-apiserver**

- It's the most important thing because it is the core of Kubernetes cluster that is being used to talk to the whole cluster.
Either via **curl** (direct http REST call) or via **kubectl**, all the request comes to **kube-apiserver** only and it will be delegated from this only.
- **kube-apiserver** is both *server* and *client* component.
Client: it talks to **kubelet**, **etcd**
Server: serves **kubectl** and other components.
- It uses 2 different **CA** for its different certificates.

When it acts as a server (for **kubectl**, **kubelet**, ..etc), it gets certificate by **Kubernetes CA**. [*apiserver.crt*]

When it acts as a client (talking to **etcd**), it gets certificate by **etcd CA** [*apiserver-etcd-client.crt*]


```
--tls-cert-file=/etc/kubernetes/pki/apiserver.crt
--tls-private-key-file=/etc/kubernetes/pki/apiserver.key
--client-ca-file=/etc/kubernetes/pki/ca.crt
```

```
--etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt
--etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt
--etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key
```

- Now the question is, how Kubernetes trust 2 different CAs at once?

- Kubernetes doesn't have a single global trust store.
Each connection explicitly defines which CA to trust.

- Trust is not global,
Trust is per connection.

```
Connection 1 (incoming clients):
    → use Kubernetes CA

Connection 2 (to etcd):
    → use etcd CA
```

- You can see the certificates (client and server) and their respective CAs

```
vagrant@controlplane: /etc/kubernetes/pki$ ls
apiserver-etcd-client.crt  apiserver-kubelet-client.crt  apiserver.crt  ca.crt  etcd
apiserver-etcd-client.key  apiserver-kubelet-client.key  apiserver.key  ca.key  front-proxy-ca.crt  fr
vagrant@controlplane: /etc/kubernetes/pki$ openssl x509 -in apiserver-etcd-client.crt -noout -issuer
issuer=CN = etcd-ca
vagrant@controlplane: /etc/kubernetes/pki$ openssl x509 -in apiserver-kubelet-client.crt -noout -issuer
issuer=CN = kubernetes
vagrant@controlplane: /etc/kubernetes/pki$ openssl x509 -in apiserver.crt -noout -issuer
issuer=CN = kubernetes
```

- | | | | | |
|------------------------------|-----|----------------------|-------|------------|
| apiserver.crt | --- | server crt | ----- | kubernetes |
| apiserver-kubelet-client.crt | --- | client crt (kubelet) | ----- | kubernetes |
| apiserver-etcd-client.crt | --- | client crt (etcd) | ----- | etcd-ca |

[`openssl -in <certificate file> -noout -issuer`] is used to see the CA who signed the certificate.

- API Server is accessed via *kubectl, kubelet, scheduler, controller manager, external users* so it is accessed via
DNS names, IP addresses, service names
so the certificate must include all i.e.
kubernetes, kubernetes.default, kubernetes.default.svc, kubernetes.default.svc.cluster.local, cluster IP, node IP

- ⌘ That's why one OpenSSL configuration file has to be created to write the *aliases* (alternate names) and to be used in to create **apiserver.crt** certificate.

```
openssl x509 -req -in apiserver.csr \
  -CA ca.crt -CAkey ca.key \
  -out apiserver.crt \
  -extensions v3_req \
  -extfile openssl.cnf
```

```
[req]
req_extensions = v3_req
distinguished_name = req_distinguished_name

[ v3_req ]
basicConstraints = CA:FALSE
keyUsage = nonRepudiation, digitalSignature, keyEncipherment
subjectAltName = @alt_names

[alt_names]
DNS.1 = kubernetes
DNS.2 = kubernetes.default
DNS.3 = kubernetes.default.svc
DNS.4 = kubernetes.default.svc.cluster.local
IP.1 = 10.96.0.1
IP.2 = 172.17.0.87
```

⌘ Server Certificate: **kubelet**

- ⌘ **kubelet** will also have 2 kind of certificates, **client** and **server**.
- ⌘ **kubelet.crt** and **kubelet.key** are the server certificate and key for kubelet.
kube-apiserver talks to *kubelet* server to get the details about the pods and send the commands.
- ⌘ For all the *kubelets* as client, they talks to API Server.
So, for the API Server to know about the *kubelet*, it needs follow a specific naming convention.
system:node:<nodeName> [this should be the name in the *ssl certificate*]
for example: **system:node:node01**
- ⌘ To verify this, open worker node or controlplane node (kubelet will be there in both if installed by *kubeadm*)

- Execute **systemctl cat kubelet** [because *kubelet* doesn't run inside a pod unlike other resources] and get the **config.yaml** path.

```
Environment="KUBELET_KUBECONFIG_ARGS=--bootstrap-kubeconfig=/etc/kubernetes/t
Environment="KUBELET_CONFIG_ARGS=--config=/var/lib/kubelet/config.yaml"
# This is a file that "kubeadm init" and "kubeadm join" generates at runtime
```

- In that path i.e. */var/lib/kubelet/pki* the *certificates and keys* will be present.

```
vagrant@node01:/var/lib/kubelet/pki$ pwd
/var/lib/kubelet/pki
vagrant@node01:/var/lib/kubelet/pki$ ls
kubelet-client-2026-04-30-15-26-19.pem  kubelet-client-current.pem  kubelet.crt  kubelet.key
```

- You can see the *name* and *group* of this *kubelet-client-current.pem* by the command

openssl x509 -in <file name> -noout -text [it'll give full details]

openssl x509 -in <file name> -noout -subject [it'll give only subjects]

```
vagrant@node01:/var/lib/kubelet/pki$ sudo openssl x509 -in kubelet-client-current.pem -noout -subject
subject=O = system:nodes, CN = system:node:node01
```

system:node:node01 [node's name is *node01*]

[due to permission and all, use **sudo** with this command]

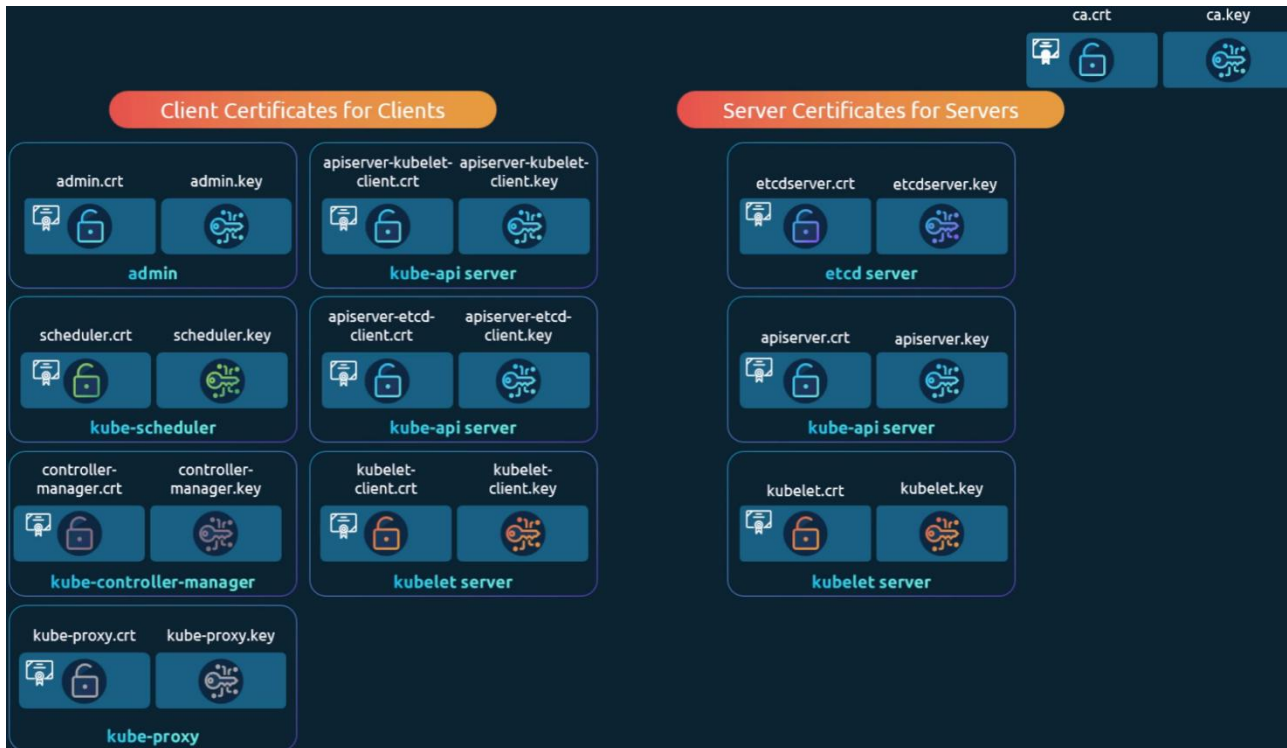
- F
- F
-

🔸 If you want to see the details of the certificate, then execute this command

```
cat <crt file name> | openssl x509 -text -noout
```

```
vagrant@controlplane: /etc/kubernetes$ kubectl config view --raw -o json
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number: 3081854885164004134 (0x2ac4f2b08f80b326)
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: CN = kubernetes
    Validity
      Not Before: Apr 30 15:01:07 2026 GMT
      Not After : Apr 30 15:06:07 2027 GMT
    Subject: O = kubeadm:cluster-admins, CN = kubernetes-admin
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:aa:e8:cd:0c:fa:e9:00:1f:fd:f5:65:e5:8c:f7:
```

々 The below is details about all the servers and clients.



々 F

々 LEGACY NOTES (JUST FOR MORE INFO)

々 Now, we'll generate the client certificates.

々 Admin User (kubectl) [client certificate]

- Generate admin.key using the below command.

```
openssl genrsa -out admin.key 2048
```

- Generate admin.csr using the below command

```
openssl req -new -key admin.key -subj "/CN=kube-admin" -out admin.csr
```

[the name can be anything, but *kube-admin* is the name *kubectl* client authenticates with]

if you want to group the keys and certificates (system, server ...etc)

```
openssl req -new -key admin.key -subj "/CN=kube-admin/O=system:masters" -out admin.csr
```

[/O means *Organization (group* in Kubernetes)]

here the group is **system:masters** , we want a **admin user**, not a normal user. Keeping the group *system:masters* allow the user the admin privileges.

There is something called **RoleBinding**, in that the *user* is mapped. That user's name must be same as /CN of the *certificate* otherwise it'll fail.

Inside the *config file*, the user's name can be different it doesn't create any issue.

```
users:
- name: system:kube-scheduler
  user:
    client-certificate-data: LS...
```

[/etc/kubernetes/scheduler.conf]

here the user's name can be different, no problem.

If you see the below, the group name of *cluster-admins* is

system:masters (that's why the group was added to the admin user)

KubeConfig File

- 🔗 To talk to API Server, you need to attach the *certificates* and all to authenticate yourself.

```
curl https://my-kube-playground:6443/api/v1/pods \
  --key admin.key \
  --cert admin.crt \
  --cacert ca.crt
```

Same for **kubectl** command as well, you have to provide these

```
kubectl get pods \
  --server my-kube-playground:6443 \
  --client-key admin.key \
  --client-certificate admin.crt \
  --certificate-authority ca.crt
```

- ♣ So, rather than writing all these things every time, you can just add the details in **kubeconfig** file and just execute the **kubectl** command without providing all those keys and certificates.
- ♣ If the **kubeconfig** file is present at **~/.kube/config** then **kubectl** automatically take that.
Otherwise, you need to use the flag **--kubeconfig** to provide the kubeconfig file's path.

```
kubectl get pods --kubeconfig config
```

- 🔗 Structure of **kubeconfig** file

- ♣ **Clusters:** Represent various Kubernetes clusters (e.g., development, testing, production, or cloud-based clusters).
- ♣ **Users:** Contain credentials and client certificate information for users (e.g., admin, dev, prod).
- ♣ **Contexts:** Link clusters and users to define which user accesses which cluster. For example, a context named "admin@production" may indicate that the admin account is used to access the production cluster. It is not necessary that the name will be **<user name>@<cluster name>**, its just a CONVENTION.

- ✧ To change the current context

kubectl config use-context <context name> [it updates the config file's current-context field]

```
apiVersion: v1
kind: Config
clusters:
- name: my-kube-playground
  cluster:
    certificate-authority: ca.crt
    server: https://my-kube-playground:6443
contexts:
- name: my-kube-admin@my-kube-playground
  context:
    cluster: my-kube-playground
    user: my-kube-admin
users:
- name: my-kube-admin
  user:
    client-certificate: admin.crt
    client-key: admin.key
```

々

Instead of the paths of certificates and key files, you can directly pass the content of those (crt & key) encoded in **base64** format.

```
cat ca.crt | base64
```

```
apiVersion: v1
clusters:
- cluster:
    certificate-authority-data: LS0tLS1CRUdJ
    server: https://192.168.56.11:6443
  name: kubernetes
contexts:
- context:
    cluster: kubernetes
    user: kubernetes-admin
  name: kubernetes-admin@kubernetes
current-context: kubernetes-admin@kubernetes
kind: Config
users:
- name: kubernetes-admin
  user:
    client-certificate-data: LS0tLS1CRUdJTiB
    client-key-data: LS0tLS1CRUdJTiBSU0EgU
```

々 To access the API Server using curl (direct http request) ---- **JUST FOR INFO**

々 In my case I couldn't find the **admin.crt** and **admin.key** files, so I had to convert the *client-certificate-data* and *client-key-data* fields (present inside `~/.kube/config`) to **base64** (BEGIN, END) format.

```
kubectrl config view --raw -o jsonpath='{.users[0].user.client-key-data}' |  
base64 -d | sudo tee admin.key
```

- [**base64 -d** convert the content to *BEGIN*, *END* type of content]
- Similarly extract the *client-cert-data* as well.
- Then just execute the **curl** command giving the **ca certificate**, **admin certificate**, **admin key** and it'll work.

```
curl https://192.168.56.11:6443 --cert admin.crt --key admin.key --  
cacert ca.crt
```

[If it doesn't work, then check the permissions of the keys and certificate files]

[NOTE: It is **https** not *http*]

- The above base url (**https://<IP>:6443**) give all the available paths that **kube-apiserver** accepts.

```
{  
  "paths": [  
    "/.well-known/openid-configuration",  
    "/api",  
    "/api/v1",  
    "/apis",  
    "/apis/",  
    "/apis/admissionregistration.k8s.io",  
    "/apis/admissionregistration.k8s.io/v1",  
    "/apis/apiextensions.k8s.io",  
    "/apis/apiextensions.k8s.io/v1",  
    "/apis/apiregistration.k8s.io",  
    "/apis/apiregistration.k8s.io/v1",  
    "/apis/apps",  
    "/apis/apps/v1",  
    "/apis/authentication.k8s.io",  
    "/apis/authentication.k8s.io/v1"
```

- There are 2 ways you can talk to **kube-apiserver**, one with **kubectrl** another is directly **http request**.

➤

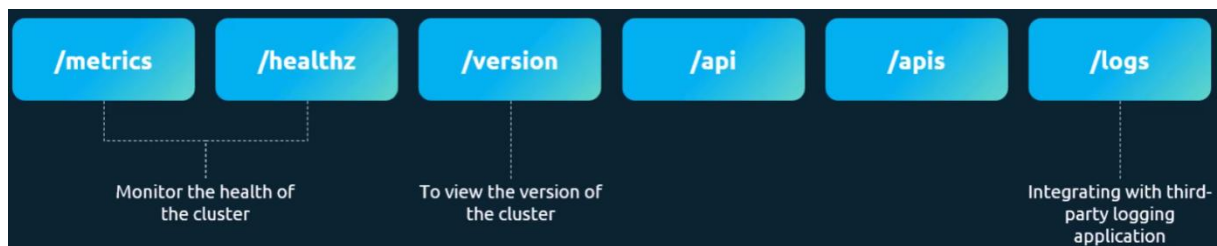
API Server

When you trigger this http request

```
curl https://192.168.56.11:6443 --cert admin.crt --key admin.key --cacert ca.crt
```

you'll get the list of *api paths* of the kube apiserver.

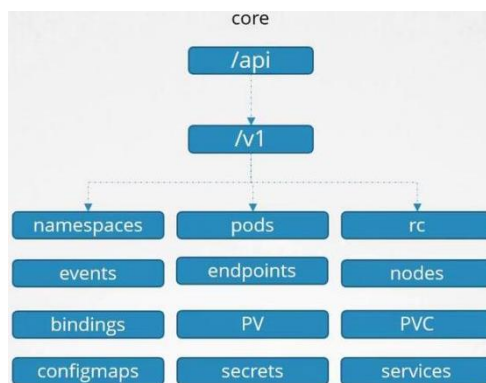
```
vagrant@controlplane:/etc/kubernetes/temp$ sudo curl https://controlplane:6443 --cert admin.crt --key admin.key --cacert ca.crt
{
  "paths": [
    "/.well-known/openid-configuration",
    "/api",
    "/api/v1",
    "/apis",
    "/apis/",
    "/apis/admissionregistration.k8s.io",
    "/apis/admissionregistration.k8s.io/v1",
    "/apis/apixtensions.k8s.io"
```

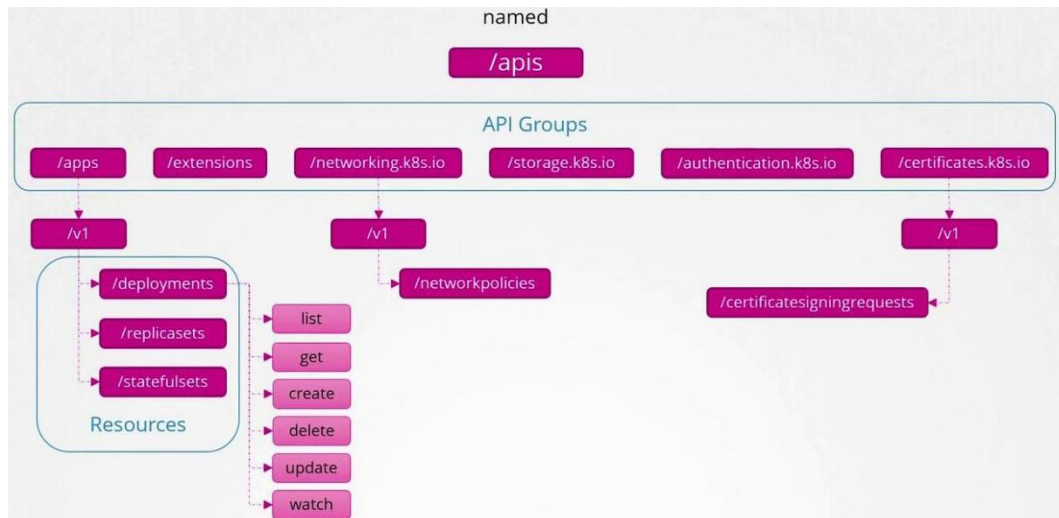


The cluster functionality is divided in 2 major categories.

/api : Core Api root Path

/apis : Named Api root Path





- Whatever things will be added, those will be added in **named** API groups because there are properly organized.

/api is the core API Group.

- /apis** is named API root path because it contains *API groups*.
where **/api** doesn't contain any *API group*, rather contains the *version* directly.

When you write the *yaml* file, you write either “**v1**” or “**<api group name>/<version>**” like this.

- If you write only version i.e. “**v1**” then it'll take the core API root path [**/api**], because it doesn't contain any API group.
/api/v1/pod, **/api/v1/services** ...etc
- If you write something like “**apps/v1**” then it'll take the Named API root path [**/apis**] because it contains the groups.
/apis/apps/v1/replicaset, ..etc

Names of the components:

- /apps**, **/extensions** ...etc present inside **/apis** (Named API groups) are called **API Groups**.
- Inside those **Versions** are there (**v1**).
- Inside each version, **Resources** are there.
Ex: **apps/v1** contains *deployments*, *replicasets*, *statefulsets* ...etc.
- Each Resources contains **Verbs** like *list*, *create*, *delete* ..etc
ex: *list* the deployments, *create* deployment, *delete* deployment ..etc.

々 To see all the named API groups, execute this command

`curl https://controlplane:6443/apis -k --cert admin.crt --key admin.key --cacert ca.crt | grep name` [NOTE: *admin.crt*, *admin.key*, *ca.crt* should be given]

```
9530 0 6930 0 0 84687 0
"name": "apiregistration.k8s.io",
"name": "apps",
"name": "events.k8s.io",
"name": "authentication.k8s.io",
"name": "authorization.k8s.io",
"name": "autoscaling",
"name": "batch",
"name": "certificates.k8s.io",
"name": "networking.k8s.io",
"name": "policy",
"name": "rbac.authorization.k8s.io",
"name": "storage.k8s.io",
"name": "admissionregistration.k8s.io",
"name": "apiextensions.k8s.io",
"name": "scheduling.k8s.io",
"name": "coordination.k8s.io",
"name": "node.k8s.io",
"name": "discovery.k8s.io",
"name": "resource.k8s.io",
"name": "flowcontrol.apiserver.k8s.io",
"name": "projectcalico.org",
"name": "crd.projectcalico.org",
"name": "operator.tigera.io",
"name": "policy.networking.k8s.io",

```

Authorization

々 There are multiple types of Authorization mechanisms

- ♣ Node
- ♣ ABAC [Attribute Based Access Control]
- ♣ RBAC [Role Based Access Control]
- ♣ Webhook

々 Node Authorizer

- ♣ We saw before, the **kubelet** certificate's name was in the form of **system:node:<node name>**
- ♣ This Node Authorizer is an authorization method of **kube-apiserver** that grant permissions to **kubelets** (nodes), limit their access to the resources linked with their *respective node only*, prevents nodes from other node's or pod's data.
- ♣ If you see the *subject* of the certificate of **kubelet** in any node, you can see the CN and group

```
vagrant@node01:/var/lib/kubelet/pki$ ls
kubelet-client-2026-04-30-15-26-19.pem  kubelet-client-current.pem  kubelet.crt  kubelet.key
vagrant@node01:/var/lib/kubelet/pki$ sudo openssl x509 -in kubelet-client-current.pem -noout -subject
subject=O = system:nodes, CN = system:node:node01
```

[CN: system:node:node01, O: system:nodes]

[node01 is then worker node's name; O means group, CN means common name or name]

- ♣ This certificate is signed by Kubernetes itself.

```
vagrant@node01:/var/lib/kubelet/pki$ sudo openssl x509 -in kubelet-client-current.pem -noout -issuer
issuer=CN = kubernetes
```

- ♣ You **CAN'T CHANGE** this format of **system:node:<node name>**
It is HARD-CODED inside Kubernetes.
- ♣ This method is enforced by **kube-apiserver** ; it expects the CN and group in that format.
If the format is not proper, then the request is treated as a *normal user*, **NOT a Node**.
- ♣ If you see the **kube-apiserver.yaml** (Kube Api Server's pod config yaml file)
[**/etc/kubernetes/manifests/kube-apiserver.yaml**]
then you can see the **--authorization-mode** flag would have set to **Node**

along with some other authorization mode.

[if kube-apiserver is running as a process, not Pod, then you can see inside **service** file using *systemctl cat kube-apiserver*]

```
spec:
  containers:
  - command:
    - kube-apiserver
    - --advertise-address=192.168.56.11
    - --allow-privileged=true
    - --authorization-mode=Node,RBAC
```

々 ABAC [Attribute Based Access Control]

- You provide access to a set of users or group a specific set of permissions.
- In here, you create json file of **kind: Policy** and provide that to *kube-apiserver*

```
{"kind": "Policy", "spec": {"user": "dev-user", "namespace": "*", "resource": "pods", "apiGroup": "*"}}
{"kind": "Policy", "spec": {"user": "dev-user-2", "namespace": "*", "resource": "pods", "apiGroup": "*"}}
{"kind": "Policy", "spec": {"group": "dev-users", "namespace": "*", "resource": "pods", "apiGroup": "*}}
```

- When a request hits the kube-apiserver, its attributes are extracted:
user, verb, resources, namespace
[alok, get, pods, default (For Example)]
and apiserver checks ABAC Policy file; if match found then *allow* else *reject*.
- It is **not-preferable** because for each change you need to update the file for each user/group.

々 RBAC [Role Based Access Control]

- Instead of writing rules in a file (like ABAC), RBAC uses **Kubernetes Objects** [just like pods, replicaset ..etc; you can see in *kubectl api-resources*],
you can write *yaml* and create the objects.

- The object that RBAC provides are

Role, RoleBinding, ClusterRole, ClusterRoleBinding

```
vagrant@controlplane:/etc/kubernetes/manifests$ kubectl api-resources | grep -i role
clusterrolebindings      rba
clusterroles              rba
rolebindings             rba
roles                    rba
```

- ♣ Its kind of, create a role, and associate users/groups to that role.
If you want to update any permissions, then **only need to change in the Role object** unlike ABAC where you need to update each Policy.
- ♣ Role: permissions within a **single namespace**.
ClusterRole: permissions across the whole **cluster** or cluster level resources (like nodes)
 - ✦ Give access to pod in a namespace: *Role*
Give access to pod in ALL namespaces: *ClusterRole*
Give access to nodes: *ClusterRole*
 - ✦ **all things that can be done using role can also be done with clusterrole.**
- ♣ RoleBinding: grant access within a **specific namespace**.
ClusterRoleBinding: grant access across the **entire cluster**.
 - ✦ **RoleBinding:**
binds a user/group to **Role, or ClusterRole**
scope: **one namespace only**
 - ✦ **ClusterRoleBinding:**
binds a user/group to **ClusterRole ONLY**
scope: **entire cluster**

🔗 Webhook

- ♣ It'll make Kubernetes to query an third party REST service to get to know if the user is authorized for that specific request or not.
- ♣ One third party provider is: *Open Policy Client*
- 🔗 There are 2 more: **AlwaysAllow, AlwaysDeny**
 - ♣ When no authorization mechanism is provided, by default it is *AlwaysAllow*.

```
ExecStart=/usr/local/bin/kube-apiserver \
  --advertise-address=${INTERNAL_IP} \
  --allow-privileged=true \
  --apiserver-count=3 \
  --authorization-mode=AlwaysAllow \
  --bind-address=0.0.0.0 \
```

- You can give multiple authorization mechanism here (comma separated)

```
ExecStart=/usr/local/bin/kube-apiserver \
  --advertise-address=${INTERNAL_IP} \
  --allow-privileged=true \
  --apiserver-count=3 \
  --authorization-mode=Node,RBAC,Webhook \
```

- When multiple authorization mechanism is given, the execution happens sequentially (from left to right)

in above case,

first **Node** authorization will happen;

if it rejects, then RBAC will happen;

if it rejects, then Webhook will happen.

- It'll keep trying till the authorization is allowed.

If it is allowed, then it'll go to the next authorization mechanism to the right.

- If all the authorization reject the request, then user will get forbidden.

- 々 To check if you are authorized for doing something, then use **auth can-i** like the below

```
vagrant@controlplane:~$ kubectl auth can-i create deployments
yes
```

- 々 If you want to check the authorization of another user, then use **--as <user name>**

```
vagrant@controlplane:~$ kubectl auth can-i create deployments --as dev-user
no
```

- 々 F

RBAC

々 Role

- It is present in “`rbac.authorization.k8s.io/v1`”
- To create a role, you need to write the rules.
- Each rule contains *apiGroups*, *resources*, *verbs*

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: pod-reader
rules:
- apiGroups:
  - ""
  resources:
  - pods
  verbs:
  - get
  - list
  - watch
```

- **NOTE:** apiGroups “” doesn’t mean all api-groups, it means Core Api Group (because in core api group is not there)

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: developer
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["list", "get", "create", "update", "delete"]
- apiGroups: [""]
  resources: ["ConfigMap"]
  verbs: ["create"]
```

- The Role is *namespace* scoped.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  namespace: default
  name: pod-reader
rules:
- apiGroups: [""] # "" indicates the core API group
  resources: ["pods"]
  verbs: ["get", "watch", "list"]
```

[namespace is

provided here]

- ♣ Lets say in the *namespace* so many pods are there, but you just want to give access of some specific pods in that role, other pods should not be accessible.

In that case, you can use **resourceNames** field.

```
kind: Role
metadata:
  name: developer
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "create", "update"]
  resourceNames: ["blue", "orange"]
```

々 RoleBinding

- ♣ This is to bind a Role to a User/Group.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: devuser-developer-binding
subjects:
- kind: User
  name: dev-user
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: developer
  apiGroup: rbac.authorization.k8s.io
```

- ♣
- ♣ This is also specific to *namespace*.

ε NOTE ----- IMPORTANT -----

here you can see **kind: User** [there is nothing called User or Group object (you'll not find this when *kubectl api-resources* is executed.)]

here, this **subjects[*].kind** is not same as **.kind** (top level); it is just a key to define what kind of subject is that.

Now, you might be thinking, **apiGroup** is not needed because the *apiGroup* rbac.authorization.k8s.io doesn't contain anything called

User or Group because if it contains, then it would be a Kubernetes Object (like pod, replicaset, role, clusterrole, ...etc)

kind can be User, Group or ServiceAccount ; and ServiceAccount is a real object (present in Core API Group) but User and Group are not object.

The RBAC matching logic is written using the combination of **kind** and **apiGroup**, so if you don't give *apiGroup* it'll fail.

```
(kind + apiGroup)
```

it'll be always

```
apiGroup: rbac.authorization.k8s.io
```

 (for User, Group) or

```
apiGroup: ""
```

 (for ServiceAccount)

```
-----
apiVersion: rbac.authorization.k8s.io/v1
# This role binding allows "jane" to read pods in the "default" namespace.
# You need to already have a Role named "pod-reader" in that namespace.
kind: RoleBinding
metadata:
  name: read-pods
  namespace: default
subjects:
# You can specify more than one "subject"
- kind: User
  name: jane # "name" is case sensitive
  apiGroup: rbac.authorization.k8s.io
roleRef:
# "roleRef" specifies the binding to a Role / ClusterRole
kind: Role #this must be Role or ClusterRole
name: pod-reader # this must match the name of the Role or ClusterRole you wish to bind to
apiGroup: rbac.authorization.k8s.io
```

- One confusion is **namespace**, if we have given *namespace* field inside **Role**, then why do we need to give it again in **RoleBinding**?

namespace in **Role** defines “In which namespace the Role should be created”

namespace in **RoleBinding** defines “I'll bind to the role present in that namespace”

so basically, if you have given **namespace: ns1** in the **RoleBinding**, then

the specific role must be present in the *namespace ns1*. if the role is present in some other namespace then it'll not work.

* [its only for **Role**. Not for **ClusterRole**]

々 **Role** and **RoleBinding** are *namespace scoped*.

If *namespace* is not provided, then the objects (*Role*, *RoleBinding*) will be created in the same *namespace* where they are being called.

Not the *default namespace*.

々 **ClusterRole**

♣ It is same as **Role**, but it is not limited to any *namespace*.

Its *cluster scoped*.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  # "namespace" omitted since ClusterRoles are not namespaced
  name: secret-reader
rules:
- apiGroups: ["" ]
  #
  # at the HTTP level, the name of the resource for accessing Secret
  # objects is "secrets"
  resources: ["secrets"]
  verbs: ["get", "watch", "list"]
```

♣ unlike *Role*, *namespace* field will not be here, because it'll be available *cluster wide*.

♣ One more difference between *Role* and *ClusterRole* is:

Role can have *namespace* resources.

ClusterRole can have both *namespace* and *cluster-scoped* resources.

♣ To check namespace and non-namespaced resources, execute the command

kubectl api-resources --namespaced=true [namespace resources]

kubectl api-resources --namespaced=false [non-namespaced]

resources]

```
vagrant@controlplane:~$ kubectl api-resources --namespaced=false
NAME                                SHORTNAMES          APIVERSION
componentstatuses                   cs                  v1
namespaces                          ns                  v1
nodes                              no                  v1
persistentvolumes                  pv                  v1
mutatingadmissionpolicies           adm                 v1
mutatingadmissionpolicybindings     adm                 v1
mutatingwebhookconfigurations       adm                 v1
validatingadmissionpolicies         adm                 v1
```

々 ClusterRoleBinding

- It is used to bind the **ClusterRole** to any User or Group or ServiceAccount.
- If you bind a **ClusterRole** using **ClusterRoleBinding** then the authorized resources (mentioned in **ClusterRole**) will be authorized to the User or Group or ServiceAccount with which the **ClusterRole** is bound to.
- **ClusterRole** can be bound with **RoleBinding** as well, in this case the **authorization** will be applied to a specific namespace where **RoleBinding** will be created.

々 After creating a binding, if you want to change the **roleRef** you'll get a validation error.

If you really want to do it, then you need to delete the **binding** object and create a new one with the new **roleRef**.

々

ServiceAccount

々 There are 2 types of accounts: **User, Service**.

- User account is used by the users like admin, developer.
- Service account is used by applications to interact with kubernetes cluster like *Jenkins, Prometheus* ..etc.
 - [Prometheus pulls kubernetes api for the performance matrix]
 - [Jenkins used Service Account to deploy the application in Kubernetes cluster]

々 In all namespace, there will be a default *ServiceAccount* whose name is “default”.

calico-apiserver	default	6d5h
calico-system	default	6d5h
default	default	6d5h
kube-node-lease	default	6d5h
kube-public	default	6d5h
kube-system	default	6d5h
test	default	5d
tigera-operator	default	6d5h

[namespace ServiceAccount AGE] (columns)

- When a pod is created, by-default the *ServiceAccount* present in that namespace (where pod is created) will be attached to the pod *IF NOT EXPLICITLY MENTIONED*.

```
vagrant@controlplane:~/.kube$ kubectl describe pods dev-pod
Name:          dev-pod
Namespace:     default
Priority:       0
Service Account: default
```

- Pod needs the **ServiceAccount** to talk to **API Server**.
 - [there are multiple reasons for which Pod talks to API Server.
 - But, ~~pod-to-pod communication doesn't happen through API Server~~]

々 Both **internal components** (like pods, ..etc) and *external services* (like Jenkins ..etc) uses **ServiceAccount** to communicate with API Server.

- ServiceAccount is required to give access to the resources via **Role** or **ClusterRole**.
- For external services, you need to create a token linked to the ServiceAccount.
 - With that token (Authorization Bearer) the external service can

communicate to API server.

kubectrl create token <service account name>

```
kubectrl create token external-sa -n dev
```

NOTE: Token **doesn't have name**, and it is **not stored anywhere**.

When you execute this command, a token will be generated and displayed in the screen using which the Service can trigger REST request using

Bearer <generated token>.

- For internal components (like Pod), the *Service Account* is directly attached to it. But that doesn't mean it doesn't require token.

For internal components, a **token is generated automatically** using which they can talk to API Server.

- When a *Service Account* is attached to a Pod, kubernetes creates a virtual volume and mounts that to the Pod. [if

spec.automountServiceAccountToken: true inside Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: my-kubernetes-dashboard
spec:
  containers:
    - name: my-kubernetes-dashboard
      image: my-kubernetes-dashboard
  automountServiceAccountToken: false
```

(or)

automountServiceAccountToken: true inside ServiceAccount]

This mount path is:

/var/run/secrets/kubernetes.io/serviceaccount/token [inside Pod]

```
Mounts:
  /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-rdvhr (ro)
```

this contains *token*, *ca.crt*, *namespace* details.

[NOTE: volume's lifecycle = pod's life cycle; **if pod is gone, volume is also destroyed**]

- In short,
ServiceAccount is required to provide specific authorization in

Role or ClusterRole.

After that, a token will be generated linked to that Service Account (automatic for internal components, manually for external services) and that token is not stored anywhere.

For Pods, the token is generated and stored inside that mounted path and it gets refreshed when it is expired.

Virtual volume's is destroyed once pod is gone.

Image Security

- ✧ All the official images like nginx/nginx, redis/redis ..etc will be having a verified badge “Docker Official Image”.

these all are official images maintained by Docker.

When you execute **docker pull nginx**,

it pulls from **library/nginx** (library stores all the official images)

- ✧ When you provide the image name, it actually translates that to **library/<image name>** because it is where the official images are present in docker hub.

```
image: library/nginx
```

- ✧ It pulls from Docker Registry (Docker Hub) by default,

```
image: docker.io/library/nginx
```

- ✧ But, if you are having a private container registry and want to pull the image from there, then you need to give full path against the *image* key.

```
image: private-registry.io/apps/internal-app
```

- ✧ Now, as this is private registry and it needs to be authenticated to access that, you need to create a resource called **secret** .

```
kubectl create secret docker-registry regcred \
  --docker-server=private-registry.io \
  --docker-username=registry-user \
  --docker-password=registry-password \
  --docker-email=registry-user@org.com
```

docker-registry: type of secret

regcred: name of secret (just an identifier)

々 And mention that *secret* in the Pod's yaml file inside the field

spec.imagePullSecrets

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
  - name: nginx
    image: private-registry.io/apps/internal-app
    imagePullSecrets:
    - name: regcred
```

```
spec:
  imagePullSecrets:
  - name: my-registry-secret-1
  - name: my-registry-secret-2
```

々

pod.spec.imagePullSecrets is a list; it'll try one by one until it finds the required image.

[containers and imagePullSecrets: both are siblings]

々 Create **secret** from yaml

```
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJl
kind: Secret
metadata:
  name: my-secret
type: kubernetes.io/dockerconfigjson
```

[yaml file looks something like this]

♣ *secret.data.\.dockerconfigjson* (\. means to keep . as a string)

♣ That is **base64** encoded.

On decoding that:

```
{
  "auths": {
    "DOCKER_REGISTRY_SERVER": {
      "username": "DOCKER_USER",
      "password": "DOCKER_PASSWORD",
      "email": "DOCKER_EMAIL",
      "auth": "RE9DS0VSX1VTRVI6RE9DS0VSX1BBU1NXT1JE"
    }
  }
}
```

[something like this

will come]

々

Security Context

- ✧ In docker, while creating a container, you can specify if you don't want the **root** user as default.

[root is risky, if Pod is compromised, then it'll have full access because of root user]

you can specify **--user=<user id>** while running Docker container.

```
docker run --user=1001 ubuntu
```

- ✧ When a container runs, it doesn't get full root privileges *even if it is root user*.

Docker uses a Linux feature called **capabilities** to split root power into smaller pieces.

Using **--cap-add <capability name>** you can provide the specific capability to the user.

```
docker run --cap-add MAC_ADMIN ubuntu
```

```
NET_ADMIN → control networking
SYS_ADMIN → system-level control
CHOWN → change file ownership
MAC_ADMIN → control security policies
```

- ✧ In Kubernetes also you can configure all of these.

- As, Containers run inside a Pod in Kubernetes.

So, if you configure the security settings at Pod level, it'll be **applied to all the containers** running inside it.

If you configure the security settings at both Pod and Container level, Pod's security settings will be **overridden by Container's security settings**.

- You can add a field `spec.containers[*].securityContext` to specify the security configurations *at container level*.

```
apiVersion: v1
kind: Pod
metadata:
  name: web-pod
spec:
  containers:
    - name: ubuntu
      image: ubuntu
      command: ["sleep", "3600"]
      securityContext:
        runAsUser: 1000
        capabilities:
          add: ["MAC_ADMIN"]
```

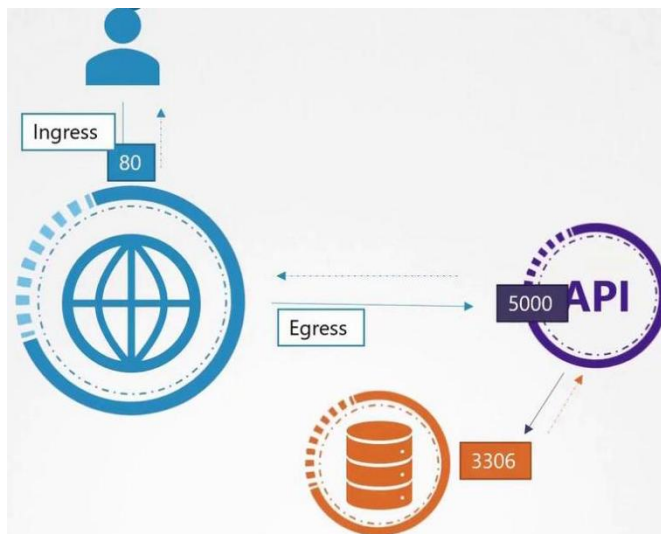
[Container Level]

- If you want to configure it in *pod level* then, add the field `spec.securityContext`.

```
apiVersion: v1
kind: Pod
metadata:
  name: web-pod
spec:
  securityContext:
    runAsUser: 1000
    capabilities:
      add: ["MAC_ADMIN"]
  containers:
```

Network Policies

- ✧ In Kubernetes, there is no restriction for inter Pod communication by default.
Any pod can talk to any other pod using the target Pod's service (ClusterIP).
- ♣ To restrict this *Network Policies* are used.
- ♣ Its kind of ABAC (Attribute Based Access Control), because here also Policy was there which had to be written in a JSON format.
[Not same as ABAC, because ABAC is of course used for Authorization; and Network Policy is used for restrict the access of Pods]
- ♣ For example, there are 3 types of pods running.
 - **Frontend, Backend, DB**
 - By default, Frontend can access the DB pods which should not happen.
- ✧ Consider the below image



- ♣
- ♣ Web Server:
 - Ingress: accept HTTP traffic on Port 80
 - Egress: sends traffic on port 5000 to Backend API server.
- ♣ Backend API Server
 - Ingress: Port 5000
 - Egress: Port 3306 to DB
- ♣ DB
 - Ingress: Port 3306

々 NetworkPolicy

- ♣ It is **namespace scoped**.
- ♣ controls traffic to and from pods within that namespace, optionally including traffic from other namespaces via selectors

```
apiVersion: networking.k8s.io/
kind: NetworkPolicy
metadata:
  name: test-network-policy
  namespace: default
spec:
  podSelector:
    matchLabels:
      role: db
  policyTypes:
    - Ingress
    - Egress
  ingress:
```

```
    ingress:
      - from:
          - ipBlock:
              cidr: 172.17.0.0/16
              except:
                - 172.17.1.0/24
          - namespaceSelector:
              matchLabels:
                project: myproject
          - podSelector:
              matchLabels:
                role: frontend
        ports:
          - protocol: TCP
            port: 6379
    egress:
      - to:
          - ipBlock:
              cidr: 10.0.0.0/24
        ports:
          - protocol: TCP
            port: 5978
```

- ♣ **spec.podSelector** selects the pod in which NetworkPolicy will be applied. (here **db**)
- ♣ **spec.ingress** and **spec.egress** contains all the configs (which pod's traffic at which port will be allowed)
- ♣ What is the use of **spec.policyTypes** then?

It enforces the ingress and egress rules.

- If *policyTypes* field is not there, Kubernetes will add the values (Ingress, Egress) based on presence of [*networkpolicy.spec.ingress*](#) and [*networkpolicy.spec.egress*](#) fields.

If only ingress is present, it becomes Ingress; if only egress is present, it becomes Egress; if both are present, it becomes both

Ingress and Egress.

In the live configuration (**kubectl get networkpolicy <name> -o yaml**), the *policyTypes* field will always be present, either defined by the user or added by Kubernetes.

- ✧ In the live config, whatever values are there [Ingress or Egress or both] inside *networkpolicy.spec.policyTypes*, only those will be enforced.

If a direction [Ingress or Egress] is listed in *policyTypes* but no corresponding rules are defined, then that direction becomes **completely blocked (default deny)**.

For example, if *policyTypes* contains Ingress but no ingress rules are defined, then all incoming traffic is blocked.

✧ namespaceSelector

- ✧ if provided, namespaces with those labels will be selected.
- ✧ If not provided, pods within the same namespace (not all namespaces) will be allowed.

Cluster Networking

✧ In a Kubernetes cluster, all the nodes must be having at least one network interface configured with

- ♣ Unique IP address
- ♣ Unique hostname
- ♣ Unique MAC address.

✧ **hostname** is system level, not specific to a network interace.

- ♣ But, in */etc/hosts* file, the IP and hostnames are written.

```
192.168.56.11 controlplane
192.168.56.21 node01
192.168.56.22 node02
```

[in *controlplane* node]

- ♣ In that controlplane node (where this */etc/hosts* file is present), if I specify **node02** then it'll take the linked IP only which is *192.168.56.22*
- ♣ In pure networking, if the node having IP *192.168.56.22* (which name is written as *node02* in */etc/hosts* of controlplane) is having some different hostname (let say **n2**) then also it should work.
Because anyhow the IP is being extracted from the controlplane node only while writing *node02*.

- ♣ But, In Kubernetes **it'll break the things**.

✧ Some ports needs to be accessible on the node.

[any port can be called from a port. But the **firewall** must let the incoming traffic in]

- ♣ **Control plane nodes**

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	6443	Kubernetes API server	All
TCP	Inbound	2379-2380	etcd server client API	kube-apiserver, etcd
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	10259	kube-scheduler	Self
TCP	Inbound	10257	kube-controller-manager	Self

Worker nodes

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	10256	kube-proxy	Self, Load balancers
TCP	Inbound	30000-32767	NodePort Services†	All
UDP	Inbound	30000-32767	NodePort Services†	All

- Why do the ports of *controller manager*, *scheduler* are opened because they are never called by others, instead they call api servers?

They exposes ports for health checks, monitoring, logging ..etc.

- For etcd, 2 ports are mentioned: 2379 & 2380

2379: client traffic (kube api server, etcdctl ..etc)

2380: peer communication (used by other etcd nodes)

Networking internals:

- Cluster requires to allocate non-overlapping IPs to nodes, services and pods.

Network plugin (CNI) allocate IP to Pods.

API Server allocate IP to Services.

Kubelet or cloud-controller-manager (CCM) allocate IP to Nodes.

Some important commands

- ip link**

displays network interfaces (etho, eth1 ..), MAC addresses, interface state (UP/DOWN)

```
vagrant@controlplane:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UP
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 02:aa:f8:fc:c7:1b brd ff:ff:ff:ff:ff:ff
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 08:00:27:c4:fe:66 brd ff:ff:ff:ff:ff:ff
4: calibf08b1dd620i52: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 02:00:0c:00:00:00 brd ff:ff:ff:ff:ff:ff
```

- ♣ **ip addr** [legacy one is: *ifconfig*]

displays IP addresses assigned to interfaces (etho 192.168.1.0)

```
vagrant@controlplane:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue sta
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdi
    link/ether 02:aa:f8:fc:c7:1b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope glo
        valid_lft 80720sec preferred_lft 80720sec
    inet6 fd17:625c:f037:2:aa:f8ff:fe7c:c71b/64 scope glo
        valid_lft 85921sec preferred_lft 13921sec
```

- ♣ **ip addr add 192.168.1.10/24 dev etho**

assigns IP address to an interface. [here etho --> 192.168.1.10]

- ♣ **ip route** [legacy one is: *route -n*]

shows routing table

ex: *default via 192.168.1.1*

```
vagrant@controlplane:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp s
10.0.2.0/24 dev enp0s3 proto kernel scope li
10.0.2.2 dev enp0s3 proto dhcp scope link sr
blackhole 10.244.49.64/26 proto 80
10.244.49.65 dev calibfd98b1dd62 scope link
10.244.49.66 dev cali93d2b2b0737 scope link
10.244.49.67 dev cali0240531d65d scope link
10.244.49.68 dev calia26e734911e scope link
10.244.49.69 dev cali0970f975c8a scope link
10.244.49.70 dev cali5fdb66d5490 scope link
10.244.49.71 dev calife997dd1838 scope link
10.244.49.72 dev calif1ce4f5f5f2 scope link
10.244.140.64 via 192.168.56.22 dev enp0s8 p
10.244.140.64/26 via 192.168.56.22 dev enp0s8
```

- ♣ **ip route add 192.168.1.0/24 via 192.168.2.1**

adds a route [here, to reach 192.168.1.x, go via gateway 192.168.2.1]

- ♣ **arp**

displays ARP table (IP -> MAC mapping)

- ♣ **route**

older version of *ip route* command.

- ♣ **netstat -plnt**

shows listening ports and processes.

[ex: which service is running at port 6443]

♣ **cat /proc/sys/net/ipv4/ip_forward**

checks if IP forwarding is enable.

0: disable; 1: enable

々 F

Pod Networking

々 [pre-requisite; Linux Namespace & bridge networking]

々 Consider the below scenario:

Node	Host IP	Pod CIDR	Bridge IP	Pods
Node-1	192.168.1.11	10.244.1.0/24	10.244.1.1	10.244.1.2, 10.244.1.3
Node-2	192.168.1.12	10.244.2.0/24	10.244.2.1	10.244.2.2
Node-3	192.168.1.13	10.244.3.0/24	10.244.3.1	10.244.3.2

々 To create these

- create a **bridge** on every node and assign it a IP (here 10.244.1.1, 10.244.2.1 ..etc) [**Bridge IP**]
- Create veth-pair (one per pod) and connect pod and bridge in the respective nodes.
- Assign IP to each pods in 10.244.1.x, 10.244.2.x ..etc. [**Pod IPs**]
So, Pod CIDR (Bridge's network) will be 10.244.1.0/24, 10.244.2.0/24, ..etc.

々 To make all pods communicate with each other across nodes.

- Configure default gateway inside each pods (Pod namespace)
10.244.x.1 [x: 1, 2, 3 .. etc for respective nodes]
[**bridge's IP**]
 - ☞ Means, if IP address is out of its subnet, then it'll go via *bridge*.
- Update **gateway** in all the **nodes** (not pods).
i.e.
network: 10.244.1.0/24, gateway: 192.168.1.11
network: 10.244.2.0/24, gateway: 192.168.1.12
network: 10.244.3.0/24, gateway: 192.168.1.13
 - ☞ Now, each node knows where to send the packet to reach the respective pods.
- Enable IP forwarding on each Node. Because for going out of the node,
--- just example ---
src IP: 10.244.1.2 (node-1 pod)
dest IP: 10.244.2.2 (node-2 pod)

⌘ For this, it should go from 10.244.1.1 (node-1 bridge) to 192.168.1.11 (node-1's LAN ip)

⌘ To forward packet from one network (10.244 ...) to another (192.168. ...) or *vice-versa*, **node-1 OS** acts as a router here.

So, IP forwarding needs to be enabled.

➤ NOW, **PACKET FORWARD CAN HAPPEN SMOOTHLY.**

Pod can reach its host's LAN port (default gateway is bridge, and IP forwarding inside host)

from host to another node's pod (pod's gateways are their respective nodes)

another node can send packet to its pod (private network inside it only and IP of bridge is configured)

that pod can now see the src address (host's pod) and send response back via its respective bridge -> node -> host -> bridge -> host's pod.

々 In the previous approach, we were writing the *gateways* in all the nodes which is not maintainable in large scale setup.

➤ Connect all the nodes to a router [IP: **192.168.1.1**]

➤ Add router's IP as the default gateway in all the nodes.

[so for every pod's IP, node will not able to find the gateway, so they'll forward packet to default gateway which is their router]

➤ Router will have all the route gateways (which we were writing in all the nodes)

Network	Gateway
10.244.1.0/24	192.168.1.11
10.244.2.0/24	192.168.1.12
10.244.3.0/24	192.168.1.13

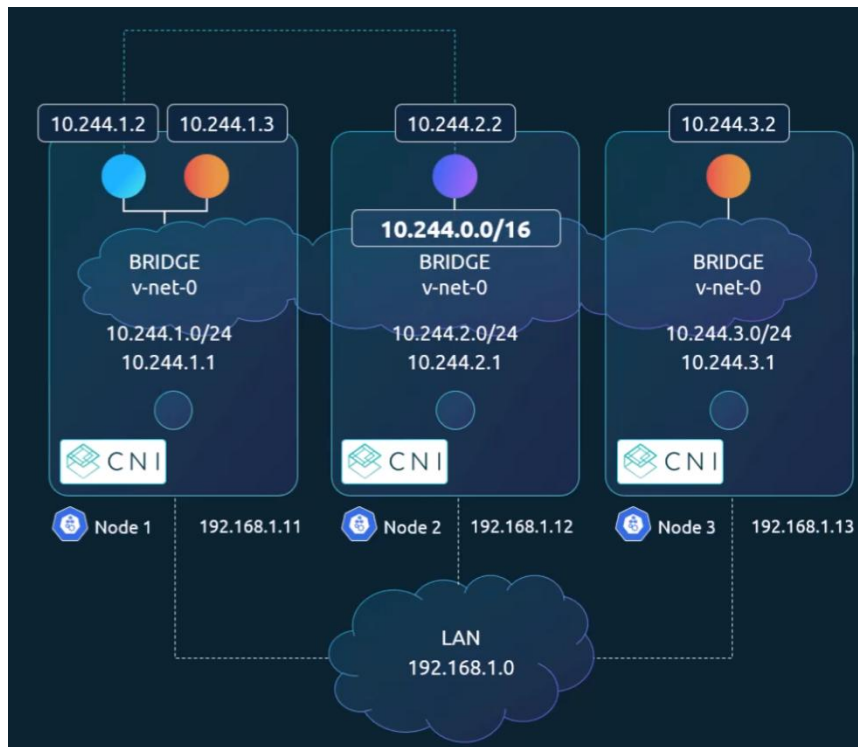
all at one place.

➤ Now, router will forward the packet to respective node; and that node to its respective pod.

➤ now, the individual pod namespaces will become a single entity throughout the cluster having network ip

10.244.0.0/16

[10.244.1.0/24 -- node-1's pod namespaces;
10.244.2.0/24 -- node-2's pod namespaces] ... etc



[individual namespace became cluster wide namespace now having
subnet **10.244.0.0/16**]

々 F

ConfigMap

々 Kubernetes object [apiVersion: "v1"] to keep env variables which can be used

- ♣ *Inside a container's command (cmd) and args.*
- ♣ *Environment variables for a container.*

々 Inside a container's command (cmd) and args

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-config

data:
  app_mod: production
```

♣ [ConfigMap]

here the variables are written under ConfigMap.data field (not inside spec unlike others)

- ♣ **app_mod: production** (this is the only key value pair written here which can be used)

```
apiVersion: v1
kind: Pod
metadata:
  name: demo-pod
spec:
  containers:
    - name: app
      image: busybox
      command: ["/bin/sh"]
      args: ["-c", "echo Running in $APP_MOD; sleep 3600"]
      env:
        - name: APP_MOD
          valueFrom:
            configMapKeyRef:
              name: app-config
              key: app_mod
```

♣ [Pod]

- ♣ Here pod.spec.containers[*].args is using **APP_MOD** (Upper case)
But, in ConfigMap it was **app_mod** (Lower case)

- ✧ In `pod.spec.env`
name: **APP_MOD**
and this is derived from `ConfigMap.app_mod`
- ✧ Inside Pod, **APP_MOD** (upper case) has value of `pod_map` (of `ConfigMap`)
- ✧ So, inside `pod.spec.containers[*].args` is using **APP_MOD** instead of `app_mod`.
- ✧ Commands and args are anyways combined:

```
command: ["/bin/sh", "-c", "echo Running in $(APP_MOD); sleep 3600"]
```

this is same as the previous one

々 Environment variables for a container

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: db-config
data:
  DB_HOST: mysql-service
  DB_PORT: "3306"
```

[ConfigMap]

```
apiVersion: v1
kind: Pod
metadata:
  name: env-pod
spec:
  containers:
    - name: app
      image: nginx
      env:
        - name: DATABASE_HOST
          valueFrom:
            configMapKeyRef:
              name: db-config
              key: DB_HOST
        - name: DATABASE_PORT
          valueFrom:
            configMapKeyRef:
              name: db-config
              key: DB_PORT
```

[Pod]

- ♣ If you execute the below inside the container

```
echo $DATABASE_HOST  
echo $DATABASE_PORT
```

then you'll get the values set inside ConfigMap.

- ♣ NOTE: the ConfigMap's keys are DB_HOST, DB_PORT but in the containers config (in pod yaml config) the names are DATABASE_HOST, DATABASE_PORT so this should be used.

♣

々 F

々 F

々 F

々

々 F

々 F

々

CNI in Kubernetes

々 Container runtime (containerd, cri-o) calls the cli plugins to assign ips and all to the pods.

♣ There are 2 main directories

/etc/cni/net.d/

/opt/cni/bin/

♣ /etc/cni/net.d/

✧ This contains CNI Network Configuration files.

✧ It tells kubelet/container runtime about which CNI plugin to use.

And how to configure pod networking.

✧ Files can be of extension **.conf**, **.json**, **.conflist**

✧ Ex: 10-calico.conflist, 10-flannel.conflist, 10-weave.conflist

```
root@controlplane:~# ls /etc/cni/net.d/
10-calico.conflist  calico-kubeconfig
```

✧ If multiple config files are there, it choose in alphabetical order.

✧ It just contains configurations about which plugin to use and with what configurations.

It doesn't contains any executables.

✧ *type* : which CNI plugins to execute (calico, flannel, ..etc)

ipam (ip add management) : how pod IP addresses will be allocated?

name : logical name of CNI network (just an identifier)

k8s api details : how CNI plugin communicate with k8s api server

plugins : (list) run multiple plugins one after another

mtu (maximum transmission unit) : maximum packet size allowed

port mapping settings : allow container ports to be exposed on host.

♣ /opt/cni/bin/

✧ This directly contains actual CNI plugin executables.

```
-rwxr-xr-x 1 root root 5698763 Apr 24 10:38 bridge
-rwxr-xr-x 1 root root 79605982 Apr 30 15:14 calico
-rwxr-xr-x 1 root root 79605982 Apr 30 15:14 calico-ipam
-rwxr-xr-x 1 root root 13725422 Apr 24 10:38 dhcp
-rwxr-xr-x 1 root root 5251069 Apr 24 10:38 dummy
-rwxr-xr-x 1 root root 5702145 Apr 24 10:38 firewall
-rwxr-xr-x 1 root root 2862657 Apr 30 15:14 flannel
-rwxr-xr-x 1 root root 5150067 Apr 24 10:38 host-device
```

⚡ These binaries can *create veth pairs, create bridges, assign pod IPs, configure routes, configure iptables, configure NAT, connect pod namespace, setup overlays, enforce policies.*

⚡ So basically

/etc/cni/net.d : instructions

/opt/cni/bin : actual workers

々

Ingress

々 Consider the below scenario:

- You have a deployment having multiple replicas.
To access them you created a NodePort service which is exposed at port 30080.
let the service name is: *wear-service*
- Now, you want the users to access it via a domain, but the port here is 30080 (NodePort's port must be ≥ 30000).
so, users can access it via
<http://my-online-store.com:30080>
- But, this is not proper. User should not enter the port directly.
But to avoid port, it needs to be a valid HTTP (80) or HTTPS(443) port.
[<http://....> will use 80 port, <https://....> will use 443 port]
- So, now you need to add a reverse proxy (load balancer) outside of the cluster whose IP will be added in the domain and the traffic can be routed properly with the url
<http://my-online-store.com>
- If you are deploying in a public cloud environment like AWS, Azure, GCP, then instead of creating a service of type *NodePort*, you can create a service of type *LoadBalancer*.
It'll create a NodePort and also a Load balancer in the cloud by itself.
- Now, you are deploying one more types of pods, and a service for those.
Let say the service name s: *watch-service*
and you want it to be accessed at
<http://my-online-store.com/watch> or <http://watch.my-online-store.com>
- Now this service will also be of type *LoadBalancer*, so the cloud platform will create another load balancer and NodePort service for this new deployment.
- So for each new deployment, one more load balancer will be created.
- Now all the load balancers will have separate IPs, and to direct the request to the respective load balancers, you again need to create a Load Balancer.

/wear : load-balancer-1

/watch : load-balancer-2

- Each time a new deployment comes, one new load balancer has to be created.

Then the main load balancer will have to be configured again.

And multiple load balancer will cost more.

々 How to fix this issue?

- You can ask the developers to handle these at application level. (/wear, /watch routes)
but this is not the right way. Because, it should not be done at application level as the features might be completely different.
- In this case you can use Ingress,

々 Ingress

- It's a *layer 7 load balancing solution* that provides a single end-point that routes the traffic to specific services depending upon the path or url user has provided.
- It can be configured like
my-online-store.com/wear, my-online-store.com/watch : it'll go to respective wear or watch services [path based]
wear.my-online-store.com, watch.my-online-store.com [host based]
- It also provides SSL security, load balancing along with HTTP routing.
- But still you need to expose the Ingress port with a NodePort or LoadBalancer service.
But this is just a one-time effort. Even if a new deployment comes, that will be handled at Ingress only without any need of changing the load balancer or node port service.

々 **Service : forwards request to a Pod**

Ingress : forwards request to a Service

々 Ingress in Kubernetes

- Kubernetes doesn't provide any Ingress controller by default.
You need to deploy by your own which will route the traffic to a specific service.
- Multiple options are there like *nginx-ingress, HAPROXY, traefik* ..etc.

- Then you need to maintain a config file which contains set of rules about the traffic routing.
Its called Ingress Resources.
- **Ingress Resources**: it defines the rule about what to do and how.
Ingress Controllers : it performs the actions according to those rules.
- nginx provide load balancing.
But in **nginx-ingress** (Ingress Controller): load balancing is just a part of it. Along with it, it has built-in additional intelligence to monitor kubernetes cluster for new definitions or ingress resources and configure the nginx server accordingly.
- *nginx-ingress* is deployed in a Deployment just like others in a Pod.

✧ Configuring Ingress Controller

- First create a deployment of *nginx-ingress* , keeping replicas=1 only.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-ingress-controller
spec:
  replicas: 1
  selector:
    matchLabels:
      name: nginx-ingress
  template:
    metadata:
      labels:
        name: nginx-ingress
    spec:
      containers:
        - name: nginx-ingress-controller
          image: quay.io/kubernetes-ingress-controller/nginx-ingress-controller:0.21.0
          args:
            - /nginx-ingress-controller
            - --configmap=$(POD_NAMESPACE)/nginx-configuration
          env:
            - name: POD_NAME
              valueFrom:
                fieldRef:
                  fieldPath: metadata.name
            - name: POD_NAMESPACE
              valueFrom:
                fieldRef:
                  fieldPath: metadata.namespace
      ports:
        - name: http
          containerPort: 80
        - name: https
          containerPort: 443
```

[image is provided by kubernetes itself]

image name is *nginx-ingress-controller*:<some version>

[it is not a normal *nginx* image, it is made specific for kubernetes ingress]

- ⌘ In the args, **--configmap=\$(POD_NAMESPACE)/nginx-configuration** and `POD_NAMESPACE` is a env variable which is equal to *metadata.namespace* means, it is the namespace where *Deployment* of *nginx-ingress* is present.
- ⌘ So it is just saying the nginx ingress controller to see the ConfigMap (Kubernetes resource) present inside the given namespace.
Default/nginx-configuration (if deployment is present in default namespace)

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: nginx-configuration
```

ConfigMap should look something like this. Name is ***nginx-configuration***

- ⌘ So its like
default/nginx-configuration

- ⌘ Create a Service of type NodePort to expose the deployment

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-ingress
spec:
  type: NodePort
  ports:
    - port: 80
      targetPort: 80
      protocol: TCP
      name: http
    - port: 443
      targetPort: 443
      protocol: TCP
      name: https
  selector:
    name: nginx-ingress
```

- ⚡ Ingress controller needs to have some specific permissions to do something in the cluster.

So, it needs to have a **ServiceAccount** with specific permissions.

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: nginx-ingress-serviceaccount
```

🔗 Creating Ingress Resource (kind: Ingress)

it specifies rules to route according to path or domain itself.

- ⚡ In case of a single service (wear for example in our case)

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: ingress-wear
spec:
  defaultBackend:
    service:
      name: wear-service
      port: 80
```

- ⚡ [ingress.spec.defaultBackend](#) means when no route rule is specified, this is the service where request is to be routed.

- ⚡ You can add as many rules as you want.

🔗 Adding multiple rules to Ingress resource

- ⚡ Rule-1, path based routing i.e.

my-online-store.com, my-online-store.com/wear, my-online-store.com/watch ..etc.

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: ingress-wear-watch
spec:
  rules:
    - http:
        paths:
          - path: /wear
            backend:
              service:
                name: wear-service
                port: 80
          - path: /watch
            backend:
              service:
                name: watch-service
                port: 80
```

- Rule-2, based on domain names

wear.my-online-store.com, www.my-online-store.com, watch.my-online-store.com ..etc.

```
kind: Ingress
metadata:
  name: ingress-host-based
spec:
  rules:
  - host: wear.my-online-store.com
    http:
      paths:
      - path: /
        backend:
          serviceName: wear-service
          servicePort: 80
  - host: watch.my-online-store.com
    http:
      paths:
      - path: /
        backend:
          serviceName: watch-service
          servicePort: 80
```

- The difference between the 2 is:

first one has only 1 rule, but multiple paths.

Second one has 2 rules, each having only one path that is / means the domain itself. no further paths.

々 NOTE:

- **Rules:** one rule per host (or per domain)

first one had only one rule because only one domain was there.

Multiple paths (.../wear, .../watch)

- **Paths:** multiple paths per rule (or per host)

second one had 2 rules because 2 domains were there (wear..., watch...)

々 How ingress class, ingress controller, ingress resources are linked?

- IngressClass contains the controller type.
nginx-ingress.

Let say 2 different controllers are present.

nginx-ingress and *traefik*

- Now you need to create 2 **IngressClass** (each one pointing to one controller).

```
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
  name: nginx
spec:
  controller: k8s.io/ingress-nginx
```

```
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
  name: traefik
spec:
  controller: traefik.io/ingress-controller
```

- Here, controller's values are hard-coded. Its not name of ingress controller's deployment or something else.

For **one type of controller**, one **hard-coded value** is there.

- Now, mention the IngressClass in [ingress.spec.ingressClassName](#)

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-app
spec:
  ingressClassName: nginx
```

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: traefik-app
spec:
  ingressClassName: traefik
```

- IngressClass will automatically selects the controller (deployed controller running in Pods) because both the controllers are of different types (nginx-ingress, traefik)
- But in case of same types of controllers, for example 2 *nginx-ingress* controllers are present then

- ⌘ IngressClass will be same only, because the [IngressClass.spec.controller](#) will contain hard-coded value only according to the type of controller.

[here only one type of controller is there *nginx-ingress*]

- ⌘ Create 2 IngressClass having different names, but same [IngressClass.spec.controller](#)

```
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
  name: public-nginx
spec:
  controller: k8s.io/ingress-nginx
```

```
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
  name: internal-nginx
spec:
  controller: k8s.io/ingress-nginx
```

[only [IngressClass.metadata.name](#) in both are different]

- ⌘ Create 2 Ingress (ingress resource) giving [Ingress.spec.ingressClassName](#) as the IngressClass's names.

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: internal-app
spec:
  ingressClassName: internal-nginx
rules:
```

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: public-app
spec:
  ingressClassName: public-nginx
rules:
```

- ⌘ Create 2 deployments of controllers (image of type *nginx-ingress* on both)

```
spec:
  containers:
    - name: nginx-ingress
      image: registry.k8s.io/ingress-nginx/controller:v1.12.0
      args:
        - /nginx-ingress-controller
        - --ingress-class=public-nginx
        - --controller-class=k8s.io/ingress-nginx
```

```
spec:
  containers:
    - name: nginx-ingress
      image: registry.k8s.io/ingress-nginx/controller:v1.12.0
      args:
        - /nginx-ingress-controller
        - --ingress-class=internal-nginx
        - --controller-class=k8s.io/ingress-nginx
```

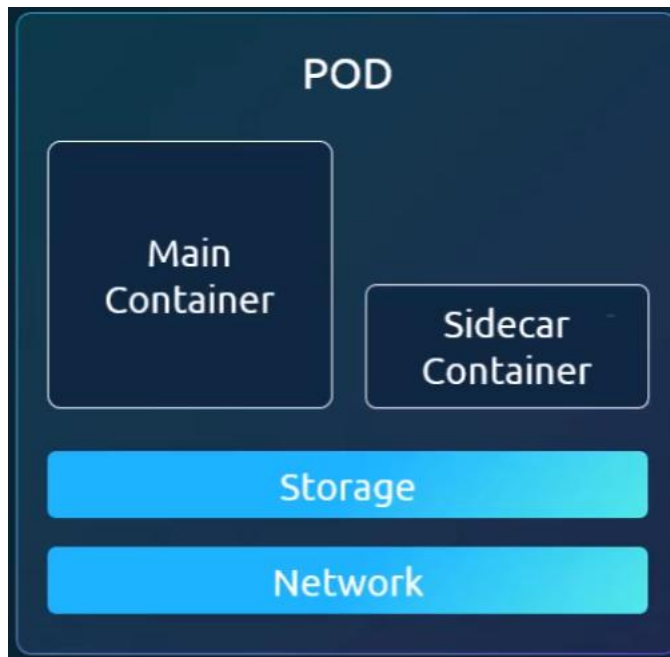
- ⌘ Here we need **--controller-class=<hard coded value of ingress controller>**
- ⌘ Also you can write these on *ConfigMap* as well just like the before.

⌘

々

Service Mesh

々 Sidecars



- One Pod can have multiple containers, one Main container and others sidecar containers.
- Sidecars are responsible for Log shipping, Monitoring, File loading ..etc so that Main container can only focus on Business logic.

```
containers:  
  - name: nginx-container  
    image: nginx  
    volumeMounts:  
      - name: shared-data  
        mountPath: /usr/share/nginx/html  
  - name: sidecar-container  
    image: fluent/fluentd  
    volumeMounts:  
      - name: shared-data  
        mountPath: /pod-data
```

々 Envoy

- It is a proxy.
 - In a application, along with Business Logic, other things will be there like *TLS*, *Auth*, *RETRY* .
- These should be maintained in a nother layer called **Proxy** so that Developers will focus on writing Business Logic.

- Proxy receives the request and forwards to the application.



- envoy** is a proxy or communication bus that runs as a Sidecar container.

Microservices

- When you convert a monolith to microservice, for all the services you need to handle the below things
authentication, authorization, networking, logging, monitoring, tracing.
- Networking part like rate limiting, circuit breakers, retry & timeouts are handled in application level.
- Proxy handles all this so that developers will focus on writing business logic only.
- Application (services) will call the other services, that cannot be changed because proxy doesn't know which service the request will be forwarded to.

Without Proxy:

service-1 --- **Service** of service-2(Kubernetes Service) --- **service-2**

with Proxy:

service-1 --- **Proxy** --- **Service** of service-2(Kubernetes Service) --- **service-2**

istio

- We don't add Sidecar container (*envoy*) by ourself, just install **istio**, it automatically inject the Sidecar to each pod.
- After running a deployment or pod, *istio* receives the PodSpec and update that so kubelet create the proxy container (sidecar) inside the pod.
- It consists of **dataplane** (which contains the *envoy* only) and **control plane** (runs on master nodes).

previously control plane was consisting of **3 daemons**: *Citadel* (certificate generation), *Pilot*(service discovery), *Galley*(validating configuration files)

- ♣ Currently, those 3 daemons combined to a single daemon called *istiod* (it's a Deployment)
- ♣ You need to *enable istio sidecar injection in the namespace* otherwise it'll not inject envoy proxy as sidecar on those namespace.

々 In Mesh architecture, role of control plane: manages all traffic into and out of services via proxies.

々 Istio agent delivers configuration secrets to Envoy proxies.

々 In mesh architecture, proxies communicate with each other through **data plane**.

々 Observability: logs, metrics, tracing

Dynamic configuration: updating routing/config without restarting

Manual TLS (mTLS): secure encrypted communication between services

Service discovery: finding services automatically.

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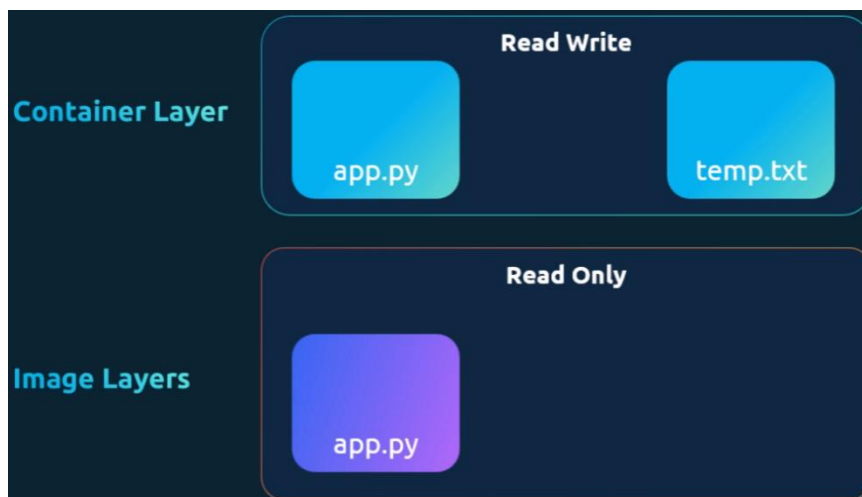
Storage

々 Docker image and container

- ✦ Docker image consists of **read-only** layers. (Image Layers)
layers are created based on the Dockerfile
- ✦ When a container is created using the Docker image, a new layer is created on top of those image layers.
This new layer is **writable**. (Container Layer)
- ✦ Container layer's life is same as the container.
When container is destroyed, this container layer will also be destroyed.
- ✦ Lets say in the image layer (in Dockerfile you do copy some files) a file was copied, so it'll be available inside all the containers created using that image.

That file will be **read-only** and that is being shared among all the containers.

You can edit that file, but when you'll save that, *Docker creates a copy of that file in Read Write layer (Container Layer) and whatever modification you'll do, that'll be there in the Container layer. Image layer will not be affected.*



々 Docker volume

- ✦ The data present inside the container will be destroyed once the container is destroyed.
To persist those data, you can create Docker volume and attach that to the container.

- Create a volume using the command
docker volume create <volume name>
docker volume create data_volume
- Add the volume while running a container
docker run -v <volume name>:<mount directory path> <docker image name>
docker run -v data_volume:/var/lib/mysql mysql
- Docker volumes are present inside the directory
/var/lib/docker/volumes
in linux host.
- Even if you don't create a volume, but mention in **docker run** command, then Docker will automatically create a volume and attach it.
- This is called **Volume Mounting**.

✧ Bind mount

- Mount a *directory* instead of a *docker volume*.
- **docker run -v <host directory path>:<container directory path> <docker image>**
both *volume* and *bind mount* has same command format.
If it contains `./` or `../` or `/` then it'll take this as a directory, otherwise as a volume.

✧ Using -v flag to mount is old, nowadays --mount is used.

- **docker run --mount type=<mount type>,source=<volume name or host directory>,target=<container mount directory> mysql**
docker run --mount
type=bind,source=/data/mysql,target=/var/lib/mysql mysql
for volume mount, *type* need not to be provided. Docker by default takes the *type* as *volume* if not provided.

✧ Storage Drivers

- It manages all the things like create a writable layer (container layer), copy files to that layer for modification, ..etc.
- Selection of storage drivers depends upon the underlying OS.
- Ex: Ubuntu uses AUFS by default.
- Docker chooses the best storage driver available based on the OS.

々 Volume Drivers

- ♣ *Volumes* are not handled by *Storage Drivers*.
- ♣ Default volume driver is **local**
- ♣ Other third party options are: Convoy, Flocker, NetApp, gcd-docker, RexRay ..etc
- ♣ You can mention the volume driver while creating a container
`docker run -it --name mysql --volume-driver resray/ebs --mount src=ebs-vol,target=/var/lib/mysql mysql`
it'll create a container and attach a AWS EBS volume to it.

々 In short,

layers and all managed by storage drivers.

Volume and mount on container is managed by volume drivers.

CSI (Container Storage Interface)

- 々 We saw **CRI** (Container Runtime Interface) [to support multiple container runtime, not only containerd], **CNI** (Container Network Interface) [any third party can create CNI plugin according to CNI and can be used in Kubernetes].

Similarly, **CSI** was developed to support multiple storage solutions.



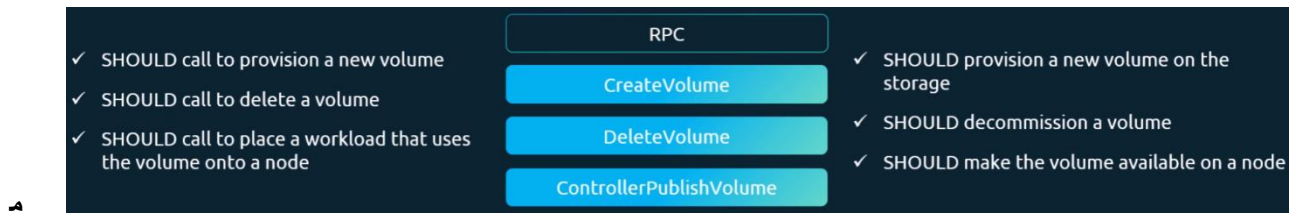
- ♣ CSI is ~~not~~ a Kubernetes specific standards, Its universal.
So that **any orchestration tool** can use any kind of Storage solutions.
- ♣ Kubernetes, Cloud Foundry, Mesos are on-board with CSI.
- 々 CSI Drivers are **volume drivers** (if you compare to docker).
These are different types of storage providers (~~not storage drivers~~) that can be attached to the containers running inside the Pods.
- 々 CSI defines a series of RPCs used by **container orchestrators** to interact with **storage providers**.

It basically uses **gRPC** (RPC developed by Google which is fast and optimized).

[RPC: Remote Procedure Call

One program calls a function/method running in another system/process over network or IPC, as if it were a local function.]

- ♣ Volume creation:
when pod is deployed and requires a storage solution, Kubernetes initiates “Create Volume” RPC call.
- ♣ Same for Volume deletion as well.



CS to Kubernetes is what MCP to Backend development frameworks.

Both provides a same type of request format, and delegate to the required providers.

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Volumes

々 Just like Docker, in kubernetes also we can attach volumes to the containers.

- ✦ In docker when you create a volume, it is automatically be stored in a fixed directory

`/var/lib/docker/volumes/`

- ✦ But in Kubernetes, you mention a Volume and provides the directory where it should be stored.

- ✦ `pod.spec.volumes`

```
volumes:
- name: data-volume
  hostPath:
    path: /data
    type: Directory
```

- ✦ Mount this container inside container at

`pod.spec.containers[*].volumeMounts`

```
spec:
  containers:
  - image: alpine
    name: alpine
    command: ["/bin/sh", "-c"]
    args: ["shuf -i 0-100 -n 1 >> /opt/number.out;"]
    volumeMounts:
    - mountPath: /opt
      name: data-volume
```

- ✦ Multiple types can be there

```
BlockDevice
CharDevice
Directory
DirectoryOrCreate
File
FileOrCreate
Socket
```

[`pod.spec.volumes.hostPath.type`]

- ✦ In this case one issue is there,

In case of multi-node cluster, each node will reference their own `/data` directory (according to the above yaml provided), which will lead to inconsistent data storage.

Kubernetes supports multiple options such as:

NFS, GlusterFS, Flocker, Fibre Channel, CephFS, ScaleIO

also public cloud storage options like:

AWS EBS, Azure Disk or File Storage, Google Persistent Disk.

- ♣ Example of AWS EBS

```
volumes:  
  - name: data-volume  
    awsElasticBlockStore:  
      volumeID: <volume-id>  
      fsType: ext4
```

- ♣ Along with hostPath, awsElasticBlockStore there are several types of volumes like

configMap, ephemeral, emptyDir ..etc.

[*emptyDir depends on Pod's lifecycle; ~~not container's lifecycle~~*]

- 々 One question might arise, AWS EBS is a disk (storage provider to a system), and container needs **volumes**, how will it work?

- ♣ EBS can be attached to a system and can be mounted at a directory.
Just like linux uses disks at a directory.

- ♣ Then that directory can be mounted to the container.

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PersistentVolume & PersistentVolumeClaim

々 In case of Volumes, we need to create the volume in each Pod and mounting that to the container.

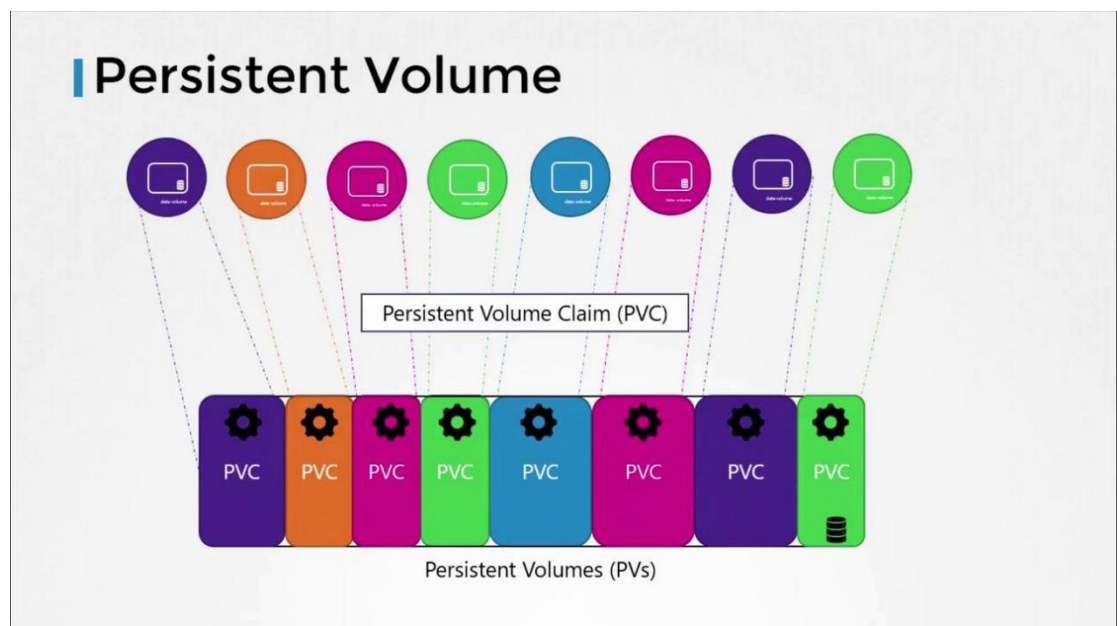
- ✦ It works for simple deployments but become cumbersome in multi-user environment.
- ✦ Each pod required its own storage configuration, making global update difficult and error-prone.

々 Persistent Volume

- ✦ The Cluster administrators create a pool of storage resources, which can be claimed by different users via **PersistentVolumeClaims** (PVCs).
- ✦ Administrator manages the storage centrally.

Global level changes becomes easier.

User claims storage without repetitive configuration.



々 **PersistentVolume (PV)** consists of the below parts:

- ✦ [pv.spec.accessModes](#)

✦ Determines how a volume can be mounted.

[ReadOnlyMany](#), [ReadWriteOnce](#), [ReadWriteMany](#),
[ReadWriteOncePod](#) (values)

[ReadOnlyMany](#), [ReadWriteOnce](#), [ReadWriteMany](#) : once means can be mounted by only one node (multiple pods present inside that nodes can access) and many means can be mounted by many nodes.

➤ `pv.spec.capacity`

◦ Allocated size of volume.

- Now what kind of storage you want to use for this, you can specify the same like `awsElasticBlockStore`, `hostPath` ..etc.

[should not use `hostPath` because it will depend upon the Nodes]

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: pv-vol1
spec:
  accessModes:
    - ReadWriteOnce
  capacity:
    storage: 1Gi
  hostPath:
    path: /tmp/data
```

- [~~`hostPath`~~ should not be used]

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: pv-vol1
spec:
  accessModes:
    - ReadWriteOnce
  capacity:
    storage: 1Gi
  awsElasticBlockStore:
    volumeID: <volume-id>
    fsType: ext4
```

- [it can be used to create volume on EBS]

々 **PersistentVolumeClaim (PVC)**

- There can be **one-to-one** mapping of *PersistentVolumeClaim* to *PersistentVolume*.

- Cluster administrators can create multiple PVs.

Users can create their PVCs.

- Kubernetes will find an ideal PV that matches the requirements of PVC. Requirements means *capacity*, *access modes*, *volume modes*, *storage class*, and *additional parameters*.

◦ If no matching PV is found, then the PVC keeps in Pending state.

- Additionally you can select the PV using their *labels* by writing inside *pvc.spec.selector* .
- Sometimes a small requirement PVC may be bound with a higher capacity PV if there is no other option.
But, even if there are resources left over in that PV, no other PVC can be bound with that because its already taken.

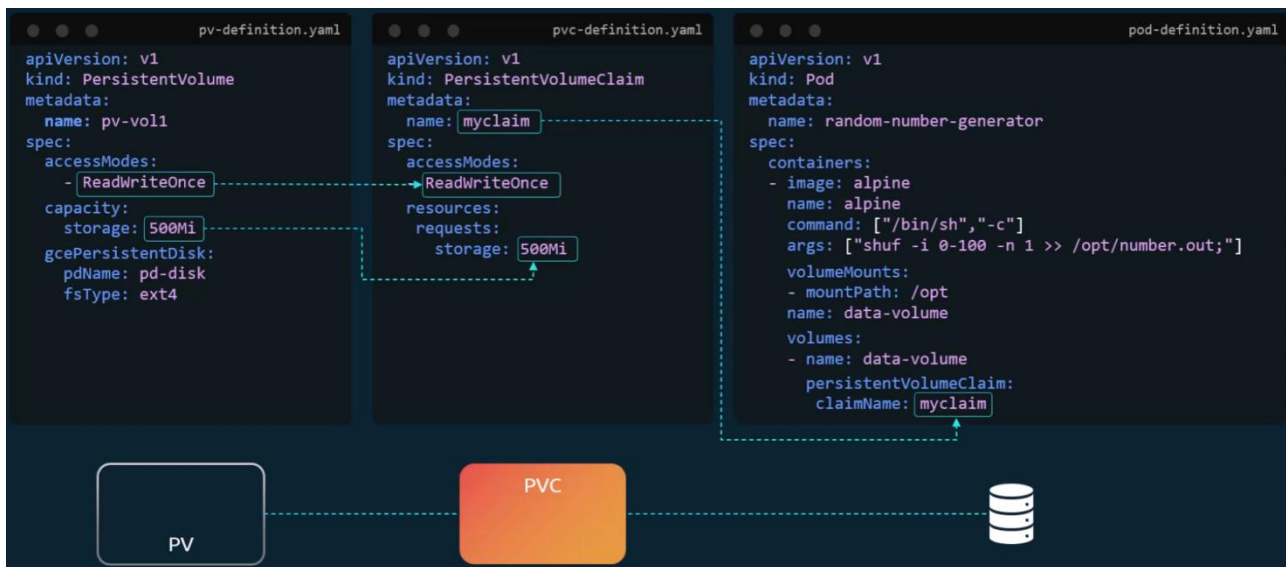
```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: myclaim
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 500Mi
```

- ↗ By default when the PVC is deleted, PV will not be deleted.
- *pvc.spec.persistentVolumeReclaimPolicy*
 - ↻ **Retain** : the volume will be left in its current phase (Released) for manual reclamation by the administrator.
but it can't be bound by other PVCs.
 - ↻ **Delete** : the volume will be deleted from Kubernetes on release from its claim.
 - ↻ **Recycle** : the volume will be recycled back into the pool of unbound persistent volumes on release from its claim.
It is deprecated because only doing *rm -rf ** will not renew the volume.

々 You can attach the PV to the Pod like the below

```
spec:
  containers:
    - image: alpine
      name: alpine
      command: ["/bin/sh", "-c"]
      args: ["shuf -i 0-100 -n 1 >> /opt/number.out;"]
      volumeMounts:
        - mountPath: /opt
          name: data-volume
  volumes:
    - name: data-volume
      persistentVolumeClaim:
        claimName: myclaim
```

- ⌘ `pod.spec.volumes` : it'll contain the volume name and the PVC it'll use.
- ⌘ `pod.spec.containers[*].volumeMounts` : use the volume mentioned in `pod.spec` level.

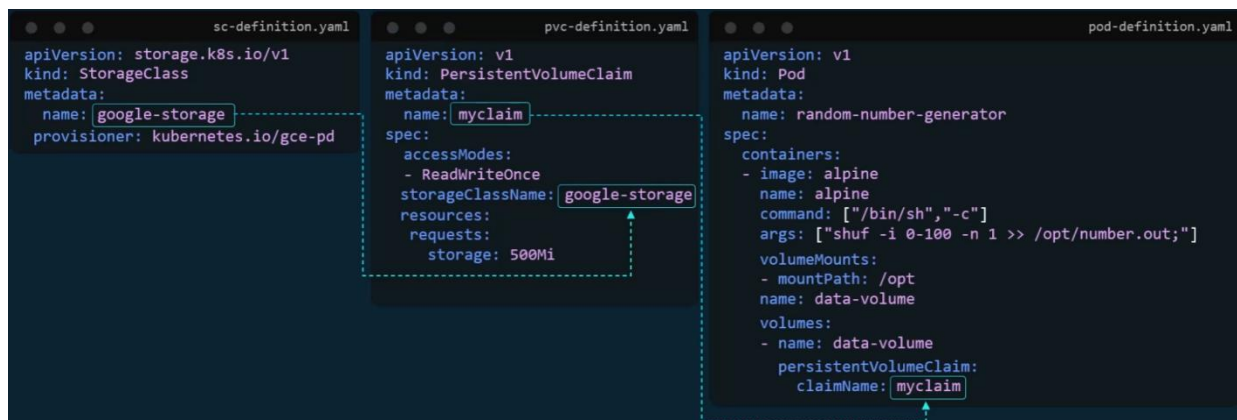


々

々

StorageClass

- ✧ In case of PV (PersistentVolume), the cluster administrator needs to manually create the PV then PVC can use those.
And later, Pods can use that PVC to use the PV.
- ✧ With StorageClass, you create a provisioner.
 - ♣ Mention that provisioner in the PVC and it'll automatically create the PV according to the requirement that can be directly used.
 - ♣ No need of cluster administrators to create the PV manually.
- ✧ When you create PV manually and using a cloud provider, you need to create the disk in the cloud provider.
 - ♣ For example, in case of EBS, you need to create a EBS in AWS Console.
 - ♣ Then when you create PV, it'll use that disk.
 - ♣ When you use StorageClass, disk creation on cloud and PV creation happens automatically.
- ✧ Steps



- ♣ You first create a **StorageClass** resource with a specific provisioner.
[provisioner are hard-coded values for different providers]

```
# sc-definition.yaml
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: google-storage
provisioner: kubernetes.io/gce-pd
```

[for gce]

- ♣ Then create a PVC and use this **StorageClass** in that [pvc.spec.storageClassName](#)

```
# pvc-definition.yaml
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: myclaim
spec:
  accessModes:
    - ReadWriteOnce
  storageClassName: google-storage
  resources:
    requests:
      storage: 500Mi
```

- Then use this PVC in the Pod same as before.

```
spec:
  containers:
    - image: alpine
      name: alpine
      command: ["/bin/sh", "-c"]
      args: ["shuf -i 0-100 -n"]
      volumeMounts:
        - mountPath: /opt
          name: data-volume
  volumes:
    - name: data-volume
      persistentVolumeClaim:
        claimName: myclaim
```

- StorageClass also creates PV after creating the disk on the cloud. It is all automatic.
- Other provisioners are like *AWSElasticBlockStore*, *AzureFile*, *AzureDisk*, *CephFS*, *Cinder* ..etc.
- Along with [storageclass.provisioner](#) other parameters can also be passed inside [storageclass.parameters](#) field.
 - it contains map<string, string> which is **completely provider specific**.
- Just like PV, here also you will get a reclaim policy field which is [storageclass.reclaimPolicy](#)

Autoscaling

- ♪ 3 types of autoscaling are there in kubernetes
 - ♣ **Horizontal Pod Autoscaler**
Automatically scales the number of pods in a deployment based on metrics such as CPU or memory usage.
 - ♣ **Vertical Pod Autoscaler**
Adjusts resource **limits and requests for pods** based on their actual consumption.
 - ♣ **Cluster Autoscaler**
Manages the overall cluster by adding or removing nodes in response to workload requirements.
- ♪ The autoscaling process must be **automated**, continuously monitoring workloads and provisioning resources without manual intervention.
Autoscaling should be bidirectional (scaling up and scaling down should happen) & it should happen automatically when the load is heavy scale up and otherwise scale down to save cost.
- ♪ 3 key factors:
 - ♣ Designed for scale
 - ♣ Automatic
 - ♣ Bidirectional
- ♪ **Horizontal Pod Autoscaler** (kind: *HorizontalPodAutoscaler*)
 - ♣ HPA monitors the application's resource utilization from the metrics collected from metrics server.
 - ♣ HPA Configuration steps:
 - Create a deployment
 - Define resources inside that Deployment (cpu, memory ..etc)
 - Create HPA pointing to that deployment.

Deployment

```
spec:
  containers:
  - name: myapp
    image: myapp:latest
    resources:
      limits:
        cpu: 500m
      requests:
        cpu: 200m
```

[[deployment.spec.template.spec.containers\[*\].resources](#)]

HPA

```
# hpa.yaml
apiVersion: autoscaling/v2beta2
kind: HorizontalPodAutoscaler
metadata:
  name: myapp-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: myapp
  minReplicas: 1
  maxReplicas: 10
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 50
```

[hpa.spec.*scaleTargetRef* will contain the target resource type; in our case it is Deployment]

This **averageUtilization** is calculated as per the below formula

$$desiredReplicas = ceil \left| currentReplicas \times \frac{currentMetricValue}{desiredMetricValue} \right|$$

- ⌘ Lets say current metric for cpu is: *200m*, and desired is *100m*.
and currently 2 pods are running.
- ⌘ It means the pods are overloaded (because instead of 100m, they are consuming 200m each).
So we need to increase the pod count so that the **load per pod will be reduced**.
- ⌘ $2 * (200/100) = 4$
so, 4 pods will run so that each will be having *100* cpu which is desired one.
- ⌘ [hpa.spec.maxReplicas](#), [hpa.spec.minReplicas](#) are the extreme pod counts.
Pod count can't be less than minReplicas and ~~can't be more than~~ maxReplicas.

々 **Vertical Pod Autoscaler** (kind: VerticalPodAutoscaler)

- ⌘ When you assign a *resource* and *limit* inside a Pod, kubernetes will enforce that much only.
i.e. if that much resource is present in the Node, then only the pod will be assigned to the node.
After the pod running, it can use resources within the *limit* configured inside the Pod.
- ⌘ But lets say one pod is getting a lot of request and it needs to use more resources otherwise it'll crash.
Here VPA comes in.
- ⌘ There are 3 components in VPA
 - ⌘ **VPA Recommender**
 - ⌘ **VPA Updater**
 - ⌘ **VPA Admission Controller**
- ⌘ When the pod is pushing its limits beyond the limit, **VPA Recommender**, watching the metrics repeatedly from *metrics server*, such as 190Mi out of 192Mi memory and 700m out of 750m CPU (means almost close to the limit), give suggestion to increase the resource request and limit.
Increasing limit makes sense, but why request?

- ⌘ Because pod needs to be evicted (currently it might have updated in new version that resources can be updated in running pod) to update the resources.
- ⌘ So, it might be scheduled to any other node, so request needs to be updated.
- ⌘ Then **VPA Updater** compares the live Pod's config and the recommended resources requirements.

[Pod's resources cannot be changed while its running. It needs to be evicted (removed) and re-built again]

In case of mismatch, it deletes the existing pod.

Now **Deployment** will create a new pod because one pod is gone.

- ⌘ **VPA Admission Controller** now updates the resources of the newly created Pod *before getting stored in etcd* according to VPA Recommender's suggestions.

- ⌘ There are some policies regarding the eviction of the pod **hpa.spec.updatePolicy** field has to be updated.

- ⌘ **hpa.spec.updatePolicy.updateMode: off**

VPA provides recommendations, but doesn't automatically evicts the pod.

Manual intervention is required.

- ⌘ **hpa.spec.updatePolicy.updateMode: Initial**

Apply recommendations only when pod is first created.

Don't disturb running pods.

- ⌘ **hpa.spec.updatePolicy.updateMode: Auto**

Continuously keep pod updated with recommended resources.

- ⌘ NTOE: HPA is not there in Kubernetes by default. It needs to be installed.

```
vagrant@controlplane:~$ kubectl explain vpa.spec --recursive
the server doesn't have a resource type "vpa"
```

- ⌘ **git clone <https://github.com/kubernetes/autoscaler.git>**
- ⌘ **cd autoscaler/vertical-pod-autoscaler/**
- ⌘ **./hack/vpa-up.sh**
- ⌘ Now it'll work.

```

apiVersion: "autoscaling.k8s.io/v1"
kind: VerticalPodAutoscaler
metadata:
  name: webapp-vpa
spec:
  targetRef:
    apiVersion: "v1"
    kind: Pod
    name: simple-webapp-color
  updatePolicy:
    updateMode: "Off"

```

- ☞ Just like HPA, VPA also has [targetRef](#) to select the resources (Deployment, Pod, ReplicaSet .etc).

🍴 In simple terms

HorizontalPodAutoscaler : keeps the request & limit same, increase pods.

VerticalPodAutoscaler : increase/decrease request & limit of the pod.

🍴 **Cluster Autoscaler**

- ☞ It doesn't create new cluster, rather add new nodes if the existing nodes are not having available resources.
- ☞ It'll scale up or scale down the nodes inside a cluster depending upon the resource utilization.
- ☞ Ex: in Google Cloud
`gcloud container clusters create my-cluster --cluster-autoscaler=min-nodes=3,max-nodes=10`
 - ☞ Define *min* and *max* number of nodes and it'll do the work.

Serverless

- 々 In traditional computing, companies buy and manage and scales the servers by themselves.

In Serverless computing, cloud providers manage the things like *provisioning, maintenance, scaling*.

- ♣ Only the server infrastructure is handled by cloud provider here, But some essential tasks like *resizing image, send email, ...etc* needs to be handled by you.

々 Function as a Service (FaaS)

- ♣ Its kind of role specific hired people who will do a single specific task. Like, one will send email, one will resize image ..etc.

- ♣ It optimizes cost because it has pay-as-you-go model.

- ♣ Example:

consider a small online t-shirt store that needs to send a confirmation email after a purchase.

Instead of managing an email server or developing complex dispatch code, you could simply implement a function using a service like AWS Lambda.

This function would be triggered automatically after a customer makes a purchase, sending the confirmation email via services such as SendGrid or Amazon SES.

The FaaS platform scales the function based on incoming request volume.

- ♣ FaaS offers event-driven execution of code on a pay-as-you-go basis

- ♣ Cloud platforms providing FaaS

- ✧ AWS Lambda and AWS Fargate (Amazon Web Services)
- ✧ Azure Functions (Microsoft Azure)
- ✧ Google Cloud Functions (Google Cloud)
- ✧ IBM Cloud Functions

々 **Kubernetes Serverless**

- ♣ Kubernetes Serverless means that the running code that is not in use should automatically scales to zero so that you don't pay for the idle resources.

- ♣ Knative/OpenFaaS → Like hiring workers only when work arrives, and firing them when done.
Saves money, perfect for short, bursty tasks (e.g., image resize, webhook handler).
- ♣ VPA and HPA cannot scale down to zero when it is not required.
But Knative/OpenFaaS can do.
And also VPA and HPA are slower to react.
- ♣ It can listen to Kafka events (for example) and then
 - ✧ Start your app only when a message arrives in a Kafka topic.
 - ✧ Scale up the number of app instances based on how many messages are waiting.
 - ✧ Scale down to zero when no messages are coming in.

々

- 々 Sigs repo name: community
 keps repo name: enhancement
- 々 **Kep-processes** are present inside **enhancement/keps/sig-architecture/0000-kep-process/README.md**
- 々 Workload & Non-workload objects
 - ▲ Workload objects: that runs containers
 - Pod
 - Deployment
 - ReplicaSet
 - StatefulSet
 - DaemonSet
 - Job
 - CronJob
 - ▲ Non-workload objects: that doesn't run containers
 - Service
 - ConfigMap
 - Secret
 - Ingress
 - PVC
 - PV
 - Namespace
 - ServiceAccount
- 々 F

Observability

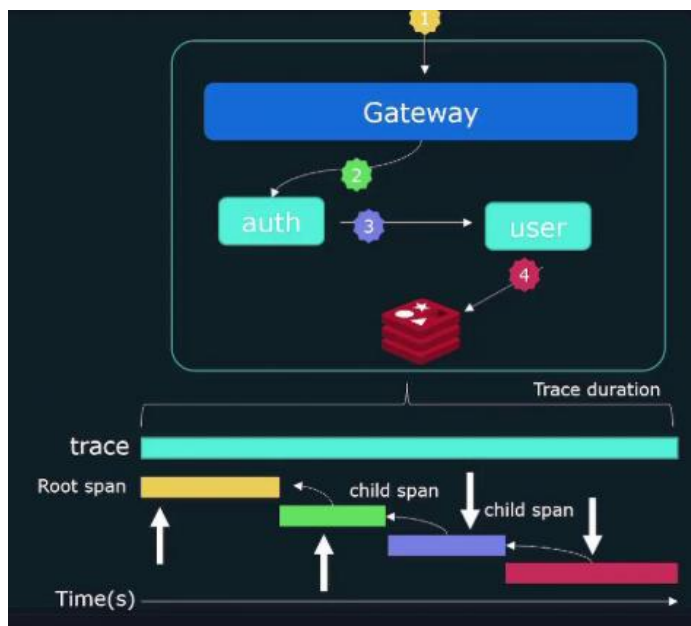
々 3 pillars of Observability

♣ **Logging**

ex: OS logs, Application logs, Database logs

- Its composed of **timestamp**, **message**
- While logs provide valuable contextual data, their high verbosity and the intertwining of processes across multiple systems can complicate the process of pinpointing issues during an outage.

♣ **Traces**



- It allow you to follow operations as they traverse through various systems & services.
- Each **trace** will be having a unique id called **trace-id**.
Individual events forming a trace are called **spans**.
- Each **span** (not trace) tracks the following
 - start time (when the span started)
 - duration (how long that span lasted)
 - parent-id (what was responsible for triggering this specific span)

♣ **Metrics**

- It provides state of a system using numerical values.
- It contains info like *CPU Load, Number of opened files, HTTP response time, Number of errors ..etc.*

- The fetched data can be visualized as dashboards or graph using tools.
- It consists of 4 things
metric name, value (most recent or current value of metric),
timestamp, dimensions (additional information about the metric).

node_filesystem_avail_bytes{fstype="vfat", mountpoint="/home"}	5000	4:30AM 12/1/22
Metric name	Value	Timestamp

々 Prometheus

- It is responsible for collecting and aggregating *metrics* (not logs or traces, only metrics)

々 SLI/SLO/SLA (Service Level Indicators/Objectives/Agreements)

- **SLI** reflects current performance of a service such as *availability, latency, or error rate*.
“How are we doing right now?”
Ex: “99.95% of requests succeeded in the last hour.”
- **SLO** is the aim to achieve SLI.
“what are we aiming for?”
Ex: “Achieve 99.9% availability over 30 days”
- **SLA** is the agreement between customer & provider.
“We promise this much. We’ll do this if we fails”
Ex: “We guarantee 99.9% uptime; if not met, you get a 10% refund.”

々 F

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々 F

々 F

々 F

々

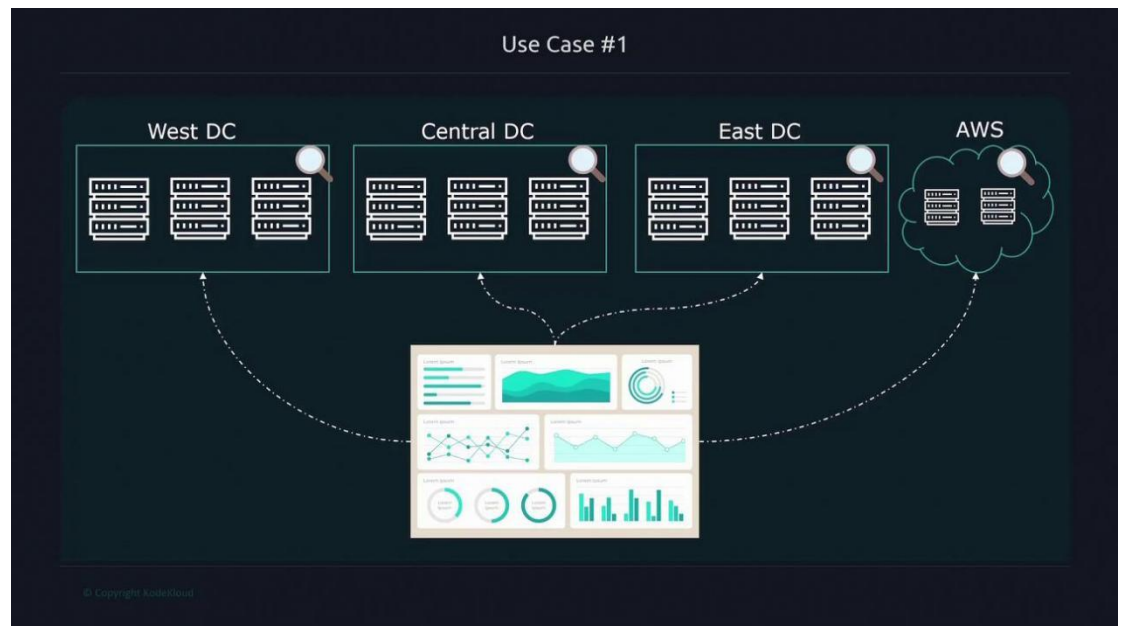
Prometheus

々 Use-cases

➤ Unified Metrics Aggregation

Prometheus is designed to scrap *metrics* from various location, whether on-premises or cloud-based, and provides built-in dashboarding utilities to display the data in a consolidated view.

It doesn't depend upon location, it scrap irrespective of the locations.



➤ Proactive Alerting for Server Health

for example, a MySQL database server that experiences high memory usage, might lead to memory outage.

Operation team needs to be *alerted* before the memory usage reaches a critical level (eg 80%).

- Prometheus has built-in mechanism to alert when a metrics exceeds the threshold.
- It can triggers notification via email, Slack, SMS ..etc.

➤ Performance Analysis for new Features

for example, video upload feature was added and when a large video is uploaded, its leading to *high latency*.

- Prometheus helps teams to correlate these values and find out the threshold at which performance issue occurred.

- ♣ In simple terms:

Unified Metrics Aggregation: trace metrics irrespective of the location or type (on-premises or cloud)

Proactive Alerting for Server Health: alert the team if anything reaches a critical level

Performance Analysis for new Features: what caused the performance issue when a new feature was introduced.

々 Prometheus Basics

- ♣ It scrapes metrics from the targets via HTTP endpoints.
- ♣ It monitors broad range of metrics like
 - ✧ *system metrics*: CPU utilization, memory usage, Disk space usage, service uptime
 - ✧ *Application-specific metrics*: exception counts, latency measurements, number of *pending* request
- ♣ It not only collect metrics from OS and Application, but also *networking devices, database, & other critical IT infrastructure components*.
- ♣ ~~It is not designed to monitor events, logs or traces.~~

Prometheus Architecture

✧ Inside the main **Prometheus Server**, mainly 3 components are there



➤ Retrieval

- ✧ This sends HTTP request to the targets and collect the metrics.
It gets the targets from Kubernetes SD, EC2 SD, *static configs*, ..etc.
- ✧ It continuously **pulls** the metrics from the targets in each intervals.
- ✧ It scraps those metrics and forwards to TSDB.

➤ TSDB (Time Series Database)

- ✧ It stores all scraped metrics received from *Retrieval*.
- ✧ Every metric is stored as *metric name, labels (or dimensions), timestamp, value*

```
node_filesystem_avail_bytes{fstype="vfat", mountpoint="/home"} 5000 4:30AM 12/1/22
```

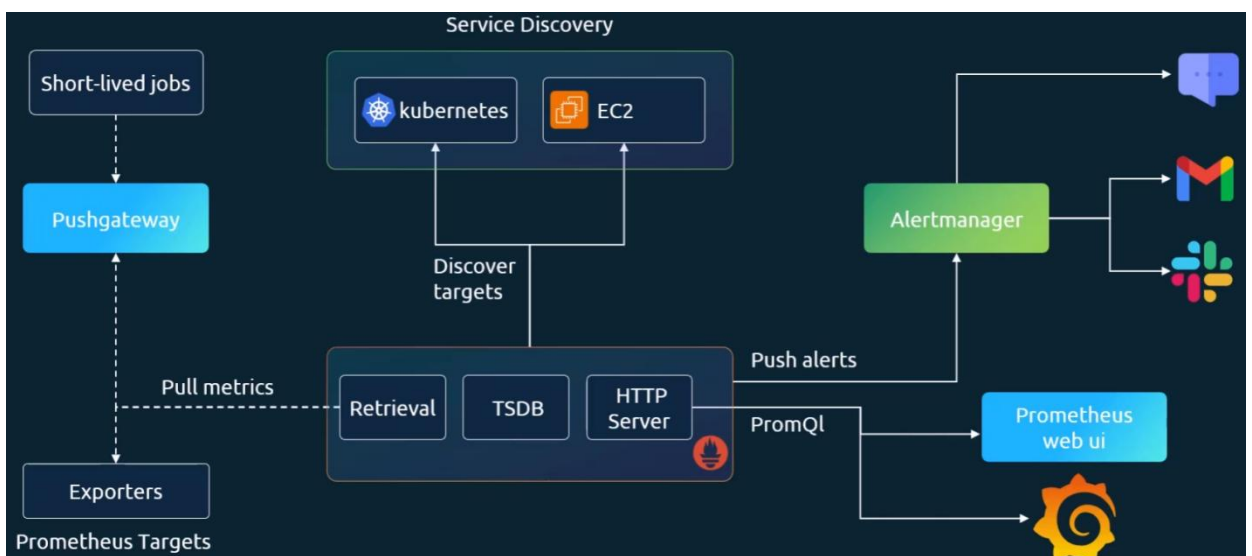
✧

Metric name	Dimensions	Value	Timestamp
node_filesystem_avail_bytes{fstype="vfat", mountpoint="/home"}		5000	4:30AM 12/1/22

➤ HTTP Server

- ✧ This component exposes UI and APIs.

✧ Overall Architecture



⚡ Exporters

- These collect *metrics* from the systems, convert them to prometheus format, and expose them to **/metrics** (can be changed) endpoint which can be collected by Retrieval (Prometheus server's component)
- It can be of multiple type like *Node Exporter*, *MySQL Exporter*, ..etc. [these systems have metrics internally, but not in prometheus format and not exposed by default; Exporter collect, convert to prometheus format, and expose those.]
- For example, Node Exporter reads the data from multiple files, like */proc/meminfo* which look like this

```
vagrant@controlplane:~$ cat /proc/meminfo
MemTotal:      2010780 kB
MemFree:       128888 kB
MemAvailable:  466464 kB
Buffers:       29360 kB
Cached:        426004 kB
```

- The target doesn't push the data, instead **Prometheus pulls the data**.

⚡ Pushgateway (for Short-lived jobs)

- For short lived jobs, pull will not be proper as it might miss the metrics. So in this case use a component called **Pushgateway**.
- *Short lived Job push the data to Pushgateway.*
Prometheus pulls the data from that Pushgateway like any other target.

⚡ Service Discovery

- Prometheus expects us to hard-code the targets in [prometheus.yml](#)

```
scrape_configs:
  - job_name: node-exporter

  static_configs:
    - targets:
      - 10.0.0.5:9100
      - 10.0.0.6:9100
```

- *Service Discovery* helps in this.
Instead of hard-coding the IPs, prometheus dynamically discovers targets.

- ☞ For example

```
api-1 → 10.244.1.5
api-2 → 10.244.1.6
api-3 → 10.244.1.7
```

now, api-2 crashes.

```
api-4 → 10.244.2.9 (new pod created)
```

```
remove 10.244.1.6
add 10.244.2.9
```

now prometheus learns:

⚡ AlertManager

- ☞ Receives alerts from Prometheus, process them and send notifications.
- ☞ Example: Slack, Email, Teams, Webhook ..etc.

🌀 Installing Node Exporter

⚡ wget <node exporter link>

wget

https://github.com/prometheus/node_exporter/releases/download/v1.11.1/node_exporter-1.11.1.linux-amd64.tar.gz

⚡ Un-tar this

tar -xzvf <tar file name>

`tar -xzvf node_exporter-1.11.1.linux-amd64.tar.gz`

⚡ run the file **node_exporter**

⚡ Now it'll expose the metrics at **/metrics** end-point on port **9100** .

```
# TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 1
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0
```

[**curl http://<IP>:9100/metrics**]

🌀 Installing Prometheus (main component consisting of *Retrieval, TSDB, HTTP Server*)

[Node Exporter also exposes end-point but to expose the metrics.

Prometheus's HTTP Server expose end-point to display all the metrics in dashboard and in other formats along with *api* end-points]

⚡ wget <prometheus link>

wget

<https://github.com/prometheus/prometheus/releases/download/v3.11.3/prometheus-3.11.3.linux-amd64.tar.gz>

- ♣ Un-tar thile

tar -xzvf <tar file name>

`tar -xzvf prometheus-3.11.3.linux-amd64.tar.gz`

- ♣ Inside that directory **prometheus.yml** will be there which is the config file.

々 prometheus.yml

- ♣ global



```
global:
  scrape_interval: 15s # Se
  evaluation_interval: 15s
  # scrape_timeout is set #

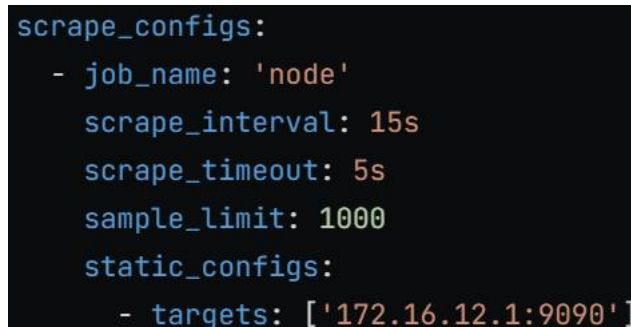
global:
  scrape_interval: 1m
  scrape_timeout: 10s
```

scrape_interval : pull metrics in each interval

evaluation_interval : applies rules (like cpu_usage > 80)

scrape_timeout : if target doesn't respond within this, scrape fails

- ♣ scrap_config



```
scrape_configs:
  - job_name: 'node'
    scrape_interval: 15s
    scrape_timeout: 5s
    sample_limit: 1000
    static_configs:
      - targets: ['172.16.12.1:9090']
```

[if the fields (scrape_interval, scrape_timeout, evaluation_interval ..etc) are not present, it'll take from global]

- ♣ sample_limit : maximum number of samples **per target**

[sample: promhttp metric handler requests_total{code="503"} 0
like this]

- ♣ static_configs : IP and Ports of all the targets from which metrics has to scraped.

- ♣ By default it'll check from **/metrics** path in each target, for any custom path add in the field metrics_path field.

Also, by default it'll use **http**, if you want **https** then write against the

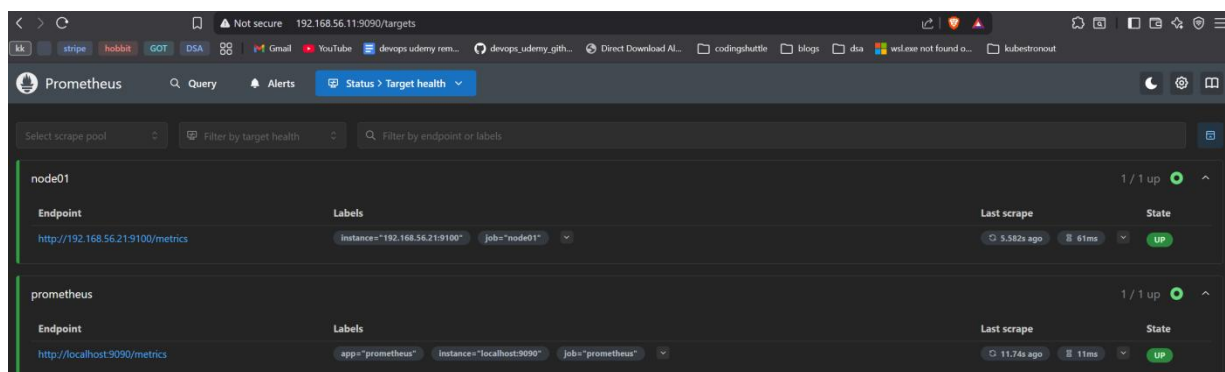
scheme field.

```
scrape_configs:
  - job_name: 'nodes'
    scrape_interval: 30s
    scrape_timeout: 3s
    scheme: https
    metrics_path: /stats/metrics
    static_configs:
      - targets: ['10.231.1.2:9090', '192.168.43.9:9090']
```

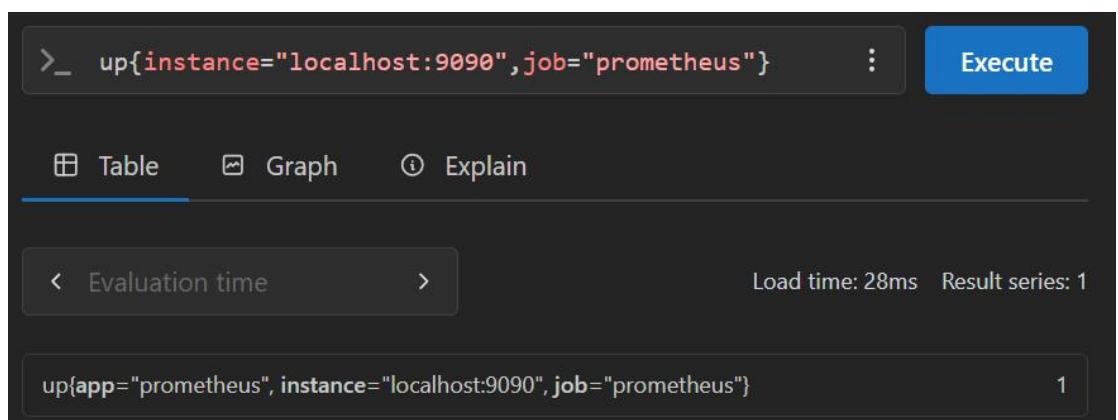
- You can also give authorization

```
basic_auth:
  [ username: <string> ]
  [ password: <secret> ]
  [ password_file: <string> ]
```

- When you run **prometheus** file, the UI and api will be exposed at **9090** port.



- Also you can execute the queries like this to see the status
[1 means active]



Prometheus Metrics

♪ You can explore clicking on the 3 dots



♪ A lot of metrics are there

Explore metrics

Metric	Type	Help
go_gc_cleanups_executed_cleanups_total	counter	Approximate total count of cleanup functions (created by runtime) approximate cleanup queue length. Useful for detecting slow cle
go_gc_cleanups_queued_cleanups_total	counter	Approximate total count of cleanup functions (created by runtime) /gc/cleanups/executed:cleanups to approximate cleanup queue /gc/cleanups/queued:cleanups.
go_gc_cycles_automatic_gc_cycles_total	counter	Count of completed GC cycles generated by the Go runtime. Sou
go_gc_cycles_forced_gc_cycles_total	counter	Count of completed GC cycles forced by the application. Sourced
go_gc_cycles_total_gc_cycles_total	counter	Count of all completed GC cycles. Sourced from /gc/cycles/total:

♪ Metrics structure

```
node_cpu_seconds_total{cpu="0",mode="idle"} 258277.86
```

node_cpu_seconds_total: **metric name**

cpu, mode : **labels** (or **dimensions**)

258277.86 : **value**

- ✦ It means “CPU core 0 has spent 258277.86 seconds in idle mode”.
- ✦ In case of multiple CPUs, it'll be having multiple entries like the below

```
node_cpu_seconds_total{cpu="0",mode="idle"} 258244.86
node_cpu_seconds_total{cpu="1",mode="idle"} 427262.54
node_cpu_seconds_total{cpu="2",mode="idle"} 283288.12
node_cpu_seconds_total{cpu="3",mode="idle"} 258202.33
```

- Each time along with the metrics, Prometheus fetches the timestamps as well to record all the things properly.
- Labels gives extra info about the metric just like the previous example. Metric name was same, but it gave info about multiple CPUs.
- Each entry will be having different label combinations, for example of different modes

```
node_cpu_seconds_total{cpu="0",mode="idle"} 258277.86
node_cpu_seconds_total{cpu="0",mode="iowait"} 61.16
node_cpu_seconds_total{cpu="0",mode="irq"} 0
node_cpu_seconds_total{cpu="0",mode="nice"} 61.12
node_cpu_seconds_total{cpu="0",mode="softirq"} 6.63
node_cpu_seconds_total{cpu="0",mode="steal"} 0
node_cpu_seconds_total{cpu="0",mode="system"} 372.46
node_cpu_seconds_total{cpu="0",mode="user"} 3270.86
```

- top** command displays all these details about CPU

```
vagrant@node02:~$ top
top - 21:08:05 up 50 min,  1 user,  load average: 0.15, 0.45, 0.47
Tasks: 151 total,   1 running, 150 sleeping,   0 stopped,   0 zombie
%Cpu(s):  2.2 us,  3.3 sy,   0.0 ni, 92.6 id,   0.0 wa,   0.0 hi,  1.8 si,   0.0 st
MiB Mem :   957.4 total,   63.4 free,   422.0 used,   472.0 buff/cache
MiB Swap:    0.0 total,    0.0 free,    0.0 used.  383.6 avail Mem
```

Metric Attributes

```
# HELP promhttp_metric_handler_requests_total Total number of scrapes by HTTP status code.
# TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 544
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0
```

- HELP** : provide a normal description of what the metrics measures.
- TYPE** : specifies the metric type (here:
promhttp_metric_handler_requests_total)

Metric Types

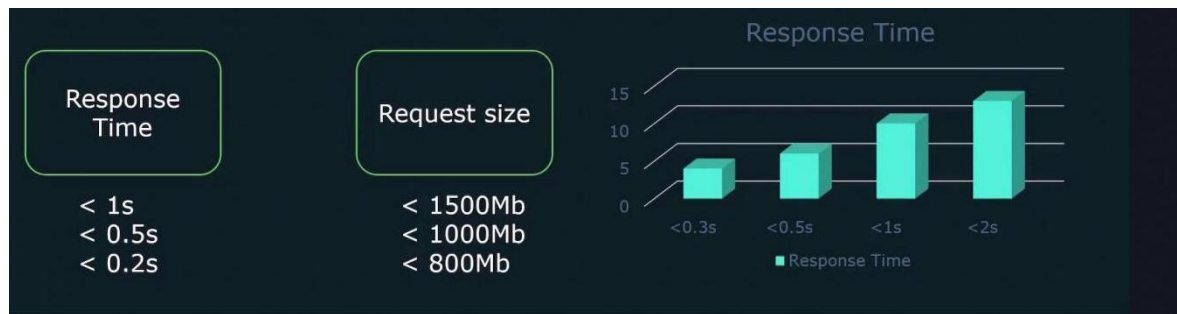
- In the previous image (metric lists), there is a Type column.
- There are the below types
Counter, Gauge, Histogram, Summary
- Counter** : how many times did something happened. It only **increases**.
[ex: total requests, error counts, job executions ..etc]
- Gauge** : what is the current value of something. it can **increase or decrease**. [ex: cpu utilization, memory usage ..etc]

- **Histogram** : How long or how big something is. Group observations into configurable **bucket sizes**.

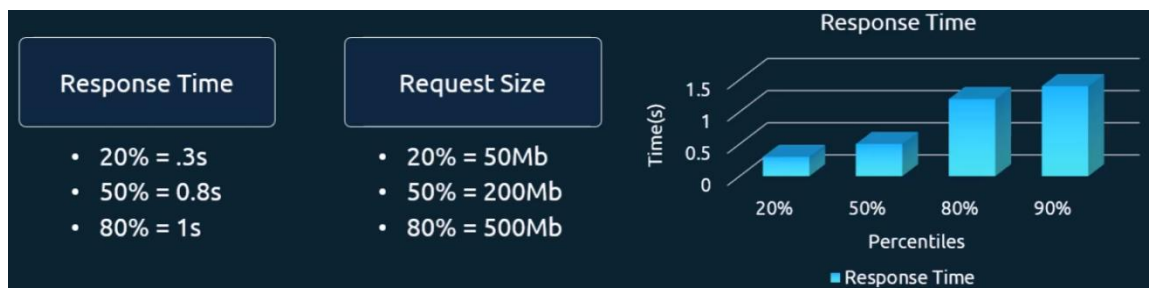
☞ Ex:

Lets say you have decided the buckets: **< 1s**, **< 0.5s**, **< 0.2s**

you want to see how many request's response time comes under these buckets. i.e. how many request completed in less than 1s, in less than 0.5s, in less than 0.2s .



- **Summary** : same as histogram. But, **Histogram deals with the numbers** where as **summary deals with percentage**.



々 Why to use **labels** ?

- Lets say you want to track the metrics of a application like requests_auth_total, requests_user_total (total user's or auth's request being triggered), in this case you are writing multiple metrics.
- Rather we can use one metric and specify labels inside that.
requests_total(path="/auth")
requests_total(path="/user")
like this.
- Its kind of a method having args. You can pass any value to the arguments.

々 **Internal Labels**

- Everything is just another label.

➤ `node_cpu_seconds_total{cpu=0}`

internally it becomes

`{__name__=node_cpu_seconds_total, cpu=0}`

⚡ Metric name is also a label against the name `__name__`

➤ Labels surrounded with double underscore (`__`) are internal to `prometheus`.

々 Every metric is assigned with 2 labels by default: **instance**, **job**

➤ **instance** is the target's **IP:PORT** (end-point in simple term)

➤ **job** is the *job_name* under which the target is present in `prometheus.yml` file.

The image shows a Prometheus UI interface on the left and a terminal window on the right. The UI displays the metric `node_boot_time_seconds` with a table view showing labels `instance="192.168.1.168:9100"` and `job="node"`. The terminal window shows the Prometheus configuration file `prometheus.yml` with the following content:

```
job_name: "node"
scheme: https
basic_auth:
  username: prometheus
  password: password
static_configs:
  - targets: ["192.168.1.168:9100"]
```

Arrows indicate the mapping from the labels in the UI to the configuration in the terminal. The `job="node"` label in the UI points to the `job_name: "node"` line in the terminal. The `instance="192.168.1.168:9100"` label in the UI points to the `targets: ["192.168.1.168:9100"]` line in the terminal.

➤

々 F

Docker Engine Metrics & Docker Container Metrics

々 Docker Engine Metrics

- ✦ It gives info about the overall docker engine like failed image builds count, cpu memory usage of docker engine ..etc.
- ✦ Inside Docker host (where Docker engine is running), update the `/etc/docker/daemon.json` file.

```
{
  "metrics-addr": "127.0.0.1:9323",
  "experimental": true
}
```

[it'll expose the metrics on **9323** port of the host]

[experimental : because some features will not be enabled in non-experimental mode]

- ✦ Then restart the docker service
systemctl restart docker
- ✦ Now it'll be accessible on 9323 port of the host.
- ✦ To scrape the metrics via *Prometheus*, you can give this end-point as the target

```
scrape_configs:
  - job_name: "docker"
    static_configs:
      - targets: ["12.1.13.4:9323"]
```

[assuming host's IP is 12.1.13.4]

々 **cAdvisor Metrics** (used for *Docker Container Metrics*)

- ✦ Its gives info at container level like cpu memory usage of containers, number of processes running inside containers, container uptime ..etc.
- ✦ All container runs inside Host itself, & all the processes and all runs by same host kernel.
So, all the details will be present in host filesystem itself.
- ✦ There are some paths that the container (running **cAdvisor**) needs access to like
 - Root directory of host (`/`)
 - `/var/run`

- `/sys`
- `/var/lib/docker`
- `/dev/disk` (it's a directory, not a device)
- Also, it needs to access the device `/dev/kmsg`
 - You can read directly using `sudo cat /dev/kmsg` but its raw logs.
Use, `dmesg -w` instead.

```
version: '3.4'
services:
  cadvisor:
    image: gcr.io/cadvisor/cadvisor
    container_name: cadvisor
    privileged: true
    devices:
      - "/dev/kmsg:/dev/kmsg"
    volumes:
      - /:/rootfs:ro
      - /var/run:/var/run:ro
      - /sys:/sys:ro
      - /var/lib/docker:/var/lib/docker:ro
      - /dev/disk:/dev/disk:ro
    ports:
      - 8080:8080
```

[docker compose]

- `ro` (each volume mount's end) is **read-only** .
- `privileged: true`
container will have all access to host. By default even if the user is root inside the container it'll not have all the root access.
- Now just run
`docker-compose up -d`
it'll run the container of *cAdvisor* .
- The cAdvisor exposes the metrics on **8080** port at path **`/metrics`** .
As we have mount same 8080 from the host as well, it can be seen using
`localhost:8080/metrics`

- You can scrape this using Prometheus.

```
scrape_configs:  
  - job_name: "cAdvisor"  
    static_configs:  
      - targets: ["12.1.13.4:8080"]
```

々

Monitoring in Kubernetes

Monitoring target in Kubernetes

- **Applications on the cluster** : application level metrics like number of requests, db connections, ..etc
- **Cluster-level components** : control-plane components like API Server, CoreDNS, Scheduler and *Kubelet*, (which provides container level metrics like *cAdvisor*)& metrics exposed by kube-state-metrics container.

Component	What it monitors	Example metrics
kube-state-metrics	Kubernetes object state	pod status, deployment replicas
cAdvisor (inside kubelet)	Container resource usage	CPU, memory, filesystem
Node Exporter	Host/node OS metrics	node CPU, RAM, disk, network
Control plane metrics	API server / scheduler / etcd internals	API latency, scheduler decisions

- You usually do NOT install *cAdvisor* separately in Kubernetes.

kubelet already embeds cAdvisor internally.

Then why kube-state-metrics if cAdvisor already exists?

- Because both solve DIFFERENT problems.
- Example:
Suppose a pod is crashing.
- **kube-state-metrics** tells:
 - ✦ Pod status = CrashLoopBackOff
 - ✦ Restart count = 15
- But it CANNOT tell:
 - ✦ CPU usage
 - ✦ memory spike
- **cAdvisor** tells:
 - ✦ Container memory usage = 950Mi
- Now you understand:
 - ✦ OOMKilled happened because memory usage increased

- **kube-state-metrics** doesn't give info about applications or containers, it gives info about Kubernetes Objects fetching from API Server.

- ♣ Ex: Pod status (running, pending, CrashLoopBackOff), deployment replicas, node status (ready or not), PVC status, HPA status ..etc.

~~♣ It doesn't monitor CPU, memory, network ..etc.~~

々 In each node of kubernetes cluster, one Node Exporter needs to be running to get Node level metrics.

- ♣ it is run as a **DaemonSet** in Kubernetes so that each node will have one Node Exporter running inside a Pod.

- ♣ Node Exporter, when runs as a resource, fetch the details from its host itself and expose those as metrics.

Just like that, when it runs inside a container, if we give access to host directories and files to this container, then Node Exporter can fetch those and expose in the metrics.

[just like cAdvisor]

- ♣ [daemonset.spec.template.spec.containers](#)

```
containers:
- name: node-exporter
  image: prom/node-exporter:latest

  args:
    - --path.procfs=/host/proc
    - --path.sysfs=/host/sys
    - --path.rootfs=/host/root
```

- ♣ In host, Node Exporter fetches the files **/proc, /sys, /root** directly
But container will also have some directory like this (/root) ..etc.
- ♣ So, inside container **/host** directory will contain all the host mounted directories inside container.
- ♣ command field is not given because its already there in node-exporter image as entry point.
If we write command here it'll overwrite that and node-exporter will never execute.
- ♣ So, we are only giving the args so that node exporter will go for /host/any-directory instead of /any-directory.

♣

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- 々 **StatefulSet** : all pods will be having their own identity. Can't be mixed up.

Ex: database.

Even if databases running on different pods contains same data (replicas), then also one should be master and other should be slave so that conflicting db update will not happen.

Stateless : pods running via Deployment are stateless. Any pod can handle any request.

- 々 **Helm**

```
mychart/
├─ chart.yaml
├─ values.yaml
└─ templates/
    ├─ deployment.yaml
    └─ service.yaml
```

♣

[consider this is the files]

- ♣ You need to execute the command

helm install myapp mychart

inside parent directory of **mychart** directory.

- ♣ myapp: release name
- mychart: chart directory name

- ♣ **values.yaml** contains the values that will be dynamically added to *yaml* files inside *template* directory.

Ex:

- ♣ deployment.yaml

```
replicas: {{ .Values.replicaCount }}
```

- ♣ values.yaml

```
replicaCount: 3
```

- ♣ So, 3 will be added in *deployment.yaml*

- ♣ **Chart.yaml** is mainly for *Helm* itself, not related to Kubernetes.

If it is not there, it'll not take the directory as a valid Helm package directory.

- ♣ **helm create mychart**

create Helm chart structure.

- ♣ **helm install** <release name> <chart directory name>
install chart.
 - ♣ **helm list**
list all installed releases.
 - ♣ **--dry-run**
this option to simulate actual installation process.
helm install myapp mychart --dry-run
 - ♣ **--set** overrides the *values.yaml* file's values.
helm install <release name> <chart name> **--set** <key>=<val>
helm install myapp mychart --set replicaCount=3
 - ♣ **helm uninstall** <release name>
helm uninstall my-release
deletes entire release.
 - ♣ **--atomic**
used with *install* and *upgrade* options.
 - ♣ **helm upgrade myapp mychart --atomic**
if upgrade fails Helm restores previous working version automatically.
 - ♣ **--install**
it is used with *helm upgrade*
 - ♣ If exists, upgrade.
Otherwise install.
- ✧ **Readiness probe** : can the pod is able to accept traffic? It must be present on any service's end-point.
- Liveness probe** : without this, once pod is up then it'll think pod is fine. It checks in each interval about status of pod.
- Startup probe** : liveness probe runs after few seconds, if application is slow then it'll restart again. So startup probe is used for for slow startup application.
- Probs starting sequence: startup -> liveness -> readiness
- ✧ Policies are **additive** (NetworkPolicy) means union of all applied policies will be considered.
- ✧ Deployment Strategies:

- ⌘ RollingUpdate : gradually scale-down prev, scale-up new
- ⌘ Recreate: remove prev, create new
- ⌘ **Blue/Green**: keep both versions (prev & new), switch traffic.
- ⌘ **Canary**: new version given to a very few user first. Then gradually increase.

々 PodSecurityAdmissions (PSA)

- ⌘ Security mechanism that controls which kind of pod can be run in a namespace.
- ⌘ Update
`namespace.metadata.labels. pod-security.kubernetes.io/enforce`
- ⌘ `pod-security.kubernetes.io/enforce: restricted`
- ⌘ **privileged, baseline, restricted**
these are the values of PSA

々 **Job**

- ⌘ It just run once.
- ⌘ Job ensures task finishes successfully, If pod fails, Job can recreate it.

々 Cloud Native Principle's immutable infrastructure:

Because you need to push a new version to update.

々 Pod Events

- ⌘ If passes:
Scheduled, Pulled, Created, Started
- ⌘ If fails:
FailedScheduling, FailedMount, BackOff, ErrImagePull

々 **kubectl patch**

- ⌘ Modify part of resource without replacing whole YAML.

```
kubectl patch deployment myapp \
-p '{"spec":{"replicas":3}}'
```

々 kubectl commands for nodes

- ⌘ **kubectl cordon <node name>**
existing pods will be there; new pods will not be scheduled.
- ⌘ **kubectl uncordon <node>**
allows scheduling pods again.

♣ **kubectl drain <node name>**

evict all existing pods. Used for node's maintenance.

々 **kubectl exec -it my-pod -- nslookup kubernetes.default**

♣ Everything after `--` belong to container command.

♣ `kubectl exec -it my-pod` → kubectl part

♣ `nslookup kubernetes.default` → run this inside pod

々 **RBAC is additive.**

♣ You can add rules like

`get pods`

`delete pods`

♣ But you can't do something like

~~`deny get pods`~~

~~`deny delete pods`~~

♣ You can add the rules which will be applied.

To not give access, simply don't write that rule.

[reduce by removing/changing bindings/roles]

々 **quantile percentile**

♣ Example:

```
<=1ms → 10 requests
<=5ms → 50 requests
<=10ms → 90 requests
<=20ms → 100 requests
```

♣ Here percentile means how many request [in percentage out of total number of requests] completed within Xms.

♣ Considering total 100 requests are there,

`<=10ms: 90 requests => 90 percentile`

```
histogram_quantile(
  0.90,
  rate(http_request_duration_seconds_bucket[5m])
)
```

♣

♣ Give the response time within which 90% requests completed.

♣ So, output will be: **10** (ms) (90 requests completed within 10ms)

♣ **pn : nth percentile**

p95 : 95th percentile

々 PromQL functions

- **rate()** : calculates per-second rate increase

`rate(http_requests_total[5m])` : requests/sec over last 5 minutes

- **irate()** : instant/smaller window rate

[more sensitive than rate]

- **increase()** : total increase over a period of time

`increase(http_requests_total[5m])` : total requests added in 5 min

- **sum(), max(), min(), avg()**

- **by()** : group results (just like sql *group by*)

it is not a standalone function, it works with **sum, avg, max, count ..etc.**

`sum by (pod) (rate(http_requests_total[5m]))` : separate sum per pod

- ✎ Suppose metrics are (request counts):

```
pod=a instance=1 → counter = 1000
pod=a instance=2 → counter = 1600
pod=b instance=1 → counter = 500
```

[NOTE: Pod a has 2 instances]

- ✎ After 5 mins it becomes

```
pod=a instance=1 → 1300
pod=a instance=2 → 1900
pod=b instance=1 → 650
```

[a-1: 300, a-2: 300, b: 150 increased]

- ✎ `http_requests_total[5m]`

Range vectors (raw metric values from last 5 min)

```
pod=a instance=1 → [1000, 1100, 1200, 1300]
pod=a instance=2 → [1600, 1700, 1800, 1900]
pod=b instance=1 → [500, 550, 600, 650]
```

- ✎ `rate(http_requests_total[5m])`

Per-second request rate

```
pod=a instance=1 → 1
pod=a instance=2 → 1
pod=b instance=1 → 0.5
```

[1000 -> 1300 in (5 mins = 300 sec); rate/sec = 1]

- ✎ ~~`by (pod) (rate(http_requests_total[5m]))`~~

~~INVALID QUERY~~

⌘ *sum by (pod) (rate(http_requests_total[5m]))*

```
pod=a → 2  
pod=b → 0.5
```

(pod a has 2 instances, so it added both and it became 2)

々 Pod needs a volume in a specific availability zone, but Kubernetes could not find a suitable node in that zone.

Ex: PVC volume created in: *us-east-1a*, pod scheduled in: *us-east-1b*

⌘ It gives error

Pod error “no available volume zone”

々 CNCF governance groups decision-making style: **Consensus-seeking**
consensus-seeking: trying to make decisions with group agreement/discussion instead of one person deciding alone.

々 **PDB** : PodDisruptionBudget (kind: PodDisruptionBudget)

⌘ It defines a minimum number of pods must be running in **cluster** (~~not per node~~)

⌘ *poddisruptionbudget.spec.minAvailable: 2*

means at least 2 pods must be there in the cluster.

⌘ Even if you do

kubect1 drain node01 --force

it'll not drain all the pods if the minimum number of pods will not be maintained in the cluster after evicting the pods from the node node01.

々 **OpenTelemetry**

⌘ Full observability: *metrics, logs, traces*

⌘ Whereas Prometheus only fetches *metrics*

々 **Helm vs Kustomize**

⌘ In both original yaml file will be written.

⌘ In helm:

replicas: {{ .Values.replicaCount }}

⌘ In kustomize:

replicas: 3

⌘ In Helm you write the variables in the **values.yaml** file.

In Kustomize, you write **patch.yaml** file.

- ♣ In patch.yaml you write all the fields that needs to be updated (not all fields, only fields needs to be modified),
it updates with the original yaml file and patch.yaml file.

々 Argo CD **ApplicationSet**

- ♣ You have multiple environments (dev for developers, test for testers, prod for public), then you need to create 3 applications in Argo CD.
- ♣ Using **ApplicationSet** you write one template and specify the environments.

It'll create the applications by itself.

```
environments:  
- dev  
- staging  
- prod
```

- ♣ it'll create

```
myapp-dev  
myapp-staging  
myapp-prod
```

々 **kubectl port-forward**

- ♣ It is used for debugging.
- ♣ If you want to bind host's port to container's port then it is useful.
- ♣ **kubectl port-forward pod/mypod 8080:3000**
host's 8080 will be bound with container's 3000
- ♣ **kubectl port-forward pod/mypod 8080 3000**
here 8080:8080 and 3000:3000
both ports are bound

- ♣ **: matters**

々 **CRD** : Custom Resource Definition

- ♣ Used to add your own Kubernetes object types.
- ♣ If you create your own type of resource like

```
apiVersion: mycompany.com/v1  
kind: Database
```

々 Ingress & TLS are commonly used for HTTPS apps.

- ♣ Browser sends HTTPS request to Ingress. (not traffic, just request)
- ♣ Ingress sends certificate (tls.crt)

- Browser verifies *CA signature* , *domain validity*.
If valid, secure encrypted connection is established.
- This is **HTTPS**
- Now browser sends encrypted traffic (password, tokens ..etc are encrypted so that attackers won't be able to read)
- Ingress **decrypt** the traffic. Using *tls.key*
this is **TLS Termination**
- Now Ingress forwards the traffic to specific service.
At this point its normal HTTP traffic.



- ✧ To get the endpoints of a service
kubectl get endpoints <service name>
- ✧ Give limited permissions just to do the job.
if something gets compromised, damage remains limited.
This is called **limiting blast radius**.
- ✧ Service mesh manages service-to-service communication in microservices architecture.
It provides: security, TLS, circuit breaking, rate limiting, etc
- ✧ **Headless Service**
 - When you create a service, it gets an IP by which other resources can access the service.
 - *service.spec.clusterIP* contains the service's IP.
 - When you create a service with
clusterIP: None
then it is **headless service**.

- Create a headless service and select the labels of some StatefulSet pods.

```
apiVersion: v1
kind: Service

metadata:
  name: mysql-headless

spec:
  clusterIP: None

  selector:
    app: mysql
```

(service)

```
mysql-0
mysql-1
mysql-2
```

- these are the statefulset pods

- Now these can be accessed like this

```
mysql-0.mysql-headless.default.svc.cluster.local
mysql-1.mysql-headless.default.svc.cluster.local
mysql-2.mysql-headless.default.svc.cluster.local
```

- In case of normal service we use like

mysql-headless.default.svc.cluster.local

but in case of headless, we need to mention the pod name as well as prefix

mysql-0.mysql-headless.default.svc.cluster.local

🔗 Gateway

- Ingress handles all the things (entry, TLS, routing) in a single resource which is

kind: Ingress

- In gateway,

Gateway handles entry point, TLS, ports

HTTPRoute handles routing.

- 🔗 If your application wants to access a database running remotely (not inside cluster) then you create a **service** of type **ExternalName**

```
apiVersion: v1
kind: Service

metadata:
  name: mydb

spec:
  type: ExternalName

  externalName: mysql.example.com
```

- Inside cluster: **mydb**

- ♣ It resolves to: mysql.example.com

々 CronJob

- ♣ CronJob creates *Job*.
Job create *Pod*.
Pod run task.
Pod exits after completion.

々 Meaning of **Reconcile** ?

- ♣ continuously checking actual state and making it match desired state

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Important Points

- Try to use *imperative commands* as much as you can in Exam.
- If you forget the keywords and resources, then execute **kubectl api-resources** you'll get the keywords.

```
vagrant@controlplane:~$ kubectl api-resources
```

NAME	SHORTNAMES	APIVERSION	NAMESPACED	KIND
bindings		v1	true	Binding
componentstatuses	cs	v1	false	ComponentStatus
configmaps	cm	v1	true	ConfigMap
endpoints	ep	v1	true	Endpoints
events	ev	v1	true	Event
limitranges	limits	v1	true	LimitRange
namespaces	ns	v1	false	Namespace
nodes	no	v1	false	Node
persistentvolumeclaims	pvc	v1	true	PersistentVolumeClaim
persistentvolumes	pv	v1	false	PersistentVolume
pods	po	v1	true	Pod
podtemplates		v1	true	PodTemplate
replicationcontrollers	rc	v1	true	ReplicationController
resourcequotas	quota	v1	true	ResourceQuota
secrets		v1	true	Secret
serviceaccounts	sa	v1	true	ServiceAccount
services	svc	v1	true	Service
mutatingadmissionpolicies		admissionregistration.k8s.io/v1	false	MutatingAdmissionPolicy
mutatingadmissionpolicybindings		admissionregistration.k8s.io/v1	false	MutatingAdmissionPolicyBinding
mutatingwebhookconfigurations		admissionregistration.k8s.io/v1	false	MutatingWebhookConfiguration

[you can see there is a column 'KIND' which you can directly write in the yaml]

- Shortnames** are also there to use in *imperative commands* which will SAVE TIME.
- If you want to get a documentation regarding any specific resource, you can use **kubectl explain <resource name>** command.

```
vagrant@controlplane:~$ kubectl explain pods
KIND:      Pod
VERSION:   v1

DESCRIPTION:
  Pod is a collection of containers that can run on a host. This resource is
  created by clients and scheduled onto hosts.

FIELDS:
  apiVersion    <string>
    APIVersion defines the versioned schema of this representation of an object.
    Servers should convert recognized schemas to the latest internal value, and
    may reject unrecognized values. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources

  kind          <string>
    Kind is a string value representing the REST resource this object
    represents. Servers may infer this from the endpoint the client submits
    requests to. Cannot be updated. In CamelCase. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds

  metadata      <ObjectMeta>
    Standard object's metadata. More info:
```

- To go deeper into the specific fields of that resources,
kubectl explain <resource name>.<field name>

[you can nest these any number times:

resource.field.sub_field.sub_sub_field etc]

```
vagrant@controlplane:~$ kubectl explain pods.spec
KIND:      Pod
VERSION:   v1

FIELD: spec <PodSpec>

DESCRIPTION:
  Specification of the desired behavior of the pod. More info:
  https://git.k8s.io/community/contributors/devel/sig-architecture/api-convent
  PodSpec is a description of a pod.

FIELDS:
  activeDeadlineSeconds <integer>
    Optional duration in seconds the pod may be active on the node relative to
```

If you

want, more comprehensive, then use

kubectl explain <resource name> --recursive

```
vagrant@controlplane:~$ kubectl explain pods --recursive
KIND:      Pod
VERSION:   v1

DESCRIPTION:
  Pod is a collection of containers that can run on a host.
  created by clients and scheduled onto hosts.

FIELDS:
  apiVersion <string>
  kind <string>
  metadata <ObjectMeta>
    annotations <map[string]string>
    creationTimestamp <string>
    deletionGracePeriodSeconds <integer>
```

々 To get a basic yaml file instead of writing from scratch, use ***--dry-run=client -o yaml*** with the command.

[with *run*, *create* for static] [with *get* if the object is already running]

kubectl create deployment my-dep --image=nginx --dry-run=client -o yaml

kubectl run my-pod --image nginx --dry-run client -o yaml

```
vagrant@controlplane:~$ kubectl run my-pod --image=nginx --dry-run=client -o yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: my-pod
  name: my-pod
spec:
  containers:
  - image: nginx
    name: my-pod
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
```

```

vagrant@controlplane:~$ kubectl create deployment my-dep --image=nginx --dry-run=client -o yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: my-dep
    name: my-dep
spec:
  replicas: 1
  selector:
    matchLabels:
      app: my-dep
  strategy: {}
  template:
    metadata:
      labels:
        app: my-dep
    spec:
      containers:
      - image: nginx
        name: nginx
        resources: {}

```

- ✧ **config** file is present in `~/.kube/config` which contains all the details about *users, clusters, contexts*.
 - ✦ **cluster** contains the list of *kube-api-server endpoint & certificate* (for identity).
 - ✦ **users** contains the list of users each having a *unique name, key, certificate*.
 - ✦ **context** contains the combination of triplet i.e. *user, cluster, namespace*.
[default namespace]
 - ✦ When you switch a context, the user will be updated, the cluster will be updated, and the default namespace will be updated.
You can override the default namespace with the flag `--namespace=<ns name>`
 - ✦ [`kubectl config --help`](#) command you can use to see the usage of different fields of configs.
- ✧ Taint, Tolerations, Node Selector, Node Affinity,
 - ✦ **taint** is to prevent a pod to be scheduled on a node (taint is applied on node).
 - ✦ **toleration** is to make a pod tolerant to get scheduled to a *taint* pod (toleration is applied on pod).

- ♣ **nodeSelector** is used to assign a pod to a node only if the selector labels matches with the node's labels. [HARD selection, if not matches then it is not scheduled].
 - ♣ **nodeAffinity** is advanced version of **nodeSelector**, here **hard** and **soft** selection can be there with advanced selection queries.
 - ♣ Sometimes you need to use both the things **taint**, **toleration** and **nodeAffinity** to solve the real-world problem.
- 々 You can give **requests** and **limits** on the pods.
- ♣ **requests** means the pod's minimum requirement (cpu and memory) to be scheduled to a node.
 - ♣ **limits** means the pod can use that much amount of resources at max.
 - ✧ If **cpu** usage exceeds the **limit**, then it **slows down**.
 - ✧ If **memory** usage exceeds, then pod is **killed**.
 - ♣ Instead of giving this **requests** and **limits** in all the pods, you can create **LimitRange** (kind) which will contain the **request**, **limits**, **max**, **min** .
 - ✧ If any pod doesn't have the request and limit fields, then it'll inject its default values.
 - ♣ **LimitRange** works at **namespace** level.
- 々 **DaemonSet** is same as **ReplicaSet**, it just create a replica of the pod in each worker node.
- 々 **kubeadm** creates static pods on control plane nodes to run the controlplane resources.
- 々 API Server stores the details of all the pods in **etcd** (*normal pods* or *static pods*).
- in the annotation 2 extra fields are there in case of *static pods* which differentiate it from the normal pods.
- ♣ When you edit a pod using **kubectl edit** command, it updates etcd and send **kubelet** (in streaming connection) the new **PodSpec** then **kubelet** compares it with the existing **PodSpec**.
Then changes if allowed otherwise reject.
In each changes, **containerd** creates new container and the container ID is changed.

- ⚡ In case of *static pods*, you can't change using *kubectrl edit* command in control plane nodes.

You need to update the *yaml* file inside the *manifest* directory of the target node (where the static pod is running).

- 々 READ THE THEORY HAVING HEADING “Resources inside Pods vs Resources as normal Processes”; DON'T SKIP IT.

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Linux Specific

- 々 **more** <file path>
read file in scrollable manner.
- 々 **alias** <command> <alias> (in bash, not sh)
alias date dt
- 々 **\$SHELL** : this variable contains current shell.
- 々 **\$PATH** : colon (:) separated paths, if its not there then command can't run.
To add a path
export PATH=\$PATH:<required path>
- 々 **chsh** : to change current shell
- 々 **uname -r** : kernel current version
<kernel version>:<major version>:<minor version>:<patch release>-
<distro specific info>
(output format)
- 々 Memory management
 - ♣ Kernel space
kernel and kernel related processes can only access this.
They have access to hardware directly.
Ex: device drivers
 - ♣ User space
application/programs, make *system calls* to kernel to get hardware.
- 々 Lets say you plug a pen-drive or hard-disk.
Kernel space contains **device drivers** which detects this and emit **uevents** .
User all *uevents* is sent to *User space*'s **device manager** daemon which is **udev**.
This *udev* dynamically creates a device (ex: /dev/sdb1) and attach it to the device.
- 々 Commands
 - ♣ **dmesg** : display messages of a area of kernel called *ring buffer*. Also logs from hardware.
 - ♣ **lspci** : info about all *pci* devices configured in system.

↪ **lsblk** : list block devices

disks & their partitions.

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
sda	8:0	0	100G	0	disk	
└─sda1	8:1	0	1G	0	part	/boot
└─sda2	8:2	0	50G	0	part	/
└─sda3	8:3	0	49G	0	part	/home
sdb	8:16	1	16G	0	disk	
└─sdb1	8:17	1	16G	0	part	/media/usb

[sda: disk, sda1 to sda3

partitions]

↪ **lscpu** : info about CPU

↪ **lsmem** : info about memory

↪ **free -m** : total vs used memory

↪ **lshw** : details about the hardware

々 Everything is file in linux

3 types of files:

↪ Regular file : image, scripts, configuration/data files

↪ Directory : /home/username, /root, ..etc

↪ Special files :

it is of 5 types

↪ Character files : devices to communicate with IO devices. Present inside **/dev** directory.

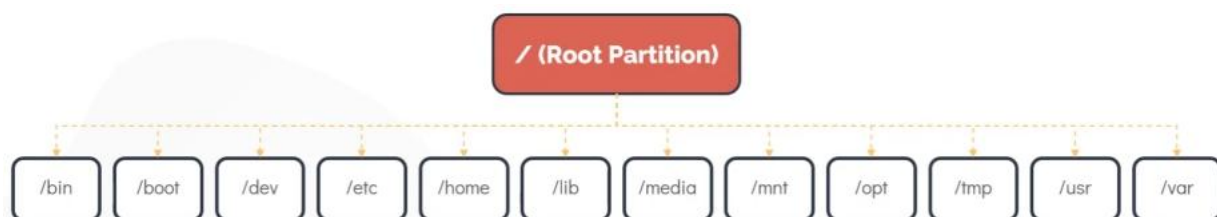
↪ Block files : block devices (hard-disk ..etc). it is also present inside **/dev** directory.

↪ Links : can be *hard links*, *soft links* .

↪ Socket: enables communication between 2 processes.

↪ Named Pipes:

々 Filesystem hierarchy



↪ **/opt** : third party programs installation directory.

↪ **/mnt** : mount filesystems *temporarily*.

- ♫ **/tmp** : keep temporary data.
- ♫ **/media** : all external media are mounted in this.
df -hp (displays all mounts)
- ♫ **/bin** : all commands like cp, mv ..etc present here.
- ♫ **/etc** : most of the configuration files are present.
- ♫ **/lib** : lib64

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